

APPENDIX A

8717644



THE UNITED STATES OF AMERICA

TO ALL TO WHOM THESE PRESENTS SHALL COME:

UNITED STATES DEPARTMENT OF COMMERCE

United States Patent and Trademark Office

May 14, 2026

**THIS IS TO CERTIFY THAT ANNEXED IS A TRUE COPY FROM THE
RECORDS OF THIS OFFICE OF THE FILE WRAPPER AND CONTENTS
OF:**

APPLICATION NUMBER: 19/088,110

FILING DATE: *March 24, 2025*

PATENT NUMBER: 12530925

ISSUE DATE: *January 20, 2026*



Certified by

DIRECTOR OF THE UNITED STATES PATENT AND TRADEMARK OFFICE

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Peter SCHILLEN : Art Unit:
Serial No.: :
Filed: 2024-03-24 : Examiner:
Confirmation No.: :
For: OPTICAL SKIN DETECTION FOR FACE :
UNLOCK

March 24, 2025

CERTIFICATION STATEMENT

In accordance with 37 C.F.R. 1.97 and 1.98 and MPEP 609, and in compliance with the duty of disclosure set forth in 37 C.F.R. 1.56, Applicants submit the attached PTO/SB/08A for consideration by the Patent and Trademark Office in the above-entitled application and to be made of record therein. Copies of the references that have not been provided in accordance with 37 C.F.R. § 1.98(d) because they are of record in a prior application (Serial Number 18/468,352) relied upon for an earlier filing date under 35 U.S.C. § 120.

Respectfully submitted,

/Christopher M. Goff/
Christopher M. Goff, Reg. No. 41,785
Armstrong Teasdale LLP
7700 Forsyth Blvd., Suite 1800
St. Louis, MO 63105

CMG/EMF/ZSW

Via EFS

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it contains a valid OMB control number.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		
The application data sheet is part of the provisional or nonprovisional application for which it is being submitted. The following form contains the bibliographic data arranged in a format specified by the United States Patent and Trademark Office as outlined in 37 CFR 1.76. This document may be completed electronically and submitted to the Office in electronic format using the Electronic Filing System (EFS) or the document may be printed and included in a paper filed application.			

Secrecy Order 37 CFR 5.2:

☐ Portions or all of the application associated with this Application Data Sheet may fall under a Secrecy Order pursuant to 37 CFR 5.2 (Paper filers only. Applications that fall under Secrecy Order may not be filed electronically.)

Inventor Information:

Inventor 1					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Peter		SCHILLEN		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE		
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 2					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Benjamin		GUTHIER		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE		
Mailing Address of Inventor:					
Address 1		Industriestrasse 35			
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
Inventor 3					Remove
Legal Name					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Prefix	Given Name	Middle Name	Family Name	Suffix
	Friedrich		SCHICK	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE	

Mailing Address of Inventor:				
Address 1	Industriestrasse 35			
Address 2				
City	Ludwigshafen am Rhein	State/Province		
Postal Code	67063	Country ⁱ	DE	
Inventor 4				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Manuel		GUENTHER	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Zuerich	Country of Residence ⁱ	CH	

Mailing Address of Inventor:				
Address 1	Andreastrasse 15			
Address 2				
City	Zuerich	State/Province		
Postal Code	8050	Country ⁱ	CH	
Inventor 5				Remove
Legal Name				

Prefix	Given Name	Middle Name	Family Name	Suffix
	Lars		DIESSELBERG	
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service				
City	Karlsruhe	Country of Residence ⁱ	DE	

Mailing Address of Inventor:				
Address 1	Geibelstr. 22			
Address 2				
City	Karlsruhe	State/Province		
Postal Code	76185	Country ⁱ	DE	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
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Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Inventor 6					Remove
Legal Name					
Prefix	Given Name	Middle Name	Family Name	Suffix	
	Christian		LENNARTZ		
Residence Information (Select One) ☐ US Residency ☒ Non US Residency ☐ Active US Military Service					
City	Ludwigshafen am Rhein	Country of Residence ⁱ	DE		
Mailing Address of Inventor:					
Address 1	Industriestrasse 35				
Address 2					
City	Ludwigshafen am Rhein	State/Province			
Postal Code	67063	Country ⁱ	DE		
All Inventors Must Be Listed - Additional Inventor Information blocks may be generated within this form by selecting the Add button. Add					

Correspondence Information:

Enter either Customer Number or complete the Correspondence Information section below. For further information see 37 CFR 1.33(a).			
☐ An Address is being provided for the correspondence Information of this application.			
Customer Number	76656		
Email Address	uspatents@atllp.com	Add Email	Remove Email

Application Information:

Title of the Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		
Attorney Docket Number	27843-2771 202595US04	Small Entity Status Claimed	☐
Application Type	Nonprovisional		
Subject Matter	Utility		
Total Number of Drawing Sheets (if any)	2	Suggested Figure for Publication (if any)	

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
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Filing By Reference:

Only complete this section when filing an application by reference under 35 U.S.C. 111(c) and 37 CFR 1.57(a). Do not complete this section if application papers including a specification and any drawings are being filed. Any domestic benefit or foreign priority information must be provided in the appropriate section(s) below (i.e., "Domestic Benefit/National Stage Information" and "Foreign Priority Information").

For the purposes of a filing date under 37 CFR 1.53(b), the description and any drawings of the present application are replaced by this reference to the previously filed application, subject to conditions and requirements of 37 CFR 1.57(a).

Application number of the previously filed application	Filing date (YYYY-MM-DD)	Intellectual Property Authority or Country

Publication Information:

☐ Request Early Publication (Fee required at time of Request 37 CFR 1.219)

☐ **Request Not to Publish.** I hereby request that the attached application not be published under 35 U.S.C. 122(b) and certify that the invention disclosed in the attached application **has not and will not** be the subject of an application filed in another country, or under a multilateral international agreement, that requires publication at eighteen months after filing.

Representative Information:

Representative information should be provided for all practitioners having a power of attorney in the application. Providing this information in the Application Data Sheet does not constitute a power of attorney in the application (see 37 CFR 1.32). Either enter Customer Number or complete the Representative Name section below. If both sections are completed the customer Number will be used for the Representative Information during processing.

Please Select One:	☒ Customer Number	☐ US Patent Practitioner	☐ Limited Recognition (37 CFR 11.9)
Customer Number	76656		

Domestic Benefit/National Stage Information:

This section allows for the applicant to either claim benefit under 35 U.S.C. 119(e), 120, 121, 365(c), or 386(c) or indicate National Stage entry from a PCT application. Providing benefit claim information in the Application Data Sheet constitutes the specific reference required by 35 U.S.C. 119(e) or 120, and 37 CFR 1.78.

When referring to the current application, please leave the "Application Number" field blank.

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
	Continuation of	18468352	2023-09-15

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
18468352	Continuation of	18264326	2023-08-04
Prior Application Status	Pending	Remove	
Application Number	Continuity Type	Prior Application Number	Filing or 371(c) Date (YYYY-MM-DD)
18264326	a 371 of international	PCT/EP22/53941	2022-02-17
Additional Domestic Benefit/National Stage Data may be generated within this form by selecting the Add button. Add			

Foreign Priority Information:

This section allows for the applicant to claim priority to a foreign application. Providing this information in the application data sheet constitutes the claim for priority as required by 35 U.S.C. 119(b) and 37 CFR 1.55. When priority is claimed to a foreign application that is eligible for retrieval under the priority document exchange program (PDX)ⁱ the information will be used by the Office to automatically attempt retrieval pursuant to 37 CFR 1.55(i)(1) and (2). Under the PDX program, applicant bears the ultimate responsibility for ensuring that a copy of the foreign application is received by the Office from the participating foreign intellectual property office, or a certified copy of the foreign priority application is filed, within the time period specified in 37 CFR 1.55(g)(1).

Application Number	Country ⁱ	Filing Date (YYYY-MM-DD)	Remove
21157800.0	EP	2021-02-18	Access Code ⁱ (if applicable)
Additional Foreign Priority Data may be generated within this form by selecting the Add button. Add			

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications

☐ This application (1) claims priority to or the benefit of an application filed before March 16, 2013 and (2) also contains, or contained at any time, a claim to a claimed invention that has an effective filing date on or after March 16, 2013.

NOTE: By providing this statement under 37 CFR 1.55 or 1.78, this application, with a filing date on or after March 16, 2013, will be examined under the first inventor to file provisions of the AIA.

Application Data Sheet 37 CFR 1.76	Attorney Docket Number	27843-2771 202595US04
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Authorization or Opt-Out of Authorization to Permit Access:

When this Application Data Sheet is properly signed and filed with the application, applicant has provided written authority to permit a participating foreign intellectual property (IP) office access to the instant application-as-filed (see paragraph A in subsection 1 below) and the European Patent Office (EPO) access to any search results from the instant application (see paragraph B in subsection 1 below).

Should applicant choose not to provide an authorization identified in subsection 1 below, applicant **must opt-out** of the authorization by checking the corresponding box A or B or both in subsection 2 below.

NOTE: This section of the Application Data Sheet is **ONLY** reviewed and processed with the **INITIAL** filing of an application. After the initial filing of an application, an Application Data Sheet cannot be used to provide or rescind authorization for access by a foreign IP office(s). Instead, Form PTO/SB/39 or PTO/SB/69 must be used as appropriate.

1. Authorization to Permit Access by a Foreign Intellectual Property Office(s)

A. Priority Document Exchange (PDX) - Unless box A in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the European Patent Office (EPO), the Japan Patent Office (JPO), the Korean Intellectual Property Office (KIPO), the State Intellectual Property Office of the People's Republic of China (SIPO), the World Intellectual Property Organization (WIPO), and any other foreign intellectual property office participating with the USPTO in a bilateral or multilateral priority document exchange agreement in which a foreign application claiming priority to the instant patent application is filed, access to: (1) the instant patent application-as-filed and its related bibliographic data, (2) any foreign or domestic application to which priority or benefit is claimed by the instant application and its related bibliographic data, and (3) the date of filing of this Authorization. See 37 CFR 1.14(h)(1).

B. Search Results from U.S. Application to EPO - Unless box B in subsection 2 (opt-out of authorization) is checked, the undersigned hereby **grants the USPTO authority** to provide the EPO access to the bibliographic data and search results from the instant patent application when a European patent application claiming priority to the instant patent application is filed. See 37 CFR 1.14(h)(2).

The applicant is reminded that the EPO's Rule 141(1) EPC (European Patent Convention) requires applicants to submit a copy of search results from the instant application without delay in a European patent application that claims priority to the instant application.

2. Opt-Out of Authorizations to Permit Access by a Foreign Intellectual Property Office(s)

☐ A. Applicant **DOES NOT** authorize the USPTO to permit a participating foreign IP office access to the instant application-as-filed. If this box is checked, the USPTO will not be providing a participating foreign IP office with any documents and information identified in subsection 1A above.

☐ B. Applicant **DOES NOT** authorize the USPTO to transmit to the EPO any search results from the instant patent application. If this box is checked, the USPTO will not be providing the EPO with search results from the instant application.

NOTE: Once the application has published or is otherwise publicly available, the USPTO may provide access to the application in accordance with 37 CFR 1.14.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Applicant Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.			
Applicant 1			Remove
If the applicant is the inventor (or the remaining joint inventor or inventors under 37 CFR 1.45), this section should not be completed. The information to be provided in this section is the name and address of the legal representative who is the applicant under 37 CFR 1.43; or the name and address of the assignee, person to whom the inventor is under an obligation to assign the invention, or person who otherwise shows sufficient proprietary interest in the matter who is the applicant under 37 CFR 1.46. If the applicant is an applicant under 37 CFR 1.46 (assignee, person to whom the inventor is obligated to assign, or person who otherwise shows sufficient proprietary interest) together with one or more joint inventors, then the joint inventor or inventors who are also the applicant should be identified in this section.			
Clear			
☒ Assignee	☐ Legal Representative under 35 U.S.C. 117	☐ Joint Inventor	
☐ Person to whom the inventor is obligated to assign.		☐ Person who shows sufficient proprietary interest	
If applicant is the legal representative, indicate the authority to file the patent application, the inventor is:			
Name of the Deceased or Legally Incapacitated Inventor:			
If the Applicant is an Organization check here. ☒			
Organization Name	trinamiX GmbH		
Mailing Address Information For Applicant:			
Address 1	Industriestrasse 35		
Address 2			
City	Ludwigshafen Am Rhein	State/Province	
Country	DE	Postal Code	67063
Phone Number		Fax Number	
Email Address			
Additional Applicant Data may be generated within this form by selecting the Add button. Add			

Assignee Information including Non-Applicant Assignee Information:

Providing assignment information in this section does not substitute for compliance with any requirement of part 3 of Title 37 of CFR to have an assignment recorded by the Office.

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

Assignee 1				
Complete this section if assignee information, including non-applicant assignee information, is desired to be included on the patent application publication. An assignee-applicant identified in the "Applicant Information" section will appear on the patent application publication as an applicant. For an assignee-applicant, complete this section only if identification as an assignee is also desired on the patent application publication.				
Remove				
If the Assignee or Non-Applicant Assignee is an Organization check here. ☐				
Prefix	Given Name	Middle Name	Family Name	Suffix
Mailing Address Information For Assignee including Non-Applicant Assignee:				
Address 1				
Address 2				
City		State/Province		
Country ⁱ		Postal Code		
Phone Number		Fax Number		
Email Address				
Additional Assignee or Non-Applicant Assignee Data may be generated within this form by selecting the Add button. Add				

Signature:					Remove
NOTE: This Application Data Sheet must be signed in accordance with 37 CFR 1.33(b). However, if this Application Data Sheet is submitted with the INITIAL filing of the application and either box A or B is not checked in subsection 2 of the "Authorization or Opt-Out of Authorization to Permit Access" section, then this form must also be signed in accordance with 37 CFR 1.14(c).					
This Application Data Sheet must be signed by a patent practitioner if one or more of the applicants is a juristic entity (e.g., corporation or association). If the applicant is two or more joint inventors, this form must be signed by a patent practitioner, all joint inventors who are the applicant, or one or more joint inventor-applicants who have been given power of attorney (e.g., see USPTO Form PTO/AIA/81) on behalf of all joint inventor-applicants.					
See 37 CFR 1.4(d) for the manner of making signatures and certifications.					
Signature	/Christopher M. Goff/			Date (YYYY-MM-DD)	2025-03-24
First Name	Christopher M.	Last Name	Goff	Registration Number	41785
Additional Signature may be generated within this form by selecting the Add button. Add					

Application Data Sheet 37 CFR 1.76		Attorney Docket Number	27843-2771 202595US04
		Application Number	
Title of Invention	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

This collection of information is required by 37 CFR 1.76. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 23 minutes to complete, including gathering, preparing, and submitting the completed application data sheet form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

- 1 The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
- 2 A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
- 3 A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
- 4 A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
- 5 A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
- 6 A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
- 7 A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
- 8 A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
- 9 A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

TRANSMITTAL FOR POWER OF ATTORNEY TO ONE OR MORE REGISTERED PRACTITIONERS

NOTE: This form is to be submitted with the Power of Attorney by Applicant form (PTO/AIA/82B) to identify the application to which the Power of Attorney is directed, in accordance with 37 CFR 1.5, unless the application number and filing date are identified in the Power of Attorney by Applicant form. If neither form PTO/AIA/82A nor form PTO/AIA/82B identifies the application to which the Power of Attorney is directed, the Power of Attorney will not be recognized in the application.

Application Number	
Filing Date	2025-03-24
First Named Inventor	Peter SCHILLEN
Title	OPTICAL SKIN DETECTION FOR FACE UNLOCK
Art Unit	
Examiner Name	
Attorney Docket Number	278432-2771 202595US04

SIGNATURE of Applicant or Patent Practitioner

Signature	/Christopher M. Goff/	Date (Optional)	2025-03-24
Name	Christopher M. Goff	Registration Number	41785
Title (if Applicant is a juristic entity)			
Applicant Name (if Applicant is a juristic entity)			

NOTE: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. If more than one applicant, use multiple forms.



*Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

POWER OF ATTORNEY BY APPLICANT

I hereby revoke all previous powers of attorney given in the application identified in either the attached transmittal letter or the boxes below.

Application Number

Filing Date

(Note: The boxes above may be left blank if information is provided on form PTO/AIA/82A.)

☒ I hereby appoint the Patent Practitioner(s) associated with the following Customer Number as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above:

76656

OR

☐ I hereby appoint Practitioner(s) named in the attached list (form PTO/AIA/82C) as my/our attorney(s) or agent(s), and to transact all business in the United States Patent and Trademark Office connected therewith for the patent application referenced in the attached transmittal letter (form PTO/AIA/82A) or identified above. (Note: Complete form PTO/AIA/82C.)

Please recognize or change the correspondence address for the application identified in the attached transmittal letter or the boxes above to:

☒ The address associated with the above-mentioned Customer Number

OR

☐ The address associated with Customer Number:

OR

☐ Firm or Individual Name

Address

City

State

Zip

Country

Telephone

Email

I am the Applicant (if the Applicant is a juristic entity, list the Applicant name in the box):

trinamiX GmbH

- ☐ Inventor or Joint Inventor (title not required below)
- ☐ Legal Representative of a Deceased or Legally Incapacitated Inventor (title not required below)
- ☒ Assignee or Person to Whom the Inventor is Under an Obligation to Assign (provide signer's title if applicant is a juristic entity)
- ☐ Person Who Otherwise Shows Sufficient Proprietary Interest (e.g., a petition under 37 CFR 1.46(b)(2) was granted in the application or is concurrently being filed with this document) (provide signer's title if applicant is a juristic entity)

SIGNATURE of Applicant for Patent

The undersigned (whose title is supplied below) is authorized to act on behalf of the applicant (e.g., where the applicant is a juristic entity).

Signature

Date (Optional)

2020/06/09

Name

Dr. Hans Graf

Title

Patent Attorney

NOTE: Signature - This form must be signed by the applicant in accordance with 37 CFR 1.33. See 37 CFR 1.4 for signature requirements and certifications. If more than one applicant, use multiple forms.

☒ Total of 1 forms are submitted.

This collection of information is required by 37 CFR 1.131, 1.32, and 1.33. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 3 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Box No. VIII (iv) DECLARATION: INVENTORSHIP (only for the purposes of the designation of the United States of America)
The declaration must conform to the following standardized wording provided for in Section 214; see Notes to Boxes Nos. VIII, VIII (i) to (v) (in general) and the specific Notes to Box No. VIII (iv). If this Box is not used, this sheet should not be included in the request.

**Declaration of inventorship (Rules 4.17(iv) and 51bis.1(a)(iv))
 for the purposes of the designation of the United States of America:**

I hereby declare that I believe I am the original inventor or an original joint inventor of a claimed invention in the application.

This declaration is directed to the international application of which it forms a part (if filing declaration with application).

This declaration is directed to international application No. (if furnishing declaration pursuant to Rule 26ter).

I hereby declare that the above-identified international application was made or authorized to be made by me.

I hereby acknowledge that any willful false statement made in this declaration is punishable under 18 U.S.C. 1001 by fine or imprisonment of not more than five (5) years, or both.

Name: SCHILLEN, Peter
Residence: Germany
Mailing Address: Industriestraße 35
 67063 Ludwigshafen

Inventor's Signature:
 (The signature must be that of the inventor, not that of the agent)

Date: 08.09.21

Name: GUTHIER, Benjamin
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 67063 Ludwigshafen

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Date: 8.9.21

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Date: 21.9.21

☐ This declaration is continued on the following sheet, "Continuation of Box No. VIII (v)".

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Name: GUENTHER, Manuel
Residence: Switzerland
Mailing Address: Leutschenbachstrasse 46
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Date: 10.05.2021

Name: DIESELBERG, Lars
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Mailing Address: Geibelstr. 22
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Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Name: LENNARTZ, Christian
Residence: Germany
Mailing Address: Industriestraße 35
67063 Ludwigshafen

Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

Date:

Continuation of Box No. VIII (i) to (v) DECLARATION

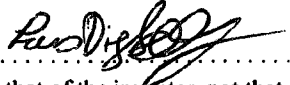
If the space is insufficient in any of Boxes Nos. VIII (i) to (v) to furnish all the information, including in the case where more than three inventors are to be named in Box No. VIII (iv), in such case, write "Continuation of Box No. VIII ..." (indicate the item number of the Box) and furnish the information in the same manner as required for the purposes of the Box in which the space was insufficient. If additional space is needed in respect of two or more declarations, a separate continuation box must be used for each such declaration. If this Box is not used, this sheet should not be included in the request.

Name: GUENTHER, Manuel
Residence: Switzerland
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Inventor's Signature:
(The signature must be that of the inventor, not that of the agent)

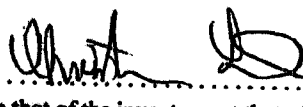
Date:

Name: DIESSELBERG, Lars
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Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 27.4.2021

Name: LENNARTZ, Christian
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Mailing Address: Industriestraße 35
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Inventor's Signature: 
(The signature must be that of the inventor, not that of the agent)

Date: 27.4.21

FIG.1

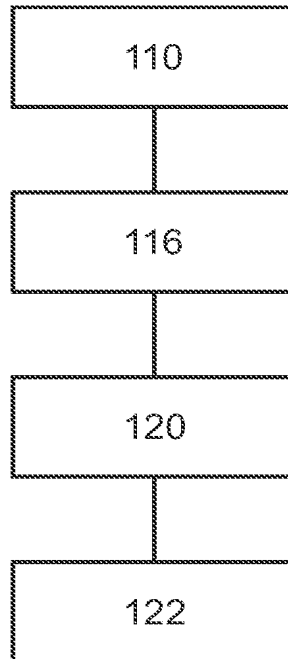


FIG.2

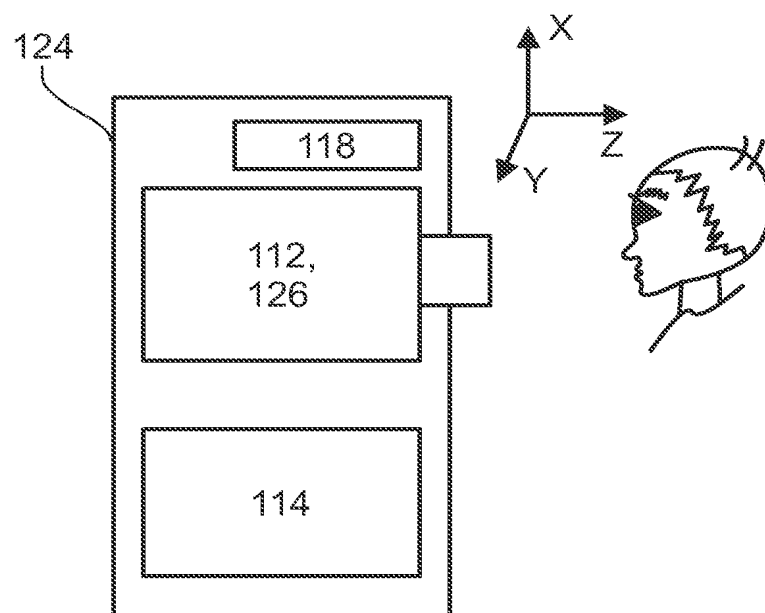
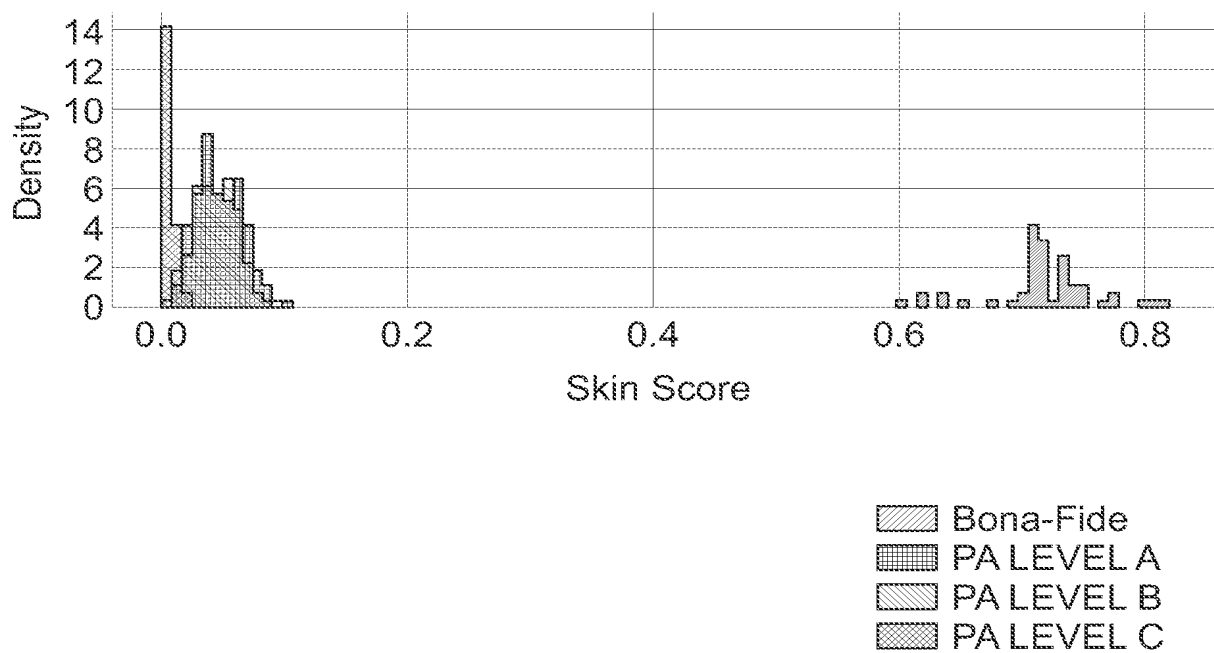


FIG.3





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APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	03/24/2025 02:49:27 PM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	69707303	FILING DATE	-
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 9

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-03-24 ADS 27843-2771.pdf	10	Application Data Sheet	2180 KB
2025-03-24 Track One Request 27843-2771.pdf	2	Track One Request	181 KB
2025-03-24 Track One Application As Filed 27843-2771-APP.TEXT.docx	69	Application body structured text document	109 KB

Warning: At least one claim appears to contain more than one sentence. Please review and revise if necessary. At least one claim appears to not end with a period. Please review and revise if necessary. The abstract should be in narrative form and generally limited to a single paragraph preferably within the range of 50 to 150 words in length. See MPEP 608.01(b).

2025-03-24 Executed Declarations 27843-2771.pdf	3	Oath or Declaration filed	198 KB
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2025-03-24 POA 27843-2771.pdf	2	Power of Attorney	1451 KB
2025-03-24 IDS 27843-2771.pdf	6	Information Disclosure Statement (IDS) Form (SB08)	1298 KB
2025-03-24 IDS Certification Statement 27843-2771.pdf	1	Transmittal Letter	93 KB
2025-03-24 Con Figures 27843-2771.pdf	2	Drawings-only black and white line drawings	25 KB

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2025-03-24 Track One Application As Filed 27843-2771.pdf	69	Auxiliary PDF of Application	952 KB
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Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-03-24 ADS 27843-2771.pdf	9726AE33722D426CE7D0524B181463338DCB3E41B09BF361B2C4CECEC03A2501BDFF34F509E41D6D650E4059CEA089020773FA87A94B7937750C56045A02DFC5
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2025-03-24 Executed Declarations 27843-2771.pdf	E9D435B101AD3F469EE3DA84A5C96C2D999F607613AA846F4D996A0FFC5D31617F10D8CF73FF603A16CA1B846220E5F9D5649DFDFEF25285E364FCE09A350C0C
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2025-03-24 IDS Certification Statement 27843-2771.pdf	6B294F0D2B8D804EFD9FB65268FDF542AAFDF6C092FC69C17C8EADD4D2E47DFBBFAEA09A3BEC7ED6597F9A80BF1B00E83D72D7F2599A4B412F41E18D62525C82

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2025-03-24 Track One Application As Filed 27843-2771.pdf	74B050359AFE663EB5EF4097B2836DF885845BDA1F4859937 CA0308546FF623CFE15C33565FB3C1438B4BFE2EF9DC0A90 D58E9E4144C853C70FAF4AA075342B7

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If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		2025-03-24
	First Named Inventor	Peter SCHILLEN	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	27843-2771 202595US04	

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	1	20150374277	A1	2015-12-31	PATWARDHAN	
	2	20160242853	A1	2016-08-25	BANGERAL et al.	
	3	20190213309	A1	2019-07-11	MORESTIN et al.	
	4	20200082155	A1	2020-03-12	KALSCHEUR et al.	
	5	20200211180	A1	2020-07-02	GILLIES et al.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number			
	Filing Date		2025-03-24	
	First Named Inventor	Peter SCHILLEN		
	Art Unit			
	Examiner Name			
	Attorney Docket Number		27843-2771 202595US04	

	6	20210049391	A1	2021-02-18	ZOU et al.	
	7	20220222464	A1	2022-07-14	DIEDRICH et al.	
	8	20240119130	A1	2024-04-11	HU et al.	
	9	20160335483	A1	2016-11-17	PFURISCH et al.	

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FOREIGN PATENT DOCUMENTS

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	1	111164610	CN	A	2020-05-15	SHENZHEN GOODIX TECH CO LTD.	English-language translation attached	☒

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NON-PATENT LITERATURE DOCUMENTS

Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion for PCT/EP2022/053941 dated 2022-06-09, 12 pages	☐

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EXAMINER SIGNATURE

Examiner Signature		Date Considered	
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*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number		
Filing Date		2025-03-24
First Named Inventor	Peter SCHILLEN	
Art Unit		
Examiner Name		
Attorney Docket Number		27843-2771 202595US04

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		2025-03-24
	First Named Inventor	Peter SCHILLEN	
	Art Unit		
	Examiner Name		
	Attorney Docket Number	27843-2771 202595US04	

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached timing statement under 37 CFR 1.97(e).

☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☒ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2025-03-24
Name/Print	Christopher M. Goff	Registration Number	41785

INFORMATION DISCLOSURE STATEMENT BY APPLICANT

Application Number	
Filing Date	2025-03-24
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2771 202595US04

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

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5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
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7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Optical Skin Detection for Face UnlockCROSS-REFERENCE TO RELATED APPLICATIONS

5 This application is a continuation application of U.S. Application No. 18/468352, filed
September 15, 2023, which is a continuation application of U.S. Application No. 18/264326,
filed August 4, 2023, which is a U.S. National Phase Application of International Patent
Application No. PCT/EP22/053941, filed February 17, 2022, which claims priority to EP
Patent Application No. 21157800.0, filed February 18, 2021, each of which is hereby
10 incorporated by reference herein.

Description

15 Field of the invention

The invention relates to a method for face authentication, a mobile device and various uses
of the method. The devices, methods and uses according to the present invention
specifically may be employed for example in various areas of daily life, security technology,
20 gaming, traffic technology, production technology, photography such as digital photography
or video photography for arts, documentation or technical purposes, safety technology,
information technology, agriculture, crop protection, maintenance, cosmetics, medical
technology or in the sciences. However, other applications are also possible.

25 Prior art

In today's digital world secure access to information technology is an essential requirement
of any state-of-the-art system. Standard concepts like passphrases or PIN codes are
currently extended, combined or even replaced by biometric methods like fingerprint or face
30 recognition. Although passphrases can provide a high level of security if chosen carefully
and at adequate length, this step requires attention and the need of memorizing several
potentially long phrases depending on the IT landscape. Furthermore, it is never guaranteed
that the person providing the passphrase is authorized by the owner of the passphrase or
the digital device. In contrast biometric features like faces or fingerprints are unique and
35 person-specific properties. Therefore, using features derived from these not only can be
more convenient than passphrases/PIN codes but also more secure since they combine
personal identity with an unlock process.

Unfortunately, similarly to being able to spy passwords, fingerprints and faces can be also
40 artificially created in order to impersonate a legitimate user. Automatic face recognition
tools of the first generation use digital camera images or image streams and apply 2D
image processing methods to extract characteristic features, while machine learning
techniques are used to generate face templates to perform identity recognition based on
those features. The second generation of face recognition algorithms uses deep
45 convolutional neuronal networks instead of hand-crafted image features to generate the
classification model.

However, both approaches can be attacked, e.g., with a high-quality photograph that is nowadays freely available to be downloaded from the Internet. Therefore, the idea of presentation attack detection (PAD) gained significance. Early approaches were designed to protect against simple attacks like presenting the photograph of a legitimate user by recording a sequence of images and test for time dependent features, for example small but natural changes in the head position or eye blinking. These methods can again be tricked by playing a pre-recorded video of the user or by careful animations generated from publicly available photographs. In order to exclude displays as potential spoof objects, near infra-red (NIR) cameras can be used, since displays emit photons only in the visible regime of the electromagnetic spectrum. As a further counteraction, 3D cameras were introduced, which can clearly distinguish between planar photos or tablets with playback videos and a 3D face. Still, these systems can be attacked with high quality masks generated, for example, by 3D printing, careful 3D-arrangement of 2D photos, or hand-crafted silicone or latex masks to name a few approaches. Since masks are typically worn by humans, liveness detection based on small motions fails as well. However, these types of masks can be rejected by a PAD system that is able to classify human skin from other materials such as described in EP application No. 20159984.2 filed on February 28, 2020 and EP application No. 20 154 961.5190679 filed on January 31, 2020, the full content of which is included by reference.

A further problem might arise from differences in skin optical properties with respect ethnic origin. A reliable recognition and PAD technology needs to be fully agnostic with respect to these different origins.

Besides security considerations, also the speed of the unlock process and the required computational power are important to provide an acceptable user experience. High-speed face recognition can be used for several tasks after the successful unlock of a device, for example to check whether the user is still in front of the display or to launch further secure applications like banking apps by performing an on the fly check of the person in front of the display. Again, speed and computational resources have great impact on the user experience.

Current 3D algorithms are computationally very demanding and presentation attack detections require several video frames to be processed. Therefore, expensive hardware needs to be involved to deliver acceptable unlock performance. Furthermore, high power consumption is a consequence.

Summarizing, current methods for face unlock do not provide the ability to reliably detect spoof attacks by 3D masks and perform this task at a speed below human detection limit.

US 2019/213309 A1 describes a system and method of authenticating a user's face with a ranging sensor. The ranging sensor includes a time of flight sensor and a reflectance sensor. The ranging sensor transmits a signal that is reflected off of a user and received back at the ranging sensor. The received signal can be used to determine distance between

the user and the sensor, and the reflectance value of the user. With the distance or the reflectivity, a processor can activate a facial recognition process in response to the distance and the reflectivity.

5 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Although simple presentation attacks with photographs and videos of the legitimate face can be detected, an
10 approach to reliably detect presentation attacks with 3D face masks is still missing. Specifically, another layer of security is needed to enable the replacement of passphrases/PIN codes by biometric features derived from faces to unlock digital devices. Moreover, a method is needed that operates fully agnostic with respect to different skin types originating from different ethnic origins.

15

Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in combination, are presented in the dependent claims and/or in the following specification and
20 detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which, besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.
25
30

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be
35 used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, notwithstanding the fact that the respective feature or element may be present once or more than once.

Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way. The invention may, as the skilled person will recognize, be performed by
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using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a method for face authentication is disclosed. The face to be authenticated may be, specifically, a human face. The term "face authentication" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

- 5 c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 10 d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

15 The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

20 The term “camera” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a device having at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera may be a digital camera. As an example, the camera may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera may be or may comprise at least one near infrared camera.

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30 As used herein, without limitation, the term “image” specifically may relate to data recorded by using a camera, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera may also comprise one or more variable lenses which may be adjusted, automatically or manually.

35 The camera may be a camera of a mobile device such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera may be part of a mobile device which, besides the at least one camera, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The term “mobile device” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device

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45 may also refer to a tablet computer or another type of portable computer.

Specifically, the camera may be or may comprise at least one optical sensor having at least one light-sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive device for detecting a light beam, such as for detecting an illumination and/or a light spot generated by at least one light beam. As further used herein, a “light-sensitive area” generally refers to an area of the optical sensor which may be illuminated externally, by the at least one light beam, in response to which illumination at least one sensor signal is generated. The light-sensitive area may specifically be located on a surface of the respective optical sensor. Other embodiments, however, are feasible. The camera may comprise a plurality of optical sensors each having a light sensitive area. As used herein, the term “the optical sensors each having at least one light sensitive area” refers to configurations with a plurality of single optical sensors each having one light sensitive area and to configurations with one combined optical sensor having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-sensitive device configured to generate one output signal. In case the camera comprises a plurality of optical sensors, each optical sensor may be embodied such that precisely one light-sensitive area is present in the respective optical sensor, such as by providing precisely one light-sensitive area which may be illuminated, in response to which illumination precisely one uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be a single area optical sensor. The use of the single area optical sensors, however, renders the setup of the camera specifically simple and efficient. Thus, as an example, commercially available photo-sensors, such as commercially available silicon photodiodes, each having precisely one sensitive area, may be used in the set-up. Other embodiments, however, are feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared spectral range. The optical sensor may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name HertzstueckTM from trinamiXTM GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably

at least one semiconductor photodiode selected from the group consisting of: a Ge: Au photodiode, a Ge: Hg photodiode, a Ge: Cu photodiode, a Ge: Zn photodiode, a Si: Ga photodiode, a Si: As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm. Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels. Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors. The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array

of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

- 10 The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the camera form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term “essentially perpendicular” refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term “essentially parallel” refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The camera further may comprise at least one transfer device. The camera may further comprise one or more additional elements such as one or more additional optical elements.

45 The camera may comprise at least one optical element selected from the group consisting

of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam, such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical fashion with respect to the optical axis. Still, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses, wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the camera may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical sensors, the optical sensors may be located in one and the same beam path or partial beam

path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

5 The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a
10 longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

15 The camera is configured for determining at least one image of the scene, in particular the first image. As used herein, the term "scene" may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to "pixels", reference is either made to the units of image information generated by the single pixels of the sensor element or to the
20 single pixels of the sensor element directly. The first image is at least one two-dimensional image. As used herein, the term "two dimensional image" may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The first image may be an RGB (red green blue) image. The term "determining at least one first image" may refer to capturing and/or recording the first image.

25 The face detection step comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit. Specifically, the face detection step comprises detecting the face in the first image by identifying in the first image at least one
30 pre-defined or pre-determined geometrical feature which is characteristic for faces by using the at least one processing unit.

As further used herein, the term "processing unit" generally refers to an arbitrary data processing device adapted to perform the named operations such as by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the
35 at least one processing unit may comprise a software code stored thereon comprising a number of computer commands. The processing unit may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named
40 operations. Operations, including evaluating the images may be performed by the at least one processing unit. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays
45 (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or

alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit and the camera may fully or partially be integrated into a single device. Thus, generally, the processing unit also may form part of the camera. Alternatively, the processing unit and the camera may fully or partially be embodied as separate devices.

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The processing unit may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the processing unit may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the processing unit may comprise one or more data storage devices. Further, the processing unit may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The processing unit may be configured to one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the camera. The processing unit, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAPTTM), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The processing unit may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The processing unit may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The processing unit may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash

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drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

5 The processing unit may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

10 The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The term "geometrical feature characteristic for faces" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

25 The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step, the skin detection step may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit. However, embodiments are feasible wherein the skin detection step is performed before the face detection step.

As used herein, the term “illumination unit”, also denoted as illumination source, may generally refers to at least one arbitrary device configured for generating at least one illumination pattern. The illumination unit may be configured for providing the illumination pattern for illumination of the scene. The illumination unit may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit may comprise at least one light source. The illumination unit may comprise a plurality of light sources. The illumination unit may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination unit may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination unit may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit may be configured for emitting light at a single wavelength. In other embodiments, the illumination unit may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the

possibility of the light beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

The illumination unit may be or may comprise at least one multiple beam light source. For example, the illumination unit may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit may be adapted to generate and/or to project a cloud of points, for example the illumination unit may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit is particularly preferred. The illumination unit may be integrated into a housing of the camera or may be separated from the camera.

Further, the illumination unit may be configured for emitting modulated or non-modulated light. In case a plurality of illumination units is used, the different illumination units may have different modulation frequencies which later on may be used for distinguishing the light beams.

The light beam or light beams generated by the illumination unit generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The illumination unit may be configured such that the light beam or light beams propagates from the illumination unit towards the scene along an optical axis of the illumination unit and/or of the camera. For this purpose, the illumination unit and/or the camera may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical

axis. As an example, the light beam or light beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

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As used herein, the term “at least one illumination pattern” refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the scene. As used herein, the term “illumination feature” refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit may be adapted to generate and/or to project a cloud of points. The illumination unit may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination unit may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination unit comprises at least one laser source and/or at least one laser diode which is designated for generating laser radiation. The illumination unit may comprise the at least one diffractive optical element (DOE). The illumination unit may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern. As further used herein, the term “projecting at least one illumination pattern” may refer to providing the at least one illumination pattern for illuminating the at least one scene.

The skin detection step comprises determining the at least one second image, also denoted as reflection image, using the camera. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c).

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The second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features. As used herein, the term “reflection feature” may refer to a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. As used herein, the term “beam profile” of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

For example, the illumination unit may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit may be configured for determining the beam profile of the respective reflection feature. As used herein, the term “determining the beam profile” refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The reflection feature may cover or may extend over at least one pixel of the second image. For example, the reflection feature may cover or may extend over plurality of pixels. The processing unit may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The processing unit may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum j \cdot r_{pixel}},$$

wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

The processing unit is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature

by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

As used herein, the term “beam profile information” may refer to arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b), beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. The feature-based methods may be explained in the following. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

Other methods for authenticating a user’s face are known, such as from US 2019/213309 A1. However, these methods use time of flight (ToF) sensors. A well-known working principle of a ToF sensor is sending out light and measuring the time span until receiving the reflected light. In contrast, the proposed beam profile analysis uses a projected illumination pattern. For ToF sensor using such a projected pattern is not possible. Using illumination pattern may be advantageous e.g. in view of covering and therefore allowing to take into account different locations on the face. This may enhance reliability and security of the authentication of the user’s face.

The skin detection step may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit. Specifically, the processing unit is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. As used herein, the term “at least one predetermined or predefined criterion” refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property.

The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. As used herein, the term “indicate” refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. As used herein, the term “biological tissue” generally refers to biological material comprising living cells. Specifically, the processing unit may be configured for skin detection. The term “identification” of being generated by biological tissue, in particular human skin, may refer to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. As used herein, the term “material property” refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term “determining at least one material property” may refer to assigning the material property to respective reflection feature, in particular to the detected face. The processing unit may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

The reflection properties of skin may be characterized by the simultaneous occurrence of direct reflection at the surface (Lambertian-like) and subsurface scattering (volume scattering). This leads to a broadening of the laser spot on skin compared to the above-mentioned materials.

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The first beam profile information may be a reflection profile. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

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Specifically, the processing unit may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a device executing the method according to the present invention. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the processing unit or the device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation, e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The processing unit may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit.

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Additionally or alternatively, the first beam profile information may be determined by applying at least one image filter to the image of the area. As further used herein, the term "image" refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are

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given for any x,y-position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a material feature. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The processing unit may be configured for determining at least one material feature ϕ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the scene having generated the reflection feature.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof. The further material dependent image filter Φ_{2other} may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.80$.

The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

- wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k : $(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as 'Material A', 'Material B', etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

- The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . **The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$.** For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

- Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, **wherein z_k is a distance value corresponding to the image f_k from the predefined data set.** This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term "not

distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J-1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-Function $B(a, b) = \int_0^1 t^{a-1} (1-t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1-t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq 0.075$, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

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where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured

10 points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

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For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

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$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

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For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

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$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x))dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)|dx}{\int f(x)dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$\varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1g_2) = [p_{g_1,g_2}]$, whereas p_{g_1,g_2} is the occurrence rate of the grey combination $(g_1,g_2)=[f(x_1,y_1),f(x_2,y_2)]$, and the relation ρ defines the distance between (x_1,y_1) and (x_2,y_2) , which is $\rho(x,y)=(x+a,y+b)$ with a and b selected from 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

- 5 For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

- 10 For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$\varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This

- 15 material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based

- 20 on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

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$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein the function f is smaller than α times the maximum of f , but larger than a threshold value ε times the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the areas may not simply correspond to an inner and an outer circle around the spot center. As an example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material feature may be determined by

$$\phi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

The processing unit may be configured for using at least one predetermined relationship between the material feature ϕ_{2m} and the material property of the surface having generated the reflection feature for determining the material property of the surface having generated the reflection feature. The predetermined relationship may be one or more of an empirical relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

While feature based approaches, as the approaches described above, are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The term “machine-learning” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The term “skin classification model” may refer to a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification

model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The term “training”, also denoted learning, as used herein, is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates “human skin”. The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an

example, be stored in at least one data storage device of the processing unit. The processing unit is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

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The 3D detection step may be performed after the skin detection step and/or the face detection step. However, other embodiments are feasible, in which the 3D detection step is performed before the skin detection step and/or the face detection step. The material property may be determined by evaluating ϕ_{2m} subsequently after determining of the longitudinal coordinate z in step d) such that the information about the longitudinal coordinate z can be considered for evaluating of ϕ_{2m} .

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The 3D detection step comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The 3D detection step may comprise determining a second beam profile information of at least four of the reflection features by analysis of their respective beam profile. The second beam profile information may comprise a quotient Q of areas of the beam profile.

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As used herein, the term “analysis of the beam profile” may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

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The processing unit may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

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The processing unit may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The processing unit is configured for integrating the first area and the second area.

The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2.

The processing unit may be configured for integrating the first area and the second area. The processing unit may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term "area of the beam profile" generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The processing unit may be adapted to integrate the first area and the second area. The processing unit may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an

analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

5 The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may
10 have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e.
15 proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a
20 low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or
25 selected as second area of the beam profile if it is close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

30 Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part,
35 wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a
40 number of photons in the second area of the beam profile. The processing unit may be configured for determining an area integral of the beam profile. The processing unit may be configured for determining the edge information by integrating and/or summing of the first area. The processing unit may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the processing unit may be configured for determining an integral of
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the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 0.4 to 0.6 of the full radius.

The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

The processing unit may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

The processing unit may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The processing unit may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

wherein x and y are transversal coordinates, $A1$ and $A2$ are the first and second area of the beam profile, respectively, and $E(x, y)$ denotes the beam profile.

- 10 Additionally or alternatively, the processing unit may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several
15 slices or cuts may result in improved distance information.

For example, in case of the optical sensor having a matrix of pixels, the processing unit may be configured for evaluating the beam profile, by

- determining the pixel having the highest sensor signal and forming at least one center
20 signal;
- evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and
- determining at least one longitudinal coordinate z of the object by evaluating the quotient Q .

- 25 As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the
30 sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as pre-
35 processing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors,
40 wherein, in the latter case, as an example, the sensor signals of the group of pixels may be

added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The processing unit may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The processing unit may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to

determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

As used herein, the term “longitudinal coordinate for the reflection feature” refers to a distance between the optical sensor and the point of the scene remitting the corresponding illumination features. The processing unit may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The processing unit may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

The processing unit may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. As used herein, the term “depth” or depth information may refer to a distance between the object and the optical sensor and

may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The processing unit may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.

3D features determined step c) may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) and c) may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

Preferably subsequent to steps a) to c) the authentication step may be performed. The authentication step may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) no face is detected and/or in step b) the reflection features is determined not to be generated by skin and/or in step c) the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

Steps a) to d) may be performed by using at least one device, for example at least one mobile device such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. As used herein, the term “enrolling” may refer to a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image,

and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

5 Besides accuracy, also execution speed and power consumption are important requirements. Further restrictions on the availability of computational resources can be introduced by security considerations. For example, steps a) to d) may have run in a secure zone of the processing unit to avoid any software-based manipulations during program execution. The compact nature of the above described material detection networks may solve this problem by showing excellent runtime behavior in the said secure zone, whereas traditional PAD so-
10 lutions require the examination of several consecutive frames which results in large computational cost and longer response times of the algorithm.

In a further aspect of the present invention a computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the
15 method according to the present invention, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to the present invention. Specifically, the computer program may be stored on a computer-readable data carrier and/or on a computer-readable storage medium.

20 As used herein, the terms “computer-readable data carrier” and “computer-readable storage medium” specifically may refer to non-transitory data storage means, such as a hardware storage medium having stored thereon computer-executable instructions. The computer-readable data carrier or storage medium specifically may be or may comprise a storage medium such as a random-access memory (RAM) and/or a read-only memory (ROM).
25

Thus, specifically, one, more than one or even all of method steps as indicated above may be performed by using a computer or a computer network, preferably by using a computer program.
30

In a further aspect a computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to the present invention.

35 Further disclosed and proposed herein is a data carrier having a data structure stored thereon, which, after loading into a computer or computer network, such as into a working memory or main memory of the computer or computer network, may execute the method according to one or more of the embodiments disclosed herein.

40 Further disclosed and proposed herein is a computer program product with program code means stored on a machine-readable carrier, in order to perform the method according to one or more of the embodiments disclosed herein, when the program is executed on a computer or computer network. As used herein, a computer program product refers to the program as a tradable product. The product may generally exist in an arbitrary format, such as in a paper

format, or on a computer-readable data carrier and/or on a computer-readable storage medium. Specifically, the computer program product may be distributed over a data network.

5 Finally, disclosed and proposed herein is a modulated data signal which contains instructions readable by a computer system or computer network, for performing the method according to one or more of the embodiments disclosed herein.

10 Referring to the computer-implemented aspects of the invention, one or more of the method steps or even all of the method steps of the methods according to one or more of the embodiments disclosed herein may be performed by using a computer or computer network. Thus, generally, any of the method steps including provision and/or manipulation of data may be performed by using a computer or computer network. Generally, these method steps may include any of the method steps, typically except for method steps requiring manual work.

15 Specifically, further disclosed herein are:

- a computer or computer network comprising at least one processor, wherein the processor is adapted to perform the method according to one of the embodiments described in this description,
- 20 - a computer loadable data structure that is adapted to perform the method according to one of the embodiments described in this description while the data structure is being executed on a computer,
- a computer program, wherein the computer program is adapted to perform the method according to one of the embodiments described in this description while the program is being executed on a computer,
- 25 - a computer program comprising program means for performing the method according to one of the embodiments described in this description while the computer program is being executed on a computer or on a computer network,
- a computer program comprising program means according to the preceding embodiment, wherein the program means are stored on a storage medium readable to a computer,
- 30 - a storage medium, wherein a data structure is stored on the storage medium and wherein the data structure is adapted to perform the methods according to one of the embodiments described in this description after having been loaded into a main and/or working storage of a computer or of a computer network, and
- 35 - a computer program product having program code means, wherein the program code means can be stored or are stored on a storage medium, for performing the method according to one of the embodiments described in this description, if the program code means are executed on a computer or on a computer network.

40 In a further aspect a mobile device comprising at least one camera, at least one illumination unit and at least one processing unit is disclosed. The mobile device is configured for performing at least steps a) to c), and optionally step d) of the method for face authentication according to the present invention. Step d) may be performed by using at least one authentication unit. The authentication unit may be a unit of the mobile device or may be an external

authentication unit. With respect to definitions and embodiments of the mobile device reference is made to definitions and embodiments described with respect to the method.

In a further aspect of the present invention, use of the method according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use for biometric presentation attack detection.

Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1 Method for face authentication comprising the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

- Embodiment 2 The method according to the preceding embodiment, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.
- Embodiment 3 The method according to the preceding embodiment, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.
- Embodiment 4 The method according to any one of the preceding embodiments, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.
- Embodiment 5 The method according to the preceding embodiment, wherein the skin classification model is parametrized by using machine learning, wherein the property characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.
- Embodiment 6 The method according to any one of the two preceding embodiments, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.
- Embodiment 7 The method according to any one of the preceding embodiments, wherein the illumination pattern comprises a periodic grid of laser spots.
- Embodiment 8 The method according to any one of the preceding embodiments, wherein the illumination features have wavelengths in a near infrared (NIR) regime.
- Embodiment 9 The method according to the preceding embodiment, wherein the illumination features have wavelengths of 940 nm.
- Embodiment 10 The method according to any one of the preceding embodiments, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).
- Embodiment 11 The method according to any one of the preceding embodiments, wherein the camera is or comprises at least one near infrared camera.

Embodiment 12 Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding embodiments, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding embodiments.

Embodiment 13 A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding embodiments referring to a method.

Embodiment 14 A mobile device comprising at least one camera, at least one illumination unit and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding embodiments referring to a method.

Embodiment 15 Use of the method according to any one of the preceding claims for biometric presentation attack detection.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a method for face authentication according to the present invention;

Figure 2 shows an embodiment of a mobile device according to the present invention; and

Figure 3 shows experimental results.

Detailed description of the embodiments:

Figure 1 shows a flow chart of a method for face authentication according to the present invention. The face authentication may comprise verifying a recognized object or part of the

recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) (reference number 110) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera 112, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit 114;
- b) (reference number 116) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit 118 and determining at least one second image using the at least one camera 112, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit 114, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- c) (reference number 120) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit 114, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) (reference number 122) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication

unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The camera 112 may comprise at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera 112 may be a digital camera. As an example, the camera 112 may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera 112 may be or may comprise at least one near infrared camera. The image may relate to data recorded by using the camera 112, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera 112, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera 112 may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera 112 may also comprise one or more variable lenses which may be adjusted, automatically or manually.

The camera 112 may be a camera of a mobile device 124 such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera 112 may be part of the mobile device 124 which, besides the at least one camera 112, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The mobile device 124 may be a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device 124 may also refer to a tablet computer or another type of portable computer. An embodiment of a mobile device according to the present invention is shown in Figure 2.

Specifically, the camera 112 may be or may comprise at least one optical sensor 126 having at least one light-sensitive area. The optical sensor 126 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 126 may be sensitive in the infrared spectral range. The optical sensor 126 may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 126 may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 126 may be sensitive in the part of the near-infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to

1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name HertzstueckTM from trinamiXTM GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 126 may comprise at least one
5 optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical
10 sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor 126 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

15 Specifically, the optical sensor 126 may be sensitive in the near infrared region. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor 126, specifically, may be sensitive in the infrared spectral range, specifically in the range
20 of 780 nm to 3.0 micrometers. For example, the optical sensor 126 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or
25 partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The camera 112 further may comprise at least one transfer device (not shown here). The
30 camera 112 may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device may be adapted to guide the light beam onto the optical sensor 126. The transfer device specifically may comprise one or more of: at
35 least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. Thus, the focal length
40 constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin
45 refractive lens to the principal focal points of the thin lens. For a converging thin refractive

lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

10 The transfer device may have an optical axis. The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

15 20 The camera 112 is configured for determining at least one image of the scene, in particular the first image. The scene may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The first image is at least one two-dimensional image having information about transversal coordinates such as the dimensions of height and width.

25 The face detection step 110 comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit 114. As an example, the at least one processing unit 114 may comprise a software code stored thereon comprising a number of computer commands. The processing unit 114 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit 114. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit 114 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit 114 and the camera 112 may fully or partially be integrated into a single device. Thus, generally, the processing unit 114 also may form part of the camera 112. Alternatively, the processing unit 114 and the camera 112 may fully or partially be embodied as separate devices.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The geometrical feature characteristic for faces may be at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit 114 may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image. Specifically subsequent to the face detection step 110, the skin detection step 116 may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit 118. However, embodiments are feasible wherein the skin detection step 116 is performed before the face detection step 110.

The illumination unit 118 may be configured for providing the illumination pattern for illumination of the scene. The illumination unit 118 may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit 118 may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit 118 may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit 118 may comprise at least one light source. The illumination unit 118 may comprise a plurality of light sources. The illumination unit 118 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. The illumination unit 118 may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit 118 may be or may comprise at least one multiple beam light source. For example, the illumination unit 118 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit 118 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit 118 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points, for example the illumination unit 118 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit 118 is particularly preferred. The illumination unit 118 may be integrated into a housing of the camera 112 or may be separated from the camera 112.

The illumination pattern comprises at least one illumination feature adapted to illuminate at least one part of the scene. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further

convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points. The illumination unit 118 may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit 118 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit 118.

The skin detection step 116 comprises determining the at least one second image, also denoted as reflection image, using the camera 112. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c). The reflection feature may be a feature in an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. The beam profile of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying

a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor 126.

For example, the illumination unit 118 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 126, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 126 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit 114 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit 114 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit 114 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit 114 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit 114 may be configured for determining the beam profile of the respective reflection feature. The determining the beam profile may comprise identifying at least one reflection feature provided by the optical sensor 126 and/or selecting at least one reflection feature provided by the optical sensor 126 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The processing unit 114 is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

The beam profile information may be or may comprise arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b) 116, beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. With respect to feature-based methods reference is made to the description above. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

The skin detection step 116 may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit 114. Specifically, the processing unit 114 is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. Specifically, the processing unit 114 may be configured for skin detection, in particular for identifying if the detected face is human skin. The identification if the material is biological tissue, in particular human skin, may comprise to determining and/or validating whether a

surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit 114 may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. The material property may be at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The determining at least one material property may comprise assigning the material property to the detected face. The processing unit 114 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit 114 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

While feature based approaches are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) 116 may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step 116 at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The machine-learning may comprise a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The skin classification model may comprise a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The training may comprise a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step 116 may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step 116, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit 114 may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular

skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit 114 is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit 114 may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step 120 may be performed after the skin detection step 116 and/or the face detection step 110. However, other embodiments are feasible, in which the 3D detection step 120 is performed before the skin detection step 116 and/or the face detection step 110.

The 3D detection step 120 comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

The analysis of the beam profile may comprise evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least one symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit 114 may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit 114 may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit 114 may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit 114 may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit 114 may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit 114 may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is

made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

5 The longitudinal coordinate for the reflection feature may be a distance between the optical sensor 126 and the point of the scene remitting the corresponding illumination features. The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit 114 may be configured for integrating the first area and the second
10 area. The processing unit 114 may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit 114 may be configured for determining at least two areas of the beam profile
15 and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit 114 may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit 114 may be configured for segmenting the light spot into at least two areas of the beam profile
20 and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit 114 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit 114 may be configured for determining at least one first area and at
25 least one second area of the beam profile. The area of the beam profile may be an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing
30 unit 114 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera 112 may comprise at least two optical
35 sensors 126, wherein the light-sensitive areas of a first optical sensor 126 and of a second optical sensor 126 may be arranged such that the first optical sensor 126 is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor 126 is adapted to determine the second area of the beam profile of the reflection feature. The processing unit 114 may be adapted to integrate the first area and the second
40 area. The processing unit 114 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit 114 may comprise at least one data storage device for storing the predetermined relationship,
45 such as a lookup list or a lookup table.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

- 5 The processing unit 114 may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The processing unit 114 may be configured for determining the depth information of the reflection features by one or more of
- 10 the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.
- 15 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) 120 may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified
- 20 geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage
- 25 of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.
- 30 3D features determined step c) 120 may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) 116 and c) 120 may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the
- 35 generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

- 40 Preferably subsequent to steps a) 110, b) 116 and c) 120 the authentication step 122 may be performed. The authentication step 122 may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) 110 no face is detected and/or in step b) 116 the reflection features is determined not to be generated by skin and/or in step c) 120 the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) 116 the

detected face is characterized as skin and in step c) 122 the detected face is characterized as 3D object.

5 Steps a) to d) may be performed by using at least one device, for example the at least one mobile device¹²⁴ such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

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The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. The enrolling may comprise a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database.

15 The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit 114, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during

20 the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

25

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

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40 Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar

45 when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm

differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Figure 3 shows experimental results, in particular density as a function of skin score. On the x-axis the score is shown and on the y-axis the frequency. The score is a measure for the classification quality and has a value range between 0 and 1, wherein 1 indicates very high skin-similarity and 0 very low skin-similarity. A threshold for decision may be around 0.5. A reference distribution of skin-scores for bona fide presentations has been generated using 10 subjects. Skin-scores have also been recorded for presentation attacks (PA) of Level A, Level B and Level C (as defined in the relevant ISO standards). The experimental setup (target of evaluation, ToE) included a proprietary hardware device, e.g. as described in Figure 2, which includes the necessary sensors as well as the computational platform on which the PAD software is executed. The ToE was tested using six species of target of Level A PAIs (presentation attack instruments), five PAI species of Level B, and one PAI species of Level C. For each PAI species, 10 PAIs have been used. The PAI species used in this study are listed in the table below. In the table, APCER is the Attack Presentation Classification Error Rate, which refers to the number of successful attacks/all attacks*100. In the table, BPCER is the Bona Fide Presentation Classification Error Rate, which refers to the number of rejected unlock-attempts/all unlock-attempts*100. Level A and Level B attacks are all based on 2D PAIs, whereas for Level C attacks, based on 3D masks. For Level C attacks, custom rigid masks constructed using a 3D printer were used. A test crew of 10 subjects has been used to obtain the reference distribution of skin-scores for bona fide presentations.

Level	#PAI species	PAI species	#objects	#presentations	Results
A	6	4 printed, 2 digital replay (normal quality)	60	300 (60x5)	APCER=0%
B	5	4 printed, 1 digital replay (optimized for spoof)	50	250 (50x5)	APCER=0%
C	1	3D printed mask	10	50 (10x5)	APCER=0%
Bona Fide	-	-	10	50 (10x5)	BPCER=0%

The experiments with these PAIs show that the two classes of presentations (bona fide or PA) are clearly distinguishable based on the skin-score. Clear distinction between paper, 3D print and skin is possible using the method according to the present invention.

List of reference numbers

110	face detection step
112	camera
114	processing unit
116	skin detection step
118	illumination unit
120	3D detection step
122	authentication step
124	mobile device
126	optical sensor

Claims

1. A system, comprising:

- 5 a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area;
- a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams;
- a second VCSEL comprising a periodic arrangement and configured to output a second plurality of NIR light beams, wherein the second VCSEL array is offset from the first
- 10 VCSEL array;
- a diffractive optical element (DOE);
- at least one lens disposed between the DOE and the first and second VCSELs,
- wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens,
- 15 wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern,
- wherein the CMOS sensor is configured to receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and output a first image
- 20 corresponding to the first reflective light pattern and a second image corresponding to the second reflective light pattern,
- wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams; and
- 25 a processor configured to:
- detect a face in a first region in the first image;

determine a distribution of light intensity of a second region in the second image, wherein the second region corresponds to the first region;

compare the distribution of light intensity of the second region of the second image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the second image and the corresponding region;

determine a depth map of the face in the scene in the second image based on a distortion in the pseudo-random structured light pattern; and

authenticate the face based on the detection of the face, the classification of the object, and the depth map.

2. The system of claim 1, wherein the first VCSEL and the second VCSEL are disposed at a different distance from the at least one lens.

3. The system of claim 1, wherein the processor is configured to detect the face by identifying at least one pre-defined facial characteristic in the region of the first image.

4. The system of claim 1, wherein the distribution of light intensity comprises a shape and a brightness of the light beam.

5. The system of claim 1, wherein the NIR light is in a range of 700 nm to 1100 nm.

6. The system of claim 1, wherein the CMOS sensor comprises a matrix of pixels, each pixel forming a NIR light sensitive area.

7. The system of claim 6, wherein the image comprises a plurality of pixels corresponding to the pixels of the matrix of the CMOS sensor and wherein the distribution of light intensity is determined in a region of the matrix.

5

8. The system of claim 1, wherein in response to detecting the face, the processor is configured to authenticate the face based on the classification and the depth map.

9. The system of claim 1, wherein the depth map is a spatial distribution of a distance between the surfaces and the CMOS sensor.

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10. The system of claim 1, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

classify the object as three-dimensional in response to determining a deviation of

15 the depth map from the pre-defined depth map.

11. The system of claim 1, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.

12. The system of claim 1, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.

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13. The system of claim 12, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.

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14. The system of claim 1, wherein the system includes a mobile device.

15. A system, comprising:

a near infrared (NIR) camera comprising a CMOS sensor and configured to capture an object, wherein the CMOS sensor comprises an NIR light sensitive area;

5 a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a plurality of light beams in a near-infrared range of 700 nm to 1100 nm;

a second VCSEL comprising a periodic arrangement and configured to output a plurality of beams of NIR light of 700 nm to 1100 nm, wherein the second VCSEL array is offset
10 from the first VCSEL;

a diffractive optical element (DOE);

at least one lens disposed between the DOE and the first and second VCSELs,

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens,

15 wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern,

wherein the CMOS sensor is configured to receive at least one reflective light pattern and output one or more images corresponding to the at least one reflective light pattern; and
20

a processor configured to:

detect a face of the object in a first region of the one or more images;

determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap;

25

compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

5 determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region;

 determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and

10 authenticate the object based on the detection of the face, the classification of the object, and the depth map.

16. The system of claim 15, wherein the first VCSEL and the second VCSEL are disposed at a different distance from the at least one lens.

15 17. The system of claim 15, wherein the processor is configured to detect the face of the object by identifying at least one pre-defined facial characteristic in the first region of the one or more images.

18. The system of claim 15, wherein the distribution of light intensity comprises a shape and
20 a brightness of the light beam.

19. The system of claim 15, wherein the system includes a mobile device.

20. The system of claim 15, wherein in response to detecting the face, the processor is configured to authenticate the object based on the classification of the object and the depth
25 map.

21. The system of claim 15, wherein the depth map is a spatial distribution of a distance between the object and the CMOS sensor.
- 5 22. The system of claim 15, wherein the processor is configured to:
- compare the depth map of the object with a pre-defined depth map; and
 - classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.
- 10 23. The system of claim 22, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.
24. The system of claim 22, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.
- 15 25. The system of claim 24, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.
26. The system of claim 15, further comprising a neural network coupled to the processor,
- 20 the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained to detect features related to classifications of the objects and output the classification of the object based on the processed images.
- 25 27. A system, comprising:
- at least one vertical cavity surface emitting laser (VCSEL) configured to:

emit a plurality of light beams in a near-infrared range (NIR), arranged in a first spatial arrangement relative to each other, towards an object;

emit at least one substantially uniform light beam in the near-infrared range towards the object;

5 a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area, wherein the CMOS sensor is configured to receive light reflected from the object and output one or more images characterizing at least an intensity of the reflected light; and

a processor configured to:

detect a face of the object in a first region of the one or more images;

10 determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap;

compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

15 determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region;

20 determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

28. The system of claim 27, wherein the system includes a mobile device.

25

29. The system of claim 27, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and
classify the object as three-dimensional in response to determining a deviation of
the depth map from the pre-defined depth map.

- 5 30. The system of claim 29, wherein the processor is configured to differentiate skin from
non-living surfaces in the classification of the object and authenticate the object when the
object is classified as skin and three-dimensional.

Optical Skin Detection for Face Unlock

Abstract

- 5 Proposed herein are systems for face authentication. The systems are configured to classify an object based on a distribution of light intensity.

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	Peter SCHILLEN	Nonprovisional Application Number (if known):	
Title of Invention:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☒ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via USPTO patent electronic filing system.

---OR---

(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.

ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☐ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- A request for continued examination has been filed with, or prior to, this form.
- If the application is a utility application, this certification and request is being filed via USPTO patent electronic filing system.
- The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature	/Christopher M. Goff/	Date	2025-03-24
Name (Print/Typed)	Christopher M. Goff	Practitioner Registration Number	41,785

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☒ *Total of 1 forms are submitted.

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- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

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Optical Skin Detection for Face UnlockCROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation application of U.S. Application No. 18/468352, filed September 15, 2023, which is a continuation application of U.S. Application No. 18/264326, filed August 4, 2023, which is a U.S. National Phase Application of International Patent Application No. PCT/EP22/053941, filed February 17, 2022, which claims priority to EP Patent Application No. 21157800.0, filed February 18, 2021, each of which is hereby incorporated by reference herein.

DescriptionField of the invention

The invention relates to a method for face authentication, a mobile device and various uses of the method. The devices, methods and uses according to the present invention specifically may be employed for example in various areas of daily life, security technology, gaming, traffic technology, production technology, photography such as digital photography or video photography for arts, documentation or technical purposes, safety technology, information technology, agriculture, crop protection, maintenance, cosmetics, medical technology or in the sciences. However, other applications are also possible.

Prior art

In today's digital world secure access to information technology is an essential requirement of any state-of-the-art system. Standard concepts like passphrases or PIN codes are currently extended, combined or even replaced by biometric methods like fingerprint or face recognition. Although passphrases can provide a high level of security if chosen carefully and at adequate length, this step requires attention and the need of memorizing several potentially long phrases depending on the IT landscape. Furthermore, it is never guaranteed that the person providing the passphrase is authorized by the owner of the passphrase or the digital device. In contrast biometric features like faces or fingerprints are unique and person-specific properties. Therefore, using features derived from these not only can be more convenient than passphrases/PIN codes but also more secure since they combine personal identity with an unlock process.

Unfortunately, similarly to being able to spy passwords, fingerprints and faces can be also artificially created in order to impersonate a legitimate user. Automatic face recognition tools of the first generation use digital camera images or image streams and apply 2D image processing methods to extract characteristic features, while machine learning techniques are used to generate face templates to perform identity recognition based on those features. The second generation of face recognition algorithms uses deep convolutional neuronal networks instead of hand-crafted image features to generate the classification model.

However, both approaches can be attacked, e.g., with a high-quality photograph that is nowadays freely available to be downloaded from the Internet. Therefore, the idea of presentation attack detection (PAD) gained significance. Early approaches were designed to protect against simple attacks like presenting the photograph of a legitimate user by recording a sequence of images and test for time dependent features, for example small but natural changes in the head position or eye blinking. These methods can again be tricked by playing a pre-recorded video of the user or by careful animations generated from publicly available photographs. In order to exclude displays as potential spoof objects, near infra-red (NIR) cameras can be used, since displays emit photons only in the visible regime of the electromagnetic spectrum. As a further counteraction, 3D cameras were introduced, which can clearly distinguish between planar photos or tablets with playback videos and a 3D face. Still, these systems can be attacked with high quality masks generated, for example, by 3D printing, careful 3D-arrangement of 2D photos, or hand-crafted silicone or latex masks to name a few approaches. Since masks are typically worn by humans, liveness detection based on small motions fails as well. However, these types of masks can be rejected by a PAD system that is able to classify human skin from other materials such as described in EP application No. 20159984.2 filed on February 28, 2020 and EP application No. 20 154 961.5190679 filed on January 31, 2020, the full content of which is included by reference.

A further problem might arise from differences in skin optical properties with respect ethnic origin. A reliable recognition and PAD technology needs to be fully agnostic with respect to these different origins.

Besides security considerations, also the speed of the unlock process and the required computational power are important to provide an acceptable user experience. High-speed face recognition can be used for several tasks after the successful unlock of a device, for example to check whether the user is still in front of the display or to launch further secure applications like banking apps by performing an on the fly check of the person in front of the display. Again, speed and computational resources have great impact on the user experience.

Current 3D algorithms are computationally very demanding and presentation attack detections require several video frames to be processed. Therefore, expensive hardware needs to be involved to deliver acceptable unlock performance. Furthermore, high power consumption is a consequence.

Summarizing, current methods for face unlock do not provide the ability to reliably detect spoof attacks by 3D masks and perform this task at a speed below human detection limit.

US 2019/213309 A1 describes a system and method of authenticating a user's face with a ranging sensor. The ranging sensor includes a time of flight sensor and a reflectance sensor. The ranging sensor transmits a signal that is reflected off of a user and received back at the ranging sensor. The received signal can be used to determine distance between the user and

the sensor, and the reflectance value of the user. With the distance or the reflectivity, a processor can activate a facial recognition process in response to the distance and the reflectivity.

5 Problem addressed by the invention

It is therefore an object of the present invention to provide devices and methods facing the above-mentioned technical challenges of known devices and methods. Although simple presentation attacks with photographs and videos of the legitimate face can be detected, an
10 approach to reliably detect presentation attacks with 3D face masks is still missing. Specifically, another layer of security is needed to enable the replacement of passphrases/PIN codes by biometric features derived from faces to unlock digital devices. Moreover, a method is needed that operates fully agnostic with respect to different skin types originating from different ethnic origins.

15 Summary of the invention

This problem is solved by the invention with the features of the independent patent claims. Advantageous developments of the invention, which can be realized individually or in
20 combination, are presented in the dependent claims and/or in the following specification and detailed embodiments.

As used in the following, the terms “have”, “comprise” or “include” or any arbitrary grammatical variations thereof are used in a non-exclusive way. Thus, these terms may both refer to a
25 situation in which, besides the feature introduced by these terms, no further features are present in the entity described in this context and to a situation in which one or more further features are present. As an example, the expressions “A has B”, “A comprises B” and “A includes B” may both refer to a situation in which, besides B, no other element is present in A (i.e. a situation in which A solely and exclusively consists of B) and to a situation in which,
30 besides B, one or more further elements are present in entity A, such as element C, elements C and D or even further elements.

Further, it shall be noted that the terms “at least one”, “one or more” or similar expressions indicating that a feature or element may be present once or more than once typically will be
35 used only once when introducing the respective feature or element. In the following, in most cases, when referring to the respective feature or element, the expressions “at least one” or “one or more” will not be repeated, non-withstanding the fact that the respective feature or element may be present once or more than once.

40 Further, as used in the following, the terms “preferably”, “more preferably”, “particularly”, “more particularly”, “specifically”, “more specifically” or similar terms are used in conjunction with optional features, without restricting alternative possibilities. Thus, features introduced by these terms are optional features and are not intended to restrict the scope of the claims in any way.

The invention may, as the skilled person will recognize, be performed by using alternative features. Similarly, features introduced by "in an embodiment of the invention" or similar expressions are intended to be optional features, without any restriction regarding alternative embodiments of the invention, without any restrictions regarding the scope of the invention and without any restriction regarding the possibility of combining the features introduced in such a way with other optional or non-optional features of the invention.

In a first aspect of the present invention a method for face authentication is disclosed. The face to be authenticated may be, specifically, a human face. The term "face authentication" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

- c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed. Further, one, more than one or even all of the method steps may be performed repeatedly.

The term “camera” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a device having at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera may be a digital camera. As an example, the camera may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera may be or may comprise at least one near infrared camera.

As used herein, without limitation, the term “image” specifically may relate to data recorded by using a camera, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera may also comprise one or more variable lenses which may be adjusted, automatically or manually.

The camera may be a camera of a mobile device such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera may be part of a mobile device which, besides the at least one camera, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The term “mobile device” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a mobile electronics device, more specifically to a mobile communication device such as a cell phone or

smart phone. Additionally or alternatively, the mobile device may also refer to a tablet computer or another type of portable computer.

Specifically, the camera may be or may comprise at least one optical sensor having at least one
5 light-sensitive area. As used herein, an “optical sensor” generally refers to a light-sensitive
device for detecting a light beam, such as for detecting an illumination and/or a light spot
generated by at least one light beam. As further used herein, a “light-sensitive area” generally
refers to an area of the optical sensor which may be illuminated externally, by the at least one
light beam, in response to which illumination at least one sensor signal is generated. The light-
10 sensitive area may specifically be located on a surface of the respective optical sensor. Other
embodiments, however, are feasible. The camera may comprise a plurality of optical sensors
each having a light sensitive area. As used herein, the term “the optical sensors each having at
least one light sensitive area” refers to configurations with a plurality of single optical sensors
each having one light sensitive area and to configurations with one combined optical sensor
15 having a plurality of light sensitive areas. The term “optical sensor” furthermore refers to a light-
sensitive device configured to generate one output signal. In case the camera comprises a
plurality of optical sensors, each optical sensor may be embodied such that precisely one light-
sensitive area is present in the respective optical sensor, such as by providing precisely one
light-sensitive area which may be illuminated, in response to which illumination precisely one
20 uniform sensor signal is created for the whole optical sensor. Thus, each optical sensor may be
a single area optical sensor. The use of the single area optical sensors, however, renders the
setup of the camera specifically simple and efficient. Thus, as an example, commercially
available photo-sensors, such as commercially available silicon photodiodes, each having
precisely one sensitive area, may be used in the set-up. Other embodiments, however, are
25 feasible.

The optical sensor specifically may be or may comprise at least one photodetector, preferably
inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most
preferably silicon photodetectors. Specifically, the optical sensor may be sensitive in the infrared
30 spectral range. The optical sensor may comprise at least one sensor element comprising a
matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix
specifically may be identical. Groups of identical pixels of the matrix specifically may be
provided for different spectral ranges, or all pixels may be identical in terms of spectral
sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or
35 optoelectronic properties. Specifically, the optical sensor may be or may comprise at least one
array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in
the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor may be sensitive in the
part of the near infrared region where silicon photodiodes are applicable specifically in the range
of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be
40 commercially available infrared optical sensors, such as infrared optical sensors commercially
available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen
am Rhein, Germany. Thus, as an example, the optical sensor may comprise at least one optical
sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode

selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode.

Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

The optical sensor may be sensitive in one or more of the ultraviolet, the visible or the infrared spectral range. Specifically, the optical sensor may be sensitive in the visible spectral range from 500 nm to 780 nm, most preferably at 650 nm to 750 nm or at 690 nm to 700 nm.

Specifically, the optical sensor may be sensitive in the near infrared region. Specifically, the optical sensor may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor each, independently, may be or may comprise at least one element selected from the group consisting of a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. For example, the optical sensor may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The optical sensor may comprise at least one sensor element comprising a matrix of pixels.

Thus, as an example, the optical sensor may be part of or constitute a pixelated optical device. For example, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device. As an example, the optical sensor may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area.

As used herein, the term "sensor element" generally refers to a device or a combination of a plurality of devices configured for sensing at least one parameter. In the present case, the parameter specifically may be an optical parameter, and the sensor element specifically may be an optical sensor element. The sensor element may be formed as a unitary, single device or as a combination of several devices. The sensor element comprises a matrix of optical sensors.

The sensor element may comprise at least one CMOS sensor. The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS

detector, such as a CMOS detector chip. Thus, generally, the sensor element may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors may form a sensor array or may be part of a sensor array, such as the above-mentioned matrix. Thus, as an example, the sensor element may comprise an array of pixels, such as a rectangular array, having m rows and n columns, with m , n , independently, being positive integers. Preferably, more than one column and more than one row is given, i.e. $n > 1$, $m > 1$. Thus, as an example, n may be 2 to 16 or higher and m may be 2 to 16 or higher. Preferably, the ratio of the number of rows and the number of columns is close to 1. As an example, n and m may be selected such that $0.3 \leq m/n \leq 3$, such as by choosing $m/n = 1:1$, $4:3$, $16:9$ or similar. As an example, the array may be a square array, having an equal number of rows and columns, such as by choosing $m=2$, $n=2$ or $m=3$, $n=3$ or the like.

The matrix may be composed of independent pixels such as of independent optical sensors. Thus, a matrix of inorganic photodiodes may be composed. Alternatively, however, a commercially available matrix may be used, such as one or more of a CCD detector, such as a CCD detector chip, and/or a CMOS detector, such as a CMOS detector chip. Thus, generally, the optical sensor may be and/or may comprise at least one CCD and/or CMOS device and/or the optical sensors of the camera form a sensor array or may be part of a sensor array, such as the above-mentioned matrix.

The matrix specifically may be a rectangular matrix having at least one row, preferably a plurality of rows, and a plurality of columns. As an example, the rows and columns may be oriented essentially perpendicular. As used herein, the term "essentially perpendicular" refers to the condition of a perpendicular orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Similarly, the term "essentially parallel" refers to the condition of a parallel orientation, with a tolerance of e.g. $\pm 20^\circ$ or less, preferably a tolerance of $\pm 10^\circ$ or less, more preferably a tolerance of $\pm 5^\circ$ or less. Thus, as an example, tolerances of less than 20° , specifically less than 10° or even less than 5° , may be acceptable. In order to provide a wide range of view, the matrix specifically may have at least 10 rows, preferably at least 500 rows, more preferably at least 1000 rows. Similarly, the matrix may have at least 10 columns, preferably at least 500 columns, more preferably at least 1000 columns. The matrix may comprise at least 50 optical sensors, preferably at least 100000 optical sensors, more preferably at least 5000000 optical sensors. The matrix may comprise a number of pixels in a multi-mega pixel range. Other embodiments, however, are feasible. Thus, in setups in which an axial rotational symmetry is to be expected, circular arrangements or concentric arrangements of the optical sensors of the matrix, which may also be referred to as pixels, may be preferred.

Thus, as an example, the sensor element may be part of or constitute a pixelated optical device. For example, the sensor element may be and/or may comprise at least one CCD and/or CMOS device. As an example, the sensor element may be part of or constitute at least one CCD and/or CMOS device having a matrix of pixels, each pixel forming a light-sensitive area. The

sensor element may employ a rolling shutter or global shutter method to read out the matrix of optical sensors.

The camera further may comprise at least one transfer device. The camera may further
5 comprise one or more additional elements such as one or more additional optical elements. The camera may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The term “transfer device”, also denoted as “transfer system”, may generally refer to one or more optical elements which are adapted to modify the light beam,
10 such as by modifying one or more of a beam parameter of the light beam, a width of the light beam or a direction of the light beam. The transfer device may be adapted to guide the light beam onto the optical sensor. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel
15 lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. As used herein, the term “focal length” of the transfer device refers to a distance over which incident collimated rays which may impinge the
20 transfer device are brought into a “focus” which may also be denoted as “focal point”. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In
25 this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least
30 one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

35 The transfer device may have an optical axis. As used herein, the term “optical axis of the transfer device” generally refers to an axis of mirror symmetry or rotational symmetry of the lens or lens system. The transfer system, as an example, may comprise at least one beam path, with the elements of the transfer system in the beam path being located in a rotationally symmetrical
40 fashion with respect to the optical axis. Still, one or more optical elements located within the beam path may also be off-centered or tilted with respect to the optical axis. In this case, however, the optical axis may be defined sequentially, such as by interconnecting the centers of the optical elements in the beam path, e.g. by interconnecting the centers of the lenses,

wherein, in this context, the optical sensors are not counted as optical elements. The optical axis generally may denote the beam path. Therein, the camera may have a single beam path along which a light beam may travel from the object to the optical sensors, or may have a plurality of beam paths. As an example, a single beam path may be given or the beam path may be split into two or more partial beam paths. In the latter case, each partial beam path may have its own optical axis. In case of a plurality of optical sensors, the optical sensors may be located in one and the same beam path or partial beam path. Alternatively, however, the optical sensors may also be located in different partial beam paths.

The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera is configured for determining at least one image of the scene, in particular the first image. As used herein, the term "scene" may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. Consequently, when referring to "pixels", reference is either made to the units of image information generated by the single pixels of the sensor element or to the single pixels of the sensor element directly. The first image is at least one two-dimensional image. As used herein, the term "two dimensional image" may generally refer to an image having information about transversal coordinates such as the dimensions of height and width. The first image may be an RGB (red green blue) image. The term "determining at least one first image" may refer to capturing and/or recording the first image.

The face detection step comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit. Specifically, the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature which is characteristic for faces by using the at least one processing unit.

As further used herein, the term "processing unit" generally refers to an arbitrary data processing device adapted to perform the named operations such as by using at least one processor and/or at least one application-specific integrated circuit. Thus, as an example, the at least one processing unit may comprise a software code stored thereon comprising a number of computer commands. The processing unit may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors

with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit and the camera may fully or partially be integrated into a single device. Thus, generally, the processing unit also may form part of the camera. Alternatively, the processing unit and the camera may fully or partially be embodied as separate devices.

The processing unit may be or may comprise one or more integrated circuits, such as one or more application-specific integrated circuits (ASICs), and/or one or more data processing devices, such as one or more computers, preferably one or more microcomputers and/or microcontrollers, Field Programmable Arrays, or Digital Signal Processors. Additional components may be comprised, such as one or more preprocessing devices and/or data acquisition devices, such as one or more devices for receiving and/or preprocessing of the sensor signals, such as one or more AD-converters and/or one or more filters. Further, the processing unit may comprise one or more measurement devices, such as one or more measurement devices for measuring electrical currents and/or electrical voltages. Further, the processing unit may comprise one or more data storage devices. Further, the processing unit may comprise one or more interfaces, such as one or more wireless interfaces and/or one or more wire-bound interfaces.

The processing unit may be configured to one or more of displaying, visualizing, analyzing, distributing, communicating or further processing of information, such as information obtained by the camera. The processing unit, as an example, may be connected or incorporate at least one of a display, a projector, a monitor, an LCD, a TFT, a loudspeaker, a multichannel sound system, an LED pattern, or a further visualization device. It may further be connected or incorporate at least one of a communication device or communication interface, a connector or a port, capable of sending encrypted or unencrypted information using one or more of email, text messages, telephone, Bluetooth, Wi-Fi, infrared or internet interfaces, ports or connections. It may further be connected to or incorporate at least one of a processor, a graphics processor, a CPU, an Open Multimedia Applications Platform (OMAPTTM), an integrated circuit, a system on a chip such as products from the Apple A series or the Samsung S3C2 series, a microcontroller or microprocessor, one or more memory blocks such as ROM, RAM, EEPROM, or flash memory, timing sources such as oscillators or phase-locked loops, counter-timers, real-time timers, or power-on reset generators, voltage regulators, power management circuits, or DMA controllers. Individual units may further be connected by buses such as AMBA buses or be integrated in an Internet of Things or Industry 4.0 type network.

The processing unit may be connected by or have further external interfaces or ports such as one or more of serial or parallel interfaces or ports, USB, Centronics Port, FireWire, HDMI, Ethernet, Bluetooth, RFID, Wi-Fi, USART, or SPI, or analogue interfaces or ports such as one or more of ADCs or DACs, or standardized interfaces or ports to further devices such as a 2D-camera device using an RGB-interface such as CameraLink. The processing unit may further be connected by one or more of interprocessor interfaces or ports, FPGA-FPGA-interfaces, or serial or parallel interfaces ports. The processing unit may further be connected to one or more of an optical disc drive, a CD-RW drive, a DVD+RW drive, a flash drive, a memory card, a disk drive, a hard disk drive, a solid state disk or a solid state hard disk.

The processing unit may be connected by or have one or more further external connectors such as one or more of phone connectors, RCA connectors, VGA connectors, hermaphrodite connectors, USB connectors, HDMI connectors, 8P8C connectors, BCN connectors, IEC 60320 C14 connectors, optical fiber connectors, D-subminiature connectors, RF connectors, coaxial connectors, SCART connectors, XLR connectors, and/or may incorporate at least one suitable socket for one or more of these connectors.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The term "geometrical feature characteristic for faces" as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a

Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step, the skin detection step may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit. However, embodiments are feasible wherein the skin detection step is performed before the face detection step.

As used herein, the term “illumination unit”, also denoted as illumination source, may generally refers to at least one arbitrary device configured for generating at least one illumination pattern.

The illumination unit may be configured for providing the illumination pattern for illumination of the scene. The illumination unit may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit may comprise at least one light source. The illumination unit may comprise a plurality of light sources. The illumination unit may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. As an example, the light emitted by the illumination unit may have a wavelength of 300 to 1100nm, especially 500 to 1100 nm. Additionally or alternatively, light in the infrared spectral range may be used, such as in the range of 780 nm to 3.0 μm . Specifically, the light in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm may be used.

The illumination unit may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime.

The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit may be configured for emitting light at a single wavelength. In other embodiments, the illumination unit may be configured for emitting light with a plurality of wavelengths allowing additional measurements in other wavelengths channels.

5 As used herein, the term “ray” generally refers to a line that is perpendicular to wavefronts of light which points in a direction of energy flow. As used herein, the term “beam” generally refers to a collection of rays. In the following, the terms “ray” and “beam” will be used as synonyms. As further used herein, the term “light beam” generally refers to an amount of light, specifically an amount of light traveling essentially in the same direction, including the possibility of the light
10 beam having a spreading angle or widening angle. The light beam may have a spatial extension. Specifically, the light beam may have a non-Gaussian beam profile. The beam profile may be selected from the group consisting of a trapezoid beam profile; a triangle beam profile; a conical beam profile. The trapezoid beam profile may have a plateau region and at least one edge region. The light beam specifically may be a Gaussian light beam or a linear combination
15 of Gaussian light beams, as will be outlined in further detail below. Other embodiments are feasible, however.

The illumination unit may be or may comprise at least one multiple beam light source. For example, the illumination unit may comprise at least one laser source and one or more
20 diffractive optical elements (DOEs). Specifically, the illumination unit may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg
25 grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit may comprise one or more diffractive optical elements (DOEs)
30 adapted to generate the illumination pattern. For example, the illumination unit may be adapted to generate and/or to project a cloud of points, for example the illumination unit may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally
35 defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit is particularly preferred. The illumination unit may be integrated into a housing of the camera or may be separated from the camera.

Further, the illumination unit may be configured for emitting modulated or non-modulated light.
40 In case a plurality of illumination units is used, the different illumination units may have different modulation frequencies which later on may be used for distinguishing the light beams.

The light beam or light beams generated by the illumination unit generally may propagate parallel to the optical axis or tilted with respect to the optical axis, e.g. including an angle with the optical axis. The illumination unit may be configured such that the light beam or light beams propagates from the illumination unit towards the scene along an optical axis of the illumination unit and/or of the camera. For this purpose, the illumination unit and/or the camera may comprise at least one reflective element, preferably at least one prism, for deflecting the illuminating light beam onto the optical axis. As an example, the light beam or light beams, such as the laser light beam, and the optical axis may include an angle of less than 10° , preferably less than 5° or even less than 2° . Other embodiments, however, are feasible. Further, the light beam or light beams may be on the optical axis or off the optical axis. As an example, the light beam or light beams may be parallel to the optical axis having a distance of less than 10 mm to the optical axis, preferably less than 5 mm to the optical axis or even less than 1 mm to the optical axis or may even coincide with the optical axis.

As used herein, the term "at least one illumination pattern" refers to at least one arbitrary pattern comprising at least one illumination feature adapted to illuminate at least one part of the scene. As used herein, the term "illumination feature" refers to at least one at least partially extended feature of the pattern. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit may be adapted to generate and/or to project a cloud of points. The illumination unit may comprise the at least one light projector adapted to generate a cloud of points such that the illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit.

A distance between two features of the illumination pattern and/or an area of the at least one illumination feature may depend on the circle of confusion in the image. As outlined above, the illumination unit may comprise the at least one light source configured for generating the at least one illumination pattern. Specifically, the illumination unit comprises at least one laser source
5 and/or at least one laser diode which is designated for generating laser radiation. The illumination unit may comprise the at least one diffractive optical element (DOE). The illumination unit may comprise at least one point projector, such as the at least one laser source and the DOE, adapted to project at least one periodic point pattern. As further used herein, the term "projecting at least one illumination pattern" may refer to providing the at least one
10 illumination pattern for illuminating the at least one scene.

The skin detection step comprises determining the at least one second image, also denoted as reflection image, using the camera. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection
15 in step b) and/or for 3D detection in step c).

The second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features. As used herein, the term "reflection feature" may refer to a feature in an image plane generated by the scene in response to
20 illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. As used herein, the term "beam profile" of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting of a trapezoid beam
25 profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit may be configured for performing at least one image
30 analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image
35 created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region
40 detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-

space extrema detector; applying a local feature detector; applying speeded up robust features algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor.

For example, the illumination unit may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit may be configured for determining the beam profile of the respective reflection feature. As used herein, the term "determining the beam profile" refers to identifying at least one reflection feature provided by the optical sensor and/or selecting at least one reflection feature provided by the optical sensor and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The reflection feature may cover or may extend over at least one pixel of the second image. For example, the reflection feature may cover or may extend over plurality of pixels. The processing

unit may be configured for determining and/or for selecting all pixels connected to and/or belonging to the reflection feature, e.g. a light spot. The processing unit may be configured for determining the center of intensity by

$$R_{coi} = \frac{1}{l \cdot \sum_j r_{pixel}},$$

- 5 wherein R_{coi} is a position of center of intensity, r_{pixel} is the pixel position and $l = \sum_j I_{total}$ with j being the number of pixels j connected to and/or belonging to the reflection feature and I_{total} being the total intensity.

10 The processing unit is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the
15 second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

20 As used herein, the term “beam profile information” may refer to arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b), beam profile analysis may be used.
25 Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP
30 application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. The feature-based methods may be explained in the following. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model.
35 Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

Other methods for authenticating a user's face are known, such as from US 2019/213309 A1. However, these methods use time of flight (ToF) sensors. A well-known working principle of a
40 ToF sensor is sending out light and measuring the time span until receiving the reflected light. In contrast, the proposed beam profile analysis uses a projected illumination pattern. For ToF

sensor using such a projected pattern is not possible. Using illumination pattern may be advantageous e.g. in view of covering and therefore allowing to take into account different locations on the face. This may enhance reliability and security of the authentication of the user's face.

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The skin detection step may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit. Specifically, the processing unit is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. As used herein, the term "at least one predetermined or predefined criterion" refers to at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. As used herein, the term "indicate" refers to an arbitrary indication such as an electronic signal and/or at least one visual or acoustic indication. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. As used herein, the term "biological tissue" generally refers to biological material comprising living cells. Specifically, the processing unit may be configured for skin detection. The term "identification" of being generated by biological tissue, in particular human skin, may refer to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. As used herein, the term "material property" refers to at least one arbitrary property of the material configured for characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. As used herein, the term "determining at least one material property" may refer to assigning the material property to respective reflection feature, in particular to the detected face. The processing unit may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined

and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

The reflection properties of skin may be characterized by the simultaneous occurrence of direct reflection at the surface (Lambertian-like) and subsurface scattering (volume scattering). This leads to a broadening of the laser spot on skin compared to the above-mentioned materials.

The first beam profile information may be a reflection profile. For example, without wishing to be bound by this theory, human skin may have a reflection profile, also denoted back scattering profile, comprising parts generated by back reflection of the surface, denoted as surface reflection, and parts generated by very diffuse reflection from light penetrating the skin, denoted as diffuse part of the back reflection. With respect to reflection profile of human skin reference is made to "Lasertechnik in der Medizin: Grundlagen, Systeme, Anwendungen", "Wirkung von Laserstrahlung auf Gewebe", 1991, pages 10 171 to 266, Jürgen Eichler, Theo Seiler, Springer Verlag, ISBN 0939-0979. The surface reflection of the skin may increase with the wavelength increasing towards the near infrared. Further, the penetration depth may increase with increasing wavelength from visible to near infrared. The diffuse part of the back reflection may increase with penetrating depth of the light. These properties may be used to distinguish skin from other materials, by analyzing the back scattering profile.

Specifically, the processing unit may be configured for comparing the reflection beam profile with at least one predetermined and/or prerecorded and/or predefined beam profile. The predetermined and/or prerecorded and/or predefined beam profile may be stored in a table or a lookup table and may be determined e.g. empirically, and may, as an example, be stored in at least one data storage device of the detector. For example, the predetermined and/or prerecorded and/or predefined beam profile may be determined during initial start-up of a device executing the method according to the present invention. For example, the predetermined and/or prerecorded and/or predefined beam profile may be stored in at least one data storage device of the processing unit or the device, e.g. by software, specifically by the app downloaded from an app store or the like. The reflection feature may be identified as to be generated by biological tissue in case the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile are identical. The comparison may comprise overlaying the reflection beam profile and the predetermined or predefined beam profile such that their centers of intensity match. The comparison may comprise determining a deviation,

e.g. a sum of squared point to point distances, between the reflection beam profile and the predetermined and/or prerecorded and/or predefined beam profile. The processing unit may be adapted to compare the determined deviation with at least one threshold, wherein in case the determined deviation is below and/or equal the threshold the surface is indicated as biological tissue and/or the detection of biological tissue is confirmed. The threshold value may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit.

Additionally or alternatively, the first beam profile information may be determined by applying at least one image filter to the image of the area. As further used herein, the term “image” refers to a two-dimensional function, $f(x,y)$, wherein brightness and/or color values are given for any x,y -position in the image. The position may be discretized corresponding to the recording pixels. The brightness and/or color may be discretized corresponding to a bit-depth of the optical sensors. As used herein, the term “image filter” refers to at least one mathematical operation applied to the beam profile and/or to the at least one specific region of the beam profile. Specifically, the image filter Φ maps an image f , or a region of interest in the image, onto a real number, $\Phi(f(x,y)) = \varphi$, wherein φ denotes a feature, in particular a material feature. Images may be subject to noise and the same holds true for features. Therefore, features may be random variables. The features may be normally distributed. If features are not normally distributed, they may be transformed to be normally distributed such as by a Box-Cox-Transformation.

The processing unit may be configured for determining at least one material feature φ_{2m} by applying at least one material dependent image filter Φ_2 to the image. As used herein, the term “material dependent” image filter refers to an image having a material dependent output. The output of the material dependent image filter is denoted herein “material feature φ_{2m} ” or “material dependent feature φ_{2m} ”. The material feature may be or may comprise at least one information about the at least one material property of the surface of the scene having generated the reflection feature.

The material dependent image filter may be at least one filter selected from the group consisting of: a luminance filter; a spot shape filter; a squared norm gradient; a standard deviation; a smoothness filter such as a Gaussian filter or median filter; a grey-level-occurrence-based contrast filter; a grey-level-occurrence-based energy filter; a grey-level-occurrence-based homogeneity filter; a grey-level-occurrence-based dissimilarity filter; a Law’s energy filter; a threshold area filter; or a linear combination thereof; or a further material dependent image filter Φ_{2other} which correlates to one or more of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area filter, or a linear combination thereof by $|\rho_{\Phi_{2other}, \Phi_m}| \geq 0.40$ with Φ_m being one of the luminance filter, the spot shape filter, the squared norm gradient, the standard deviation, the smoothness filter, the grey-level-occurrence-based energy filter, the grey-level-occurrence-based homogeneity filter, the grey-level-occurrence-based dissimilarity filter, the Law’s energy filter, or the threshold area

filter, or a linear combination thereof. The further material dependent image filter $\Phi_{2\text{other}}$ may correlate to one or more of the material dependent image filters Φ_m by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.60$, preferably by $|\rho_{\Phi_{2\text{other}}, \Phi_m}| \geq 0.80$.

- 5 The material dependent image filter may be at least one arbitrary filter Φ that passes a hypothesis testing. As used herein, the term “passes a hypothesis testing” refers to the fact that a Null-hypothesis H_0 is rejected and an alternative hypothesis H_1 is accepted. The hypothesis testing may comprise testing the material dependency of the image filter by applying the image filter to a predefined data set. The data set may comprise a plurality of beam profile images. As used herein, the term “beam profile image” refers to a sum of N_B Gaussian radial basis functions,

$$f_k(x, y) = \left| \sum_{l=0}^{N_B-1} g_{lk}(x, y) \right|,$$

$$g_{lk}(x, y) = a_{lk} e^{-(\alpha(x-x_{lk}))^2} e^{-(\alpha(y-y_{lk}))^2}$$

- 15 wherein each of the N_B Gaussian radial basis functions is defined by a center (x_{lk}, y_{lk}) , a prefactor, a_{lk} , and an exponential factor $\alpha = 1/\epsilon$. The exponential factor is identical for all Gaussian functions in all images. The center-positions, x_{lk}, y_{lk} , are identical for all images f_k : $(x_0, x_1, \dots, x_{N_B-1}), (y_0, y_1, \dots, y_{N_B-1})$. Each of the beam profile images in the dataset may correspond to a material classifier and a distance. The material classifier may be a label such as ‘Material A’, ‘Material B’, etc. The beam profile images may be generated by using the above formula for $f_k(x, y)$ in combination with the following parameter table:

Image Index	Material classifier, Material Index	Distance z	Parameters
$k = 0$	Skin, $m = 0$	0.4 m	$(a_{00}, a_{10}, \dots, a_{N_B-1\ 0})$
$k = 1$	Skin, $m = 0$	0.6 m	$(a_{01}, a_{11}, \dots, a_{N_B-1\ 1})$
$k = 2$	Fabric, $m = 1$	0.6 m	$(a_{02}, a_{12}, \dots, a_{N_B-1\ 2})$
$\vdots$		$\vdots$	
$k = N$	Material J , $m = J - 1$		$(a_{0N}, a_{1N}, \dots, a_{N_B-1\ N})$

- 25 The values for x, y , are integers corresponding to pixels with $\begin{pmatrix} x \\ y \end{pmatrix} \in [0, 1, \dots, 31]^2$. The images may have a pixel size of 32x32. The dataset of beam profile images may be generated by using the above formula for f_k in combination with a parameter set to obtain a continuous description of f_k . The values for each pixel in the 32x32-image may be obtained by inserting integer values from 0, ..., 31 for x, y , in $f_k(x, y)$. For example, for pixel (6,9), the value $f_k(6,9)$ may be computed.

Subsequently, for each image f_k , the feature value φ_k corresponding to the filter Φ may be calculated, $\Phi(f_k(x, y), z_k) = \varphi_k$, wherein z_k is a distance value corresponding to the image f_k from the predefined data set. This yields a dataset with corresponding generated feature values φ_k . The hypothesis testing may use a Null-hypothesis that the filter does not distinguish between material classifier. The Null-Hypothesis may be given by $H_0: \mu_1 = \mu_2 = \dots = \mu_J$, wherein μ_m is the expectation value of each material-group corresponding to the feature values φ_k . Index m denotes the material group. The hypothesis testing may use as alternative hypothesis that the filter does distinguish between at least two material classifiers. The alternative hypothesis may be given by $H_1: \exists m, m' : \mu_m \neq \mu_{m'}$. As used herein, the term “not distinguish between material classifiers” refers to that the expectation values of the material classifiers are identical. As used herein, the term “distinguishes material classifiers” refers to that at least two expectation values of the material classifiers differ. As used herein “distinguishes at least two material classifiers” is used synonymous to “suitable material classifier”. The hypothesis testing may comprise at least one analysis of variance (ANOVA) on the generated feature values. In particular, the hypothesis testing may comprise determining a mean-value of the feature values for each of the J materials, i.e. in total J mean values, $\bar{\varphi}_m = \frac{\sum_i \varphi_{i,m}}{N_m}$, for $m \in [0, 1, \dots, J - 1]$, wherein N_m gives the number of feature values for each of the J materials in the predefined data set. The hypothesis testing may comprise determining a mean-value of all N feature values $\bar{\varphi} = \frac{\sum_m \sum_i \varphi_{i,m}}{N}$. The hypothesis testing may comprise determining a Mean Sum Squares within:

$$mssw = \left(\sum_m \sum_i (\varphi_{i,m} - \bar{\varphi}_m)^2 \right) / (N - J).$$

The hypothesis testing may comprise determining a Mean Sum of Squares between,

$$mssb = (\sum_m (\bar{\varphi}_m - \bar{\varphi})^2 N_m) / (J - 1).$$

The hypothesis testing may comprise performing an F-Test:

- $CDF(x) = I_{\frac{d_1 x}{d_1 x + d_2}} \left(\frac{d_1}{2}, \frac{d_2}{2} \right)$, where $d_1 = N - J, d_2 = J - 1$,
- $F(x) = 1 - CDF(x)$
- $p = F(mssb/mssw)$

Herein, I_x is the regularized incomplete Beta-Function, $I_x(a, b) = \frac{B(x; a, b)}{B(a, b)}$, with the Euler Beta-

Function $B(a, b) = \int_0^1 t^{a-1} (1 - t)^{b-1} dt$ and $B(x; a, b) = \int_0^x t^{a-1} (1 - t)^{b-1} dt$ being the incomplete Beta-Function. The image filter may pass the hypothesis testing if a p-value, p , is smaller or equal than a pre-defined level of significance. The filter may pass the hypothesis testing if $p \leq$

0.075, preferably $p \leq 0.05$, more preferably $p \leq 0.025$, and most preferably $p \leq 0.01$. For example, in case the pre-defined level of significance is $\alpha = 0.075$, the image filter may pass the hypothesis testing if the p-value is smaller than $\alpha = 0.075$. In this case the Null-hypothesis H_0 can be rejected and the alternative hypothesis H_1 can be accepted. The image filter thus distinguishes at least two material classifiers. Thus, the image filter passes the hypothesis testing.

In the following, image filters are described assuming that the reflection image comprises at least one reflection feature, in particular a spot image. A spot image f may be given by a function $f: \mathbb{R}^2 \rightarrow \mathbb{R}_{\geq 0}$, wherein the background of the image f may be already subtracted. However, other reflection features may be possible.

For example, the material dependent image filter may be a luminance filter. The luminance filter may return a luminance measure of a spot as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = - \int f(x) dx \frac{z^2}{d_{ray} \cdot n},$$

where f is the spot image. The distance of the spot is denoted by z , where z may be obtained for example by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. The surface normal of the material is given by $n \in \mathbb{R}^3$ and can be obtained as the normal of the surface spanned by at least three measured points. The vector $d_{ray} \in \mathbb{R}^3$ is the direction vector of the light source. Since the position of the spot is known by using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique wherein the position of the light source is known as a parameter of the detector system, d_{ray} , is the difference vector between spot and light source positions.

For example, the material dependent image filter may be a filter having an output dependent on a spot shape. This material dependent image filter may return a value which correlates to the translucence of a material as material feature. The translucence of materials influences the shape of the spots. The material feature may be given by

$$\varphi_{2m} = \Phi(f) = \frac{\int H(f(x) - \alpha h) dx}{\int H(f(x) - \beta h) dx},$$

wherein $0 < \alpha, \beta < 1$ are weights for the spot height h , and H denotes the Heavyside function, i.e. $H(x) = 1 : x \geq 0$, $H(x) = 0 : x < 0$. The spot height h may be determined by

$$h = \int_{B_r} f(x) dx,$$

where B_r is an inner circle of a spot with radius r .

For example, the material dependent image filter may be a squared norm gradient. This material dependent image filter may return a value which correlates to a measure of soft and hard transitions and/or roughness of a spot as material feature. The material feature may be defined by

$$\varphi_{2m} = \Phi(f) = \int \|\nabla f(x)\|^2 dx.$$

For example, the material dependent image filter may be a standard deviation. The standard deviation of the spot may be determined by

$$\varphi_{2m} = \Phi(f) = \int (f(x) - \mu)^2 dx,$$

Wherein μ is the mean value given by $\mu = \int (f(x)) dx$.

For example, the material dependent image filter may be a smoothness filter such as a Gaussian filter or median filter. In one embodiment of the smoothness filter, this image filter may refer to the observation that volume scattering exhibits less speckle contrast compared to diffuse scattering materials. This image filter may quantify the smoothness of the spot corresponding to speckle contrast as material feature. The material feature may be determined by

$$\varphi_{2m} = \Phi(f, z) = \frac{\int |\mathcal{F}(f)(x) - f(x)| dx}{\int f(x) dx} \cdot \frac{1}{z},$$

wherein $\mathcal{F}$ is a smoothness function, for example a median filter or Gaussian filter. This image filter may comprise dividing by the distance z , as described in the formula above. The distance z may be determined for example using a depth-from-defocus or depth-from-photon ratio technique and/or by using a triangulation technique. This may allow the filter to be insensitive to distance. In one embodiment of the smoothness filter, the smoothness filter may be based on the standard deviation of an extracted speckle noise pattern. A speckle noise pattern N can be described in an empirical way by

$$f(x) = f_0(x) \cdot (N(X) + 1),$$

where f_0 is an image of a despeckled spot. $N(X)$ is the noise term that models the speckle pattern. The computation of a despeckled image may be difficult. Thus, the despeckled image may be approximated with a smoothed version of f , i.e. $f_0 \approx \mathcal{F}(f)$, wherein $\mathcal{F}$ is a smoothness

operator like a Gaussian filter or median filter. Thus, an approximation of the speckle pattern may be given by

$$N(X) = \frac{f(x)}{\mathcal{F}(f(x))} - 1.$$

The material feature of this filter may be determined by

$$5 \quad \varphi_{2m} = \Phi(f) = \sqrt{\text{Var}\left(\frac{f}{\mathcal{F}(f)} - 1\right)},$$

Wherein Var denotes the variance function.

For example, the image filter may be a grey-level-occurrence-based contrast filter. This material filter may be based on the grey level occurrence matrix $M_{f,\rho}(g_1 g_2) = [p_{g_1, g_2}]$, whereas p_{g_1, g_2} is the occurrence rate of the grey combination $(g_1, g_2) = [f(x_1, y_1), f(x_2, y_2)]$, and the relation ρ defines the distance between (x_1, y_1) and (x_2, y_2) , which is $\rho(x, y) = (x+a, y+b)$ with a and b selected from 0,1.

The material feature of the grey-level-occurrence-based contrast filter may be given by

$$15 \quad \varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} p_{ij}(i-j)^2.$$

For example, the image filter may be a grey-level-occurrence-based energy filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based energy filter may be given by

$$20 \quad \varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} (p_{ij})^2.$$

For example, the image filter may be a grey-level-occurrence-based homogeneity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based homogeneity filter may be given by

$$25 \quad \varphi_{2m} = \Phi(f) = \sum_{i,j=0}^{N-1} \frac{p_{ij}}{1 + |i-j|}.$$

For example, the image filter may be a grey-level-occurrence-based dissimilarity filter. This material filter is based on the grey level occurrence matrix defined above.

The material feature of the grey-level-occurrence-based dissimilarity filter may be given by

$$\varphi_{2m} = \Phi(f) = - \sum_{i,j=0}^{N-1} \sqrt{p_{ij} \log(p_{ij})}.$$

For example, the image filter may be a Law's energy filter. This material filter may be based on the laws vector $L_5=[1,4,6,4,1]$ and $E_5=[-1,-2,0,-2,-1]$ and the matrices $L_5(E_5)^T$ and $E_5(L_5)^T$.

5 The image f_k is convoluted with these matrices:

$$f_{k,L_5E_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) L_5(E_5)^T$$

and

$$f_{k,E_5L_5}^*(x,y) = \sum_{i=-2}^2 \sum_{j=-2}^2 f_k(x+i, y+j) E_5(L_5)^T.$$

$$E = \int \frac{f_{k,L_5E_5}^*(x,y)}{\max(f_{k,L_5E_5}^*(x,y))} dx dy,$$

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$$F = \int \frac{f_{k,E_5L_5}^*(x,y)}{\max(f_{k,E_5L_5}^*(x,y))} dx dy,$$

Whereas the material feature of Law's energy filter may be determined by

$$\varphi_{2m} = \Phi(f) = E/F.$$

For example, the material dependent image filter may be a threshold area filter. This material
 15 feature may relate two areas in the image plane. A first area Ω_1 , may be an area wherein the
 function f is larger than α times the maximum of f . A second area Ω_2 , may be an area wherein
 the function f is smaller than α times the maximum of f , but larger than a threshold value ε times
 the maximum of f . Preferably α may be 0.5 and ε may be 0.05. Due to speckles or noise, the
 areas may not simply correspond to an inner and an outer circle around the spot center. As an
 20 example, Ω_1 may comprise speckles or unconnected areas in the outer circle. The material
 feature may be determined by

$$\varphi_{2m} = \Phi(f) = \frac{\int_{\Omega_1} 1}{\int_{\Omega_2} 1},$$

wherein $\Omega_1 = \{x | f(x) > \alpha \cdot \max(f(x))\}$ and $\Omega_2 = \{x | \varepsilon \cdot \max(f(x)) < f(x) < \alpha \cdot \max(f(x))\}$.

25 The processing unit may be configured for using at least one predetermined relationship
 between the material feature φ_{2m} and the material property of the surface having generated the
 reflection feature for determining the material property of the surface having generated the
 reflection feature. The predetermined relationship may be one or more of an empirical
 relationship, a semi-empiric relationship and an analytically derived relationship. The processing
 30 unit may comprise at least one data storage device for storing the predetermined relationship,
 such as a lookup list or a lookup table.

While feature based approaches, as the approaches described above, are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The term “machine-learning” as used herein is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The term “skin classification model” may refer to a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may be pre-determined and/or pre-defined.

Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The term “training”, also denoted learning, as used herein, is a broad term and is to be given its ordinary and customary meaning to a person of ordinary skill in the art and is not to be limited to a special or customized meaning. The term specifically may refer, without limitation, to a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face.

For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step may be performed after the skin detection step and/or the face detection step. However, other embodiments are feasible, in which the 3D detection step is performed before the skin detection step and/or the face detection step. The material property may be determined by evaluating φ_{2m} subsequently after determining of the longitudinal coordinate z in step d) such that the information about the longitudinal coordinate z can be considered for evaluating of φ_{2m} .

The 3D detection step comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The 3D detection step may comprise determining a second beam profile information of at least four of the reflection features by analysis of their respective beam profile. The second beam profile information may comprise a quotient Q of areas of the beam profile.

As used herein, the term "analysis of the beam profile" may generally refer to evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one

normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit may filter the beam profile by removing high spatial frequencies such as by spatial frequency analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit may be configured for removing influences from background light from the beam profile, for example, by an imaging without illumination.

The processing unit may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of the beam profile of the respective reflection feature. The processing unit may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

The processing unit may be configured for determining at least one first area and at least one second area of the reflection beam profile of each of the reflection features and/or of the reflection features in at least one region of interest. The processing unit is configured for integrating the first area and the second area.

The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam profile may be an area A1 and the second area of the beam profile may be an area A2. The processing unit may be configured for integrating the first area and the second area. The processing unit may be configured to derive a combined signal, in particular a quotient Q, by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit may be configured for determining for at least two of the areas an integral of the beam profile over the

respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit may be configured for determining at least one first area and at least one second area of the beam profile. As used herein, the term “area of the beam profile” generally refers to an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q. The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera may comprise at least two optical sensors, wherein the light-sensitive areas of a first optical sensor and of a second optical sensor may be arranged such that the first optical sensor is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor is adapted to determine the second area of the beam profile of the reflection feature. The processing unit may be adapted to integrate the first area and the second area. The processing unit may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The first area of the beam profile may comprise essentially edge information of the beam profile and the second area of the beam profile comprises essentially center information of the beam profile, and/or the first area of the beam profile may comprise essentially information about a left part of the beam profile and the second area of the beam profile comprises essentially information about a right part of the beam profile. The beam profile may have a center, i.e. a maximum value of the beam profile and/or a center point of a plateau of the beam profile and/or a geometrical center of the light spot, and falling edges extending from the center. The second region may comprise inner regions of the cross section and the first region may comprise outer regions of the cross section. As used herein, the term “essentially center information” generally refers to a low proportion of edge information, i.e. proportion of the intensity distribution corresponding to edges, compared to a proportion of the center information, i.e. proportion of the intensity distribution corresponding to the center. Preferably, the center information has a proportion of edge information of less than 10%, more preferably of less than 5%, most preferably the center information comprises no edge content. As used herein, the term “essentially edge information” generally refers to a low proportion of center information compared to a proportion of the edge information. The edge information may comprise information of the whole beam profile, in particular from center and edge regions. The edge information may have a proportion of center information of less than 10%, preferably of less than 5%, more preferably the edge information comprises no center content. At least one area of the beam profile may be determined and/or selected as second area of the beam profile if it is

close or around the center and comprises essentially center information. At least one area of the beam profile may be determined and/or selected as first area of the beam profile if it comprises at least parts of the falling edges of the cross section. For example, the whole area of the cross section may be determined as first region.

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Other selections of the first area A1 and second area A2 may be feasible. For example, the first area may comprise essentially outer regions of the beam profile and the second area may comprise essentially inner regions of the beam profile. For example, in case of a two-dimensional beam profile, the beam profile may be divided in a left part and a right part, wherein the first area may comprise essentially areas of the left part of the beam profile and the second area may comprise essentially areas of the right part of the beam profile.

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The edge information may comprise information relating to a number of photons in the first area of the beam profile and the center information may comprise information relating to a number of photons in the second area of the beam profile. The processing unit may be configured for determining an area integral of the beam profile. The processing unit may be configured for determining the edge information by integrating and/or summing of the first area. The processing unit may be configured for determining the center information by integrating and/or summing of the second area. For example, the beam profile may be a trapezoid beam profile and the processing unit may be configured for determining an integral of the trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

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In one embodiment, A1 may correspond to a full or complete area of a feature point on the optical sensor. A2 may be a central area of the feature point on the optical sensor. The central area may be a constant value. The central area may be smaller compared to the full area of the feature point. For example, in case of a circular feature point, the central area may have a radius from 0.1 to 0.9 of a full radius of the feature point, preferably from 0.4 to 0.6 of the full radius.

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In one embodiment, the illumination pattern may comprise at least one line pattern. A1 may correspond to an area with a full line width of the line pattern on the optical sensors, in particular on the light sensitive area of the optical sensors. The line pattern on the optical sensor may be widened and/or displaced compared to the line pattern of the illumination pattern such that the line width on the optical sensors is increased. In particular, in case of a matrix of optical sensors, the line width of the line pattern on the optical sensors may change from one column to another column. A2 may be a central area of the line pattern on the optical sensor. The line width of the central area may be a constant value, and may in particular correspond to the line width in the illumination pattern. The central area may have a smaller line width compared to the full line width. For example, the central area may have a line width from 0.1 to 0.9 of the full line width, preferably from 0.4 to 0.6 of the full line width. The line pattern may be segmented on the

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optical sensors. Each column of the matrix of optical sensors may comprise center information of intensity in the central area of the line pattern and edge information of intensity from regions extending further outwards from the central area to edge regions of the line pattern.

- 5 In one embodiment, the illumination pattern may comprise at least point pattern. A1 may correspond to an area with a full radius of a point of the point pattern on the optical sensors. A2 may be a central area of the point in the point pattern on the optical sensors. The central area may be a constant value. The central area may have a radius compared to the full radius. For example, the central area may have a radius from 0.1 to 0.9 of the full radius, preferably from 10 0.4 to 0.6 of the full radius.

The illumination pattern may comprise both at least one point pattern and at least one line pattern. Other embodiments in addition or alternatively to line pattern and point pattern are feasible.

- 15 The processing unit may be configured to derive a quotient Q by one or more of dividing the integrated first area and the integrated second area, dividing multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area.

- 20 The processing unit may be configured to derive the quotient Q by one or more of dividing the first area and the second area, dividing multiples of the first area and the second area, dividing linear combinations of the first area and the second area. The processing unit may be configured for deriving the quotient Q by

$$Q = \frac{\iint_{A1} E(x, y) dx dy}{\iint_{A2} E(x, y) dx dy}$$

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wherein x and y are transversal coordinates, A1 and A2 are the first and second area of the beam profile, respectively, and E(x,y) denotes the beam profile.

- 30 Additionally or alternatively, the processing unit may be adapted to determine one or both of center information or edge information from at least one slice or cut of the light spot. This may be realized, for example, by replacing the area integrals in the quotient Q by a line integral along the slice or cut. For improved accuracy, several slices or cuts through the light spot may be used and averaged. In case of an elliptical spot profile, averaging over several slices or cuts may result in improved distance information.

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For example, in case of the optical sensor having a matrix of pixels, the processing unit may be configured for evaluating the beam profile, by

- determining the pixel having the highest sensor signal and forming at least one center signal;
- 40 - evaluating sensor signals of the matrix and forming at least one sum signal;
- determining the quotient Q by combining the center signal and the sum signal; and

- determining at least one longitudinal coordinate z of the object by evaluating the quotient Q .

As used herein, a “sensor signal” generally refers to a signal generated by the optical sensor and/or at least one pixel of the optical sensor in response to illumination. Specifically, the sensor signal may be or may comprise at least one electrical signal, such as at least one analogue electrical signal and/or at least one digital electrical signal. More specifically, the sensor signal may be or may comprise at least one voltage signal and/or at least one current signal. More specifically, the sensor signal may comprise at least one photocurrent. Further, either raw sensor signals may be used, or the display device, the optical sensor or any other element may be adapted to process or preprocess the sensor signal, thereby generating secondary sensor signals, which may also be used as sensor signals, such as preprocessing by filtering or the like. The term “center signal” generally refers to the at least one sensor signal comprising essentially center information of the beam profile. As used herein, the term “highest sensor signal” refers to one or both of a local maximum or a maximum in a region of interest. For example, the center signal may be the signal of the pixel having the highest sensor signal out of the plurality of sensor signals generated by the pixels of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the pixels of the matrix. The center signal may arise from a single pixel or from a group of optical sensors, wherein, in the latter case, as an example, the sensor signals of the group of pixels may be added up, integrated or averaged, in order to determine the center signal. The group of pixels from which the center signal arises may be a group of neighboring pixels, such as pixels having less than a predetermined distance from the actual pixel having the highest sensor signal, or may be a group of pixels generating sensor signals being within a predetermined range from the highest sensor signal. The group of pixels from which the center signal arises may be chosen as large as possible in order to allow maximum dynamic range. The processing unit may be adapted to determine the center signal by integration of the plurality of sensor signals, for example the plurality of pixels around the pixel having the highest sensor signal. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the trapezoid, in particular of a plateau of the trapezoid.

As outlined above, the center signal generally may be a single sensor signal, such as a sensor signal from the pixel in the center of the light spot, or may be a combination of a plurality of sensor signals, such as a combination of sensor signals arising from pixels in the center of the light spot, or a secondary sensor signal derived by processing a sensor signal derived by one or more of the aforementioned possibilities. The determination of the center signal may be performed electronically, since a comparison of sensor signals is fairly simply implemented by conventional electronics, or may be performed fully or partially by software. Specifically, the center signal may be selected from the group consisting of: the highest sensor signal; an average of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of

sensor signals from a group of pixels containing the pixel having the highest sensor signal and a predetermined group of neighboring pixels; a sum of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an average of a group of sensor signals being above a predetermined threshold; a sum of a group of sensor signals being above a predetermined threshold; an integral of sensor signals from a group of optical sensors containing the optical sensor having the highest sensor signal and a predetermined group of neighboring pixels; an integral of a group of sensor signals being within a predetermined range of tolerance from the highest sensor signal; an integral of a group of sensor signals being above a predetermined threshold.

Similarly, the term "sum signal" generally refers to a signal comprising essentially edge information of the beam profile. For example, the sum signal may be derived by adding up the sensor signals, integrating over the sensor signals or averaging over the sensor signals of the entire matrix or of a region of interest within the matrix, wherein the region of interest may be predetermined or determinable within an image generated by the optical sensors of the matrix. When adding up, integrating over or averaging over the sensor signals, the actual optical sensors from which the sensor signal is generated may be left out of the adding, integration or averaging or, alternatively, may be included into the adding, integration or averaging. The processing unit may be adapted to determine the sum signal by integrating signals of the entire matrix, or of the region of interest within the matrix. For example, the beam profile may be a trapezoid beam profile and the processing unit may be adapted to determine an integral of the entire trapezoid. Further, when trapezoid beam profiles may be assumed, the determination of edge and center signals may be replaced by equivalent evaluations making use of properties of the trapezoid beam profile such as determination of the slope and position of the edges and of the height of the central plateau and deriving edge and center signals by geometric considerations.

Similarly, the center signal and edge signal may also be determined by using segments of the beam profile such as circular segments of the beam profile. For example, the beam profile may be divided into two segments by a secant or a chord that does not pass the center of the beam profile. Thus, one segment will essentially contain edge information, while the other segment will contain essentially center information. For example, to further reduce the amount of edge information in the center signal, the edge signal may further be subtracted from the center signal.

The quotient Q may be a signal which is generated by combining the center signal and the sum signal. Specifically, the determining may include one or more of: forming a quotient of the center signal and the sum signal or vice versa; forming a quotient of a multiple of the center signal and a multiple of the sum signal or vice versa; forming a quotient of a linear combination of the center signal and a linear combination of the sum signal or vice versa. Additionally or alternatively, the quotient Q may comprise an arbitrary signal or signal combination which contains at least one item of information on a comparison between the center signal and the sum signal.

As used herein, the term “longitudinal coordinate for the reflection feature” refers to a distance between the optical sensor and the point of the scene remitting the corresponding illumination features. The processing unit may be configured for using the at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The processing unit may be configured for executing at least one depth-from-photon-ratio algorithm which computes distances for all reflection features with zero order and higher order.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

The processing unit may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. As used herein, the term “depth” or depth information may refer to a distance between the object and the optical sensor and may be given by the longitudinal coordinate. As used herein, the term “depth map” may refer to spatial distribution of depth. The processing unit may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight, shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.

3D features determined step c) may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) and c) may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

Preferably subsequent to steps a) to c) the authentication step may be performed. The authentication step may be performed partially after each of steps a) to c). The authentication may be aborted in case in step a) no face is detected and/or in step b) the reflection features is determined not to be generated by skin and/or in step c) the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

Steps a) to d) may be performed by using at least one device, for example at least one mobile device such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The method may comprise permitting access to the device if the detected face is authenticated.

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. As used herein, the term “enrolling” may refer to a process of registering and/or signing up and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may a database of the device, e.g. of the processing unit, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Besides accuracy, also execution speed and power consumption are important requirements. Further restrictions on the availability of computational resources can be introduced by security considerations. For example, steps a) to d) may have run in a secure zone of the processing unit to avoid any software-based manipulations during program execution. The compact nature of the above described material detection networks may solve this problem by showing excellent runtime behavior in the said secure zone, whereas traditional PAD solutions require the examination of several consecutive frames which results in large computational cost and longer response times of the algorithm.

In a further aspect of the present invention a computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to the present invention, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to the present invention. Specifically, the computer program may be stored on a computer-readable data carrier and/or on a computer-readable storage medium.

As used herein, the terms "computer-readable data carrier" and "computer-readable storage medium" specifically may refer to non-transitory data storage means, such as a hard-ware storage medium having stored thereon computer-executable instructions. The computer-

readable data carrier or storage medium specifically may be or may comprise a storage medium such as a random-access memory (RAM) and/or a read-only memory (ROM).

Thus, specifically, one, more than one or even all of method steps as indicated above may be performed by using a computer or a computer network, preferably by using a computer program.

In a further aspect a computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to the present invention.

Further disclosed and proposed herein is a data carrier having a data structure stored thereon, which, after loading into a computer or computer network, such as into a working memory or main memory of the computer or computer network, may execute the method according to one or more of the embodiments disclosed herein.

Further disclosed and proposed herein is a computer program product with program code means stored on a machine-readable carrier, in order to perform the method according to one or more of the embodiments disclosed herein, when the program is executed on a computer or computer network. As used herein, a computer program product refers to the program as a tradable product. The product may generally exist in an arbitrary format, such as in a paper format, or on a computer-readable data carrier and/or on a computer-readable storage medium. Specifically, the computer program product may be distributed over a data network.

Finally, disclosed and proposed herein is a modulated data signal which contains instructions readable by a computer system or computer network, for performing the method according to one or more of the embodiments disclosed herein.

Referring to the computer-implemented aspects of the invention, one or more of the method steps or even all of the method steps of the methods according to one or more of the embodiments disclosed herein may be performed by using a computer or computer network. Thus, generally, any of the method steps including provision and/or manipulation of data may be performed by using a computer or computer network. Generally, these method steps may include any of the method steps, typically except for method steps requiring manual work.

Specifically, further disclosed herein are:

- a computer or computer network comprising at least one processor, wherein the processor is adapted to perform the method according to one of the embodiments described in this description,
- a computer loadable data structure that is adapted to perform the method according to one of the embodiments described in this description while the data structure is being executed on a computer,

- a computer program, wherein the computer program is adapted to perform the method according to one of the embodiments described in this description while the program is being executed on a computer,
- 5 - a computer program comprising program means for performing the method according to one of the embodiments described in this description while the computer program is being executed on a computer or on a computer network,
- a computer program comprising program means according to the preceding embodiment, wherein the program means are stored on a storage medium readable to a computer,
- 10 - a storage medium, wherein a data structure is stored on the storage medium and wherein the data structure is adapted to perform the methods according to one of the embodiments described in this description after having been loaded into a main and/or working storage of a computer or of a computer network, and
- 15 - a computer program product having program code means, wherein the program code means can be stored or are stored on a storage medium, for performing the method according to one of the embodiments described in this description, if the program code means are executed on a computer or on a computer network.

In a further aspect a mobile device comprising at least one camera, at least one illumination unit and at least one processing unit is disclosed. The mobile device is configured for performing at least steps a) to c), and optionally step d) of the method for face authentication according to the present invention. Step d) may be performed by using at least one authentication unit. The authentication unit may be a unit of the mobile device or may be an external authentication unit. With respect to definitions and embodiments of the mobile device reference is made to definitions and embodiments described with respect to the method.

25 In a further aspect of the present invention, use of the method according to the present invention, such as according to one or more of the embodiments given above or given in further detail below, is proposed, for a purpose of use for biometric presentation attack detection.

30 Overall, in the context of the present invention, the following embodiments are regarded as preferred:

Embodiment 1 Method for face authentication comprising the following steps:

- 35 a) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit;
- 40 b) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit and determining at least one second

image using the at least one camera, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;

c) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;

d) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.

Embodiment 2 The method according to the preceding embodiment, wherein steps a) to d) are performed by using at least one device, wherein access of the device is secured by using face authentication, wherein the method comprises permitting access to the device if the detected face is authenticated.

Embodiment 3 The method according to the preceding embodiment, wherein the method comprises at least one enrollment step, wherein in the enrollment step a user of the device is enrolled, wherein at least one 2D image of the user is stored in at least one database, wherein the method comprises identifying the user by comparing the 2D image of the user with the first image.

Embodiment 4 The method according to any one of the preceding embodiments, wherein in the skin detection step at least one parametrized skin classification model is used, wherein the parametrized skin classification model is configured for classifying skin and other materials by using the second image as an input.

Embodiment 5 The method according to the preceding embodiment, wherein the skin classification model is parametrized by using machine learning, wherein the property

characteristic for skin is determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model.

Embodiment 6 The method according to any one of the two preceding embodiments, wherein the skin detection step comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face, wherein in the skin detection step at least one region specific parametrized skin classification model is used.

Embodiment 7 The method according to any one of the preceding embodiments, wherein the illumination pattern comprises a periodic grid of laser spots.

Embodiment 8 The method according to any one of the preceding embodiments, wherein the illumination features have wavelengths in a near infrared (NIR) regime.

Embodiment 9 The method according to the preceding embodiment, wherein the illumination features have wavelengths of 940 nm.

Embodiment 10 The method according to any one of the preceding embodiments, wherein a plurality of second images is determined, wherein the reflection features of the plurality of second images are used for skin detection in step b) and/or for 3D detection in step c).

Embodiment 11 The method according to any one of the preceding embodiments, wherein the camera is or comprises at least one near infrared camera.

Embodiment 12 Computer program for face authentication configured for causing a computer or a computer network to fully or partially perform the method according to any one of the preceding embodiments, when executed on the computer or the computer network, wherein the computer program is configured for performing and/or executing at least steps a) to d) of the method according to any one of the preceding embodiments.

Embodiment 13 A computer-readable storage medium comprising instructions which, when executed by a computer or computer network, cause to carry out at least steps a) to d) of the method according to any one of the preceding embodiments referring to a method.

Embodiment 14 A mobile device comprising at least one camera, at least one illumination unit and at least one processing unit, the mobile device being configured for performing at least steps a) to c), and optionally step d), of the method for face authentication according to any one of the preceding embodiments referring to a method.

Embodiment 15 Use of the method according to any one of the preceding claims for biometric presentation attack detection.

Brief description of the figures

Further optional details and features of the invention are evident from the description of preferred exemplary embodiments which follows in conjunction with the dependent claims. In this context, the particular features may be implemented in an isolated fashion or in combination with other features. The invention is not restricted to the exemplary embodiments. The exemplary embodiments are shown schematically in the figures. Identical reference numerals in the individual figures refer to identical elements or elements with identical function, or elements which correspond to one another with regard to their functions.

Specifically, in the figures:

Figure 1 shows an embodiment of a method for face authentication according to the present invention;

Figure 2 shows an embodiment of a mobile device according to the present invention; and

Figure 3 shows experimental results.

Detailed description of the embodiments:

Figure 1 shows a flow chart of a method for face authentication according to the present invention. The face authentication may comprise verifying a recognized object or part of the recognized object as human face. Specifically, the authentication may comprise distinguishing a real human face from attack materials that were produced to mimic a face. The authentication may comprise verifying identity of a respective user and/or assigning identity to a user. The authentication may comprise generating and/or providing identity information, e.g. to other devices such as to at least one authorization device for authorization for access of a mobile device, a machine, an automobile, a building or the like. The identify information may be proofed by the authentication. For example, the identity information may be and/or may comprise at least one identity token. In case of successful authentication the recognized object or part of the recognized object is verified to be a real face and/or the identity of the object, in particular a user, is verified.

The method comprises the following steps:

- a) (reference number 110) at least one face detection step, wherein the face detection step comprises determining at least one first image by using at least one camera 112, wherein the first image comprises at least one two-dimensional image of a scene suspected to comprise the face, wherein the face detection step comprises detecting the face in the first image by identifying in the first image at least one pre-defined or pre-determined geometrical feature characteristic for faces by using at least one processing unit 114;

- 5 b) (reference number 116) at least one skin detection step, wherein the skin detection step comprises projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using at least one illumination unit 118 and determining at least one second image using the at least one camera 112, wherein the second image comprises a plurality of reflection features generated by the scene in response to illumination by the illumination features, wherein each of the reflection features comprises at least one beam profile, wherein the skin detection step comprises determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile and determining at least one material property of the reflection feature from the first beam profile information by using the processing unit 114, wherein the detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin;
- 10 c) (reference number 120) at least one 3D detection step, wherein the 3D detection step comprises determining a second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles and determining at least one depth level from the second beam profile information of said reflection features by using the processing unit 114, wherein the detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects;
- 15 d) (reference number 122) at least one authentication step, wherein the authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) the detected face is characterized as skin and in step c) the detected face is characterized as 3D object.
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The method steps may be performed in the given order or may be performed in a different order. Further, one or more additional method steps may be present which are not listed.

30 Further, one, more than one or even all of the method steps may be performed repeatedly.

The camera 112 may comprise at least one imaging element configured for recording or capturing spatially resolved one-dimensional, two-dimensional or even three-dimensional optical data or information. The camera 112 may be a digital camera. As an example, the camera 112 may comprise at least one camera chip, such as at least one CCD chip and/or at least one CMOS chip configured for recording images. The camera 112 may be or may comprise at least one near infrared camera. The image may relate to data recorded by using the camera 112, such as a plurality of electronic readings from the imaging device, such as the pixels of the camera chip. The camera 112, besides the at least one camera chip or imaging chip, may comprise further elements, such as one or more optical elements, e.g. one or more lenses. As an example, the camera 112 may be a fix-focus camera, having at least one lens which is fixedly adjusted with respect to the camera. Alternatively, however, the camera 112 may also comprise one or more variable lenses which may be adjusted, automatically or manually.

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The camera 112 may be a camera of a mobile device 124 such as of notebook computers, tablets or, specifically, cell phones such as smart phones and the like. Thus, specifically, the camera 112 may be part of the mobile device 124 which, besides the at least one camera 112, comprises one or more data processing devices such as one or more data processors. Other cameras, however, are feasible. The mobile device 124 may be a mobile electronics device, more specifically to a mobile communication device such as a cell phone or smart phone. Additionally or alternatively, the mobile device 124 may also refer to a tablet computer or another type of portable computer. An embodiment of a mobile device according to the present invention is shown in Figure 2.

Specifically, the camera 112 may be or may comprise at least one optical sensor 126 having at least one light-sensitive area. The optical sensor 126 specifically may be or may comprise at least one photodetector, preferably inorganic photodetectors, more preferably inorganic semiconductor photodetectors, most preferably silicon photodetectors. Specifically, the optical sensor 126 may be sensitive in the infrared spectral range. The optical sensor 126 may comprise at least one sensor element comprising a matrix of pixels. All pixels of the matrix or at least a group of the optical sensors of the matrix specifically may be identical. Groups of identical pixels of the matrix specifically may be provided for different spectral ranges, or all pixels may be identical in terms of spectral sensitivity. Further, the pixels may be identical in size and/or with regard to their electronic or optoelectronic properties. Specifically, the optical sensor 126 may be or may comprise at least one array of inorganic photodiodes which are sensitive in the infrared spectral range, preferably in the range of 700 nm to 3.0 micrometers. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1100 nm. Infrared optical sensors which may be used for optical sensors may be commercially available infrared optical sensors, such as infrared optical sensors commercially available under the brand name Hertzstueck™ from trinamiX™ GmbH, D-67056 Ludwigshafen am Rhein, Germany. Thus, as an example, the optical sensor 126 may comprise at least one optical sensor of an intrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge photodiode, an InGaAs photodiode, an extended InGaAs photodiode, an InAs photodiode, an InSb photodiode, a HgCdTe photodiode. Additionally or alternatively, the optical sensor may comprise at least one optical sensor of an extrinsic photovoltaic type, more preferably at least one semiconductor photodiode selected from the group consisting of: a Ge:Ga photodiode, a Ge:Hg photodiode, a Ge:Cu photodiode, a Ge:Zn photodiode, a Si:Ga photodiode, a Si:As photodiode. Additionally or alternatively, the optical sensor 126 may comprise at least one photoconductive sensor such as a PbS or PbSe sensor, a bolometer, preferably a bolometer selected from the group consisting of a VO bolometer and an amorphous Si bolometer.

Specifically, the optical sensor 126 may be sensitive in the near infrared region. Specifically, the optical sensor 126 may be sensitive in the part of the near infrared region where silicon photodiodes are applicable specifically in the range of 700 nm to 1000 nm. The optical sensor

126, specifically, may be sensitive in the infrared spectral range, specifically in the range of 780 nm to 3.0 micrometers. For example, the optical sensor 126 may be or may comprise at least one element selected from the group consisting of a CCD sensor element, a CMOS sensor element, a photodiode, a photocell, a photoconductor, a phototransistor or any combination thereof. Any other type of photosensitive element may be used. The photosensitive element generally may fully or partially be made of inorganic materials and/or may fully or partially be made of organic materials. Most commonly, one or more photodiodes may be used, such as commercially available photodiodes, e.g. inorganic semiconductor photodiodes.

The camera 112 further may comprise at least one transfer device (not shown here). The camera 112 may comprise at least one optical element selected from the group consisting of: transfer device, such as at least one lens and/or at least one lens system, at least one diffractive optical element. The transfer device may be adapted to guide the light beam onto the optical sensor 126. The transfer device specifically may comprise one or more of: at least one lens, for example at least one lens selected from the group consisting of at least one focus-tunable lens, at least one aspheric lens, at least one spheric lens, at least one Fresnel lens; at least one diffractive optical element; at least one concave mirror; at least one beam deflection element, preferably at least one mirror; at least one beam splitting element, preferably at least one of a beam splitting cube or a beam splitting mirror; at least one multi-lens system. The transfer device may have a focal length. Thus, the focal length constitutes a measure of an ability of the transfer device to converge an impinging light beam. Thus, the transfer device may comprise one or more imaging elements which can have the effect of a converging lens. By way of example, the transfer device can have one or more lenses, in particular one or more refractive lenses, and/or one or more convex mirrors. In this example, the focal length may be defined as a distance from the center of the thin refractive lens to the principal focal points of the thin lens. For a converging thin refractive lens, such as a convex or biconvex thin lens, the focal length may be considered as being positive and may provide the distance at which a beam of collimated light impinging the thin lens as the transfer device may be focused into a single spot. Additionally, the transfer device can comprise at least one wavelength-selective element, for example at least one optical filter. Additionally, the transfer device can be designed to impress a predefined beam profile on the electromagnetic radiation, for example, at the location of the sensor region and in particular the sensor area. The abovementioned optional embodiments of the transfer device can, in principle, be realized individually or in any desired combination.

The transfer device may have an optical axis. The transfer device may constitute a coordinate system, wherein a longitudinal coordinate is a coordinate along the optical axis and wherein d is a spatial offset from the optical axis. The coordinate system may be a polar coordinate system in which the optical axis of the transfer device forms a z -axis and in which a distance from the z -axis and a polar angle may be used as additional coordinates. A direction parallel or antiparallel to the z -axis may be considered a longitudinal direction, and a coordinate along the z -axis may be considered a longitudinal coordinate. Any direction perpendicular to the z -axis may be

considered a transversal direction, and the polar coordinate and/or the polar angle may be considered a transversal coordinate.

The camera 112 is configured for determining at least one image of the scene, in particular the first image. The scene may refer to a spatial region. The scene may comprise the face under authentication and a surrounding environment. The first image itself may comprise pixels, the pixels of the image correlating to pixels of the matrix of the sensor element. The first image is at least one two-dimensional image having information about transversal coordinates such as the dimensions of height and width.

The face detection step 110 comprises detecting the face in the first image by identifying in the first image the at least one pre-defined or pre-determined geometrical feature characteristic for faces by using the at least one processing unit 114. As an example, the at least one processing unit 114 may comprise a software code stored thereon comprising a number of computer commands. The processing unit 114 may provide one or more hardware elements for performing one or more of the named operations and/or may provide one or more processors with software running thereon for performing one or more of the named operations. Operations, including evaluating the images may be performed by the at least one processing unit 114. Thus, as an example, one or more instructions may be implemented in software and/or hardware. Thus, as an example, the processing unit 114 may comprise one or more programmable devices such as one or more computers, application-specific integrated circuits (ASICs), Digital Signal Processors (DSPs), or Field Programmable Gate Arrays (FPGAs) which are configured to perform the above-mentioned evaluation. Additionally or alternatively, however, the processing unit may also fully or partially be embodied by hardware. The processing unit 114 and the camera 112 may fully or partially be integrated into a single device. Thus, generally, the processing unit 114 also may form part of the camera 112. Alternatively, the processing unit 114 and the camera 112 may fully or partially be embodied as separate devices.

The detecting of the face in the first image may comprise identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces. The geometrical feature characteristic for faces may be at least one geometry-based feature which describe the shape of the face and its components, in particular one or more of nose, eyes, mouth or eyebrow and the like. The processing unit 114 may comprise at least one database wherein the geometrical feature characteristic for faces are stored such as in a lookup table. Techniques for identifying the at least one pre-defined or pre-determined geometrical feature characteristic for faces are generally known to the skilled person. For example, the face detection may be performed as described in Masi, Lacopo, et al. "Deep face recognition: A survey" 2018 31st SIBGRAPI conference on graphics, patterns and images (SIBGRAPI), IEEE, 2018, the full content of which is included by reference.

The processing unit 114 may be configured for performing at least one image analysis and/or image processing in order to identify the geometrical feature. The image analysis and/or image

processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a background correction; a decomposition into color channels; a decomposition into hue, saturation, and/or brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region detector; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying an edge detector; applying a differential edge detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator; applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be determined manually by a user or may be determined automatically, such as by recognizing a feature within the first image.

Specifically subsequent to the face detection step 110, the skin detection step 116 may be performed comprising projecting at least one illumination pattern comprising a plurality of illumination features on the scene by using the at least one illumination unit 118. However, embodiments are feasible wherein the skin detection step 116 is performed before the face detection step 110.

The illumination unit 118 may be configured for providing the illumination pattern for illumination of the scene. The illumination unit 118 may be adapted to directly or indirectly illuminating the scene, wherein the illumination pattern is remitted, in particular reflected or scattered, by surfaces of the scene and, thereby, is at least partially directed towards the camera. The illumination unit 118 may be configured for illuminating the scene, for example, by directing a light beam towards the scene, which reflects the light beam. The illumination unit 118 may be configured for generating an illuminating light beam for illuminating the scene.

The illumination unit 118 may comprise at least one light source. The illumination unit 118 may comprise a plurality of light sources. The illumination unit 118 may comprise an artificial illumination source, in particular at least one laser source and/or at least one incandescent lamp and/or at least one semiconductor light source, for example, at least one light-emitting diode, in particular an organic and/or inorganic light-emitting diode. The illumination unit 118 may be configured for generating the at least one illumination pattern in the infrared region. The illumination features may have wavelengths in a near infrared (NIR) regime. The illumination features may have wavelengths of about 940 nm. At this wavelength Melanin absorption runs out so that dark and light complexion reflect light almost identical. However, other wavelength in the NIR region may be possible such as one or more of 805 nm, 830 nm, 835 nm, 850 nm, 905 nm, or 980 nm. Moreover, using light in the near infrared region allows that light is not or only weakly detected by human eyes and is still detectable by silicon sensors, in particular standard silicon sensors.

The illumination unit 118 may be or may comprise at least one multiple beam light source. For example, the illumination unit 118 may comprise at least one laser source and one or more diffractive optical elements (DOEs). Specifically, the illumination unit 118 may comprise at least one laser and/or laser source. Various types of lasers may be employed, such as semiconductor lasers, double heterostructure lasers, external cavity lasers, separate confinement heterostructure lasers, quantum cascade lasers, distributed bragg reflector lasers, polariton lasers, hybrid silicon lasers, extended cavity diode lasers, quantum dot lasers, volume Bragg grating lasers, Indium Arsenide lasers, transistor lasers, diode pumped lasers, distributed feedback lasers, quantum well lasers, interband cascade lasers, Gallium Arsenide lasers, semiconductor ring laser, extended cavity diode lasers, or vertical cavity surface-emitting lasers. Additionally or alternatively, non-laser light sources may be used, such as LEDs and/or light bulbs. The illumination unit 118 may comprise one or more diffractive optical elements (DOEs) adapted to generate the illumination pattern. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points, for example the illumination unit 118 may comprise one or more of at least one digital light processing projector, at least one LCoS projector, at least one spatial light modulator; at least one diffractive optical element; at least one array of light emitting diodes; at least one array of laser light sources. On account of their generally defined beam profiles and other properties of handleability, the use of at least one laser source as the illumination unit 118 is particularly preferred. The illumination unit 118 may be integrated into a housing of the camera 112 or may be separated from the camera 112.

The illumination pattern comprises at least one illumination feature adapted to illuminate at least one part of the scene. The illumination pattern may comprise a single illumination feature. The illumination pattern may comprise a plurality of illumination features. The illumination pattern may be selected from the group consisting of: at least one point pattern; at least one line pattern; at least one stripe pattern; at least one checkerboard pattern; at least one pattern comprising an arrangement of periodic or non periodic features. The illumination pattern may comprise regular and/or constant and/or periodic pattern such as a triangular pattern, a rectangular pattern, a hexagonal pattern or a pattern comprising further convex tilings. The illumination pattern may exhibit the at least one illumination feature selected from the group consisting of: at least one point; at least one line; at least two lines such as parallel or crossing lines; at least one point and one line; at least one arrangement of periodic or non-periodic feature; at least one arbitrary shaped featured. The illumination pattern may comprise at least one pattern selected from the group consisting of: at least one point pattern, in particular a pseudo-random point pattern; a random point pattern or a quasi random pattern; at least one Sobol pattern; at least one quasiperiodic pattern; at least one pattern comprising at least one pre-known feature at least one regular pattern; at least one triangular pattern; at least one hexagonal pattern; at least one rectangular pattern at least one pattern comprising convex uniform tilings; at least one line pattern comprising at least one line; at least one line pattern comprising at least two lines such as parallel or crossing lines. For example, the illumination unit 118 may be adapted to generate and/or to project a cloud of points. The illumination unit 118 may comprise the at least one light projector adapted to generate a cloud of points such that the

illumination pattern may comprise a plurality of point pattern. The illumination pattern may comprise a periodic grid of laser spots. The illumination unit 118 may comprise at least one mask adapted to generate the illumination pattern from at least one light beam generated by the illumination unit 118.

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The skin detection step 116 comprises determining the at least one second image, also denoted as reflection image, using the camera 112. The method may comprise determining plurality of second images. The reflection features of the plurality of second images may be used for skin detection in step b) and/or for 3D detection in step c). The reflection feature may be a feature in
10 an image plane generated by the scene in response to illumination, specifically with at least one illumination feature. Each of the reflection features comprises at least one beam profile, also denoted reflection beam profile. The beam profile of the reflection feature may generally refer to at least one intensity distribution of the reflection feature, such as of a light spot on the optical sensor, as a function of the pixel. The beam profile may be selected from the group consisting
15 of a trapezoid beam profile; a triangle beam profile; a conical beam profile and a linear combination of Gaussian beam profiles.

The evaluation of the second image may comprise identifying the reflection features of the second image. The processing unit 114 may be configured for performing at least one image
20 analysis and/or image processing in order to identify the reflection features. The image analysis and/or image processing may use at least one feature detection algorithm. The image analysis and/or image processing may comprise one or more of the following: a filtering; a selection of at least one region of interest; a formation of a difference image between an image created by the sensor signals and at least one offset; an inversion of sensor signals by inverting an image
25 created by the sensor signals; a formation of a difference image between an image created by the sensor signals at different times; a background correction; a decomposition into color channels; a decomposition into hue; saturation; and brightness channels; a frequency decomposition; a singular value decomposition; applying a blob detector; applying a corner detector; applying a Determinant of Hessian filter; applying a principle curvature-based region
30 detector; applying a maximally stable extremal regions detector; applying a generalized Hough-transformation; applying a ridge detector; applying an affine invariant feature detector; applying an affine-adapted interest point operator; applying a Harris affine region detector; applying a Hessian affine region detector; applying a scale-invariant feature transform; applying a scale-space extrema detector; applying a local feature detector; applying speeded up robust features
35 algorithm; applying a gradient location and orientation histogram algorithm; applying a histogram of oriented gradients descriptor; applying a Deriche edge detector; applying a differential edge detector; applying a spatio-temporal interest point detector; applying a Moravec corner detector; applying a Canny edge detector; applying a Laplacian of Gaussian filter; applying a Difference of Gaussian filter; applying a Sobel operator; applying a Laplace operator;
40 applying a Scharr operator; applying a Prewitt operator; applying a Roberts operator; applying a Kirsch operator; applying a high-pass filter; applying a low-pass filter; applying a Fourier transformation; applying a Radon-transformation; applying a Hough-transformation; applying a wavelet-transformation; a thresholding; creating a binary image. The region of interest may be

determined manually by a user or may be determined automatically, such as by recognizing a feature within the image generated by the optical sensor 126.

For example, the illumination unit 118 may be configured for generating and/or projecting a cloud of points such that a plurality of illuminated regions is generated on the optical sensor 126, for example the CMOS detector. Additionally, disturbances may be present on the optical sensor 126 such as disturbances due to speckles and/or extraneous light and/or multiple reflections. The processing unit 114 may be adapted to determine at least one region of interest, for example one or more pixels illuminated by the light beam which are used for determination of the longitudinal coordinate for the respective reflection feature, which will be described in more detail below. For example, the processing unit 114 may be adapted to perform a filtering method, for example, a blob-analysis and/or an edge filter and/or object recognition method.

The processing unit 114 may be configured for performing at least one image correction. The image correction may comprise at least one background subtraction. The processing unit 114 may be adapted to remove influences from background light from the beam profile, for example, by an imaging without further illumination.

The processing unit 114 may be configured for determining the beam profile of the respective reflection feature. The determining the beam profile may comprise identifying at least one reflection feature provided by the optical sensor 126 and/or selecting at least one reflection feature provided by the optical sensor 126 and evaluating at least one intensity distribution of the reflection feature. As an example, a region of the matrix may be used and evaluated for determining the intensity distribution, such as a three-dimensional intensity distribution or a two-dimensional intensity distribution, such as along an axis or line through the matrix. As an example, a center of illumination by the light beam may be determined, such as by determining the at least one pixel having the highest illumination, and a cross-sectional axis may be chosen through the center of illumination. The intensity distribution may an intensity distribution as a function of a coordinate along this cross-sectional axis through the center of illumination. Other evaluation algorithms are feasible.

The processing unit 114 is configured for determining a first beam profile information of at least one of the reflection features located inside an image region of the second image corresponding to an image region of the first image comprising the identified geometrical feature by analysis of its beam profile. The method may comprise identifying the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. Specifically, the method may comprise matching pixels of the first image and the second image and selecting the pixels of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The method may comprise considering in addition further reflection features located outside said image region of the second image.

The beam profile information may be or may comprise arbitrary information and/or property derived from and/or relating to the beam profile of the reflection feature. The first and the second beam profile information may be identical or may be different. For example, the first beam profile information may be an intensity distribution, a reflection profile, a center of intensity, a material feature. For skin detection in step b) 116, beam profile analysis may be used. Specifically, beam profile analysis makes use of reflection properties of coherent light projected onto object surfaces to classify materials. The classification of materials may be performed as described in WO 2020/187719, in EP application 20159984.2 filed on February 28, 2020 and/or EP application 20 154 961.5 filed on January 31, 2020, the full content of which is included by reference. Specifically, a periodic grid of laser spots, e.g. a hexagonal grid as described in EP application 20 170 905.2 filed on April 22, 2020, is projected and the reflection image is recorded with the camera. Analyzing the beam profile of each reflection feature recorded by the camera may be performed by feature-based methods. With respect to feature-based methods reference is made to the description above. The feature based methods may be used in combination with machine learning methods which may allow parametrization of a skin classification model. Alternatively or in combination, convolutional neuronal networks may be utilized to classify skin by using the reflection images as an input.

The skin detection step 116 may comprise determining at least one material property of the reflection feature from the beam profile information by using the processing unit 114. Specifically, the processing unit 114 is configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular human skin, in case its reflection beam profile fulfills at least one predetermined or predefined criterion. The at least one predetermined or predefined criterion may be at least one property and/or value suitable to distinguish biological tissue, in particular human skin, from other materials. The predetermined or predefined criterion may be or may comprise at least one predetermined or predefined value and/or threshold and/or threshold range referring to a material property. The reflection feature may be indicated as to be generated by biological tissue in case the reflection beam profile fulfills the at least one predetermined or predefined criterion. The processing unit is configured for identifying the reflection feature as to be non-skin otherwise. Specifically, the processing unit 114 may be configured for skin detection, in particular for identifying if the detected face is human skin. The identification if the material is biological tissue, in particular human skin, may comprise to determining and/or validating whether a surface to be examined or under test is or comprises biological tissue, in particular human skin, and/or to distinguish biological tissue, in particular human skin, from other tissues, in particular other surfaces. The method according to the present invention may allow for distinguishing human skin from one or more of inorganic tissue, metal surfaces, plastics surfaces, foam, paper, wood, a display, a screen, cloth. The method according to the present invention may allow for distinguishing human biological tissue from surfaces of artificial or non-living objects.

The processing unit 114 may be configured for determining the material property m of the surface remitting the reflection feature by evaluating the beam profile of the reflection feature. The material property may be at least one arbitrary property of the material configured for

characterizing and/or identification and/or classification of the material. For example, the material property may be a property selected from the group consisting of: roughness, penetration depth of light into the material, a property characterizing the material as biological or non-biological material, a reflectivity, a specular reflectivity, a diffuse reflectivity, a surface property, a measure for translucence, a scattering, specifically a back-scattering behavior or the like. The at least one material property may be a property selected from the group consisting of: a scattering coefficient, a translucency, a transparency, a deviation from a Lambertian surface reflection, a speckle, and the like. The determining at least one material property may comprise assigning the material property to the detected face. The processing unit 114 may comprise at least one database comprising a list and/or table, such as a lookup list or a lookup table, of predefined and/or predetermined material properties. The list and/or table of material properties may be determined and/or generated by performing at least one test measurement, for example by performing material tests using samples having known material properties. The list and/or table of material properties may be determined and/or generated at the manufacturer site and/or by a user. The material property may additionally be assigned to a material classifier such as one or more of a material name, a material group such as biological or non-biological material, translucent or non-translucent materials, metal or non-metal, skin or non-skin, fur or non-fur, carpet or non-carpet, reflective or non-reflective, specular reflective or non-specular reflective, foam or non-foam, hair or non-hair, roughness groups or the like. The processing unit 114 may comprise at least one database comprising a list and/or table comprising the material properties and associated material name and/or material group.

While feature based approaches are accurate enough to differentiate between skin and surface-only scattering materials, the differentiation between skin and carefully selected attack materials, which involve volume scattering as well, is more challenging. Step b) 116 may comprise using artificial intelligence, in particular convolutional neuronal networks. Using reflection images as input for convolutional neuronal networks may enable the generation of classification models with sufficient accuracy to differentiate between skin and other volume-scattering materials. Since only physically valid information is passed to the network by selecting important regions in the reflection image, only compact training data sets may be needed. Additionally, very compact network architectures can be generated.

Specifically, in the skin detection step 116 at least one parametrized skin classification model may be used. The parametrized skin classification model may be configured for classifying skin and other materials by using the second image as an input. The skin classification model may be parametrized by using one or more of machine learning, deep learning, neural networks, or other form of artificial intelligence. The machine-learning may comprise a method of using artificial intelligence (AI) for automatically model building, in particular for parametrizing models. The skin classification model may comprise a classification model configured for discriminating human skin from other materials. The property characteristic for skin may be determined by applying an optimization algorithm in terms of at least one optimization target on the skin classification model. The machine learning may be based on at least one neuronal network, in particular a convolutional neural network. Weights and/or topology of the neuronal network may

be pre-determined and/or pre-defined. Specifically, the training of the skin classification model may be performed using machine-learning. The skin classification model may comprise at least one machine-learning architecture and model parameters. For example, the machine-learning architecture may be or may comprise one or more of: linear regression, logistic regression, random forest, naive Bayes classifications, nearest neighbors, neural networks, convolutional neural networks, generative adversarial networks, support vector machines, or gradient boosting algorithms or the like. The training may comprise a process of building the skin classification model, in particular determining and/or updating parameters of the skin classification model. The skin classification model may be at least partially data-driven. For example, the skin classification model may be based on experimental data, such as data determined by illuminating a plurality of humans and artificial objects such as masks and recording the reflection pattern. For example, the training may comprise using at least one training dataset, wherein the training data set comprises images, in particular second images, of a plurality of humans and artificial objects with known material property.

The skin detection step 116 may comprises using at least one 2D face and facial landmark detection algorithm configured for providing at least two locations of characteristic points of a human face. For example, the locations may be eye locations, forehead or cheeks. 2D face and facial landmark detection algorithms may provide locations of characteristic points of a human face such as eye locations. Since there are subtle differences in the reflection of the different zones of a face (for example forehead or cheeks), region specific models can be trained. In the skin detection step 116, preferably at least one region specific parametrized skin classification model is used. The skin classification model may comprise a plurality of region specific parametrized skin classification models, such as for different regions, and/or the skin classification model may be trained using region specific data such as by filtering the images used for training. For example, for training two different regions may be used such as eye-cheek region to below the nose and, in particular in case not enough reflection features can be identified within this region, the region of the forehead may be used. However, other regions may be possible, too.

The detected face is characterized as skin if the material property corresponds to at least one property characteristic for skin. The processing unit 114 may be configured for identifying a reflection feature as to be generated by illuminating biological tissue, in particular skin, in case its corresponding material property fulfills the at least one predetermined or predefined criterion. The reflection feature may be identified as to be generated by human skin in case the material property indicates "human skin". The reflection feature may be identified as to be generated by human skin in case the material property is within at least one threshold and/or at least one range. The at least one threshold value and/or range may be stored in a table or a lookup table and may be determined e.g. empirically and may, as an example, be stored in at least one data storage device of the processing unit. The processing unit 114 is configured for identifying the reflection feature as to be background otherwise. Thus, the processing unit 114 may be configured for assigning each projected spot with a material property, e.g. skin yes or no.

The 3D detection step 120 may be performed after the skin detection step 116 and/or the face detection step 110. However, other embodiments are feasible, in which the 3D detection step 120 is performed before the skin detection step 116 and/or the face detection step 110.

- 5 The 3D detection step 120 comprises determining the second beam profile information of at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of their beam profiles. The second beam profile information may comprise a quotient Q of areas of the beam profile.

10 The analysis of the beam profile may comprise evaluating of the beam profile and may comprise at least one mathematical operation and/or at least one comparison and/or at least symmetrizing and/or at least one filtering and/or at least one normalizing. For example, the analysis of the beam profile may comprise at least one of a histogram analysis step, a
15 calculation of a difference measure, application of a neural network, application of a machine learning algorithm. The processing unit 114 may be configured for symmetrizing and/or for normalizing and/or for filtering the beam profile, in particular to remove noise or asymmetries from recording under larger angles, recording edges or the like. The processing unit 114 may filter the beam profile by removing high spatial frequencies such as by spatial frequency
20 analysis and/or median filtering or the like. Summarization may be performed by center of intensity of the light spot and averaging all intensities at the same distance to the center. The processing unit 114 may be configured for normalizing the beam profile to a maximum intensity, in particular to account for intensity differences due to the recorded distance. The processing unit 114 may be configured for removing influences from background light from the beam
25 profile, for example, by an imaging without illumination.

The processing unit 114 may be configured for determining at least one longitudinal coordinate z_{DPR} for reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature by analysis of
30 the beam profile of the respective reflection feature. The processing unit 114 may be configured for determining the longitudinal coordinate z_{DPR} for the reflection features by using the so called depth-from-photon-ratio technique, also denoted as beam profile analysis. With respect to depth-from-photon-ratio (DPR) technique reference is made to WO 2018/091649 A1, WO 2018/091638 A1 and WO 2018/091640 A1, the full content of which is included by reference.

35 The longitudinal coordinate for the reflection feature may be a distance between the optical sensor 126 and the point of the scene remitting the corresponding illumination features. The analysis of the beam profile of one of the reflection features may comprise determining at least one first area and at least one second area of the beam profile. The first area of the beam
40 profile may be an area $A1$ and the second area of the beam profile may be an area $A2$. The processing unit 114 may be configured for integrating the first area and the second area. The processing unit 114 may be configured to derive a combined signal, in particular a quotient Q , by one or more of dividing the integrated first area and the integrated second area, dividing

multiples of the integrated first area and the integrated second area, dividing linear combinations of the integrated first area and the integrated second area. The processing unit 114 may be configured for determining at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile, wherein overlapping of the areas may be possible as long as the areas are not congruent. For example, the processing unit 114 may be configured for determining a plurality of areas such as two, three, four, five, or up to ten areas. The processing unit 114 may be configured for segmenting the light spot into at least two areas of the beam profile and/or to segment the beam profile in at least two segments comprising different areas of the beam profile. The processing unit 114 may be configured for determining for at least two of the areas an integral of the beam profile over the respective area. The processing unit may be configured for comparing at least two of the determined integrals. Specifically, the processing unit 114 may be configured for determining at least one first area and at least one second area of the beam profile. The area of the beam profile may be an arbitrary region of the beam profile at the position of the optical sensor used for determining the quotient Q . The first area of the beam profile and the second area of the beam profile may be one or both of adjacent or overlapping regions. The first area of the beam profile and the second area of the beam profile may be not congruent in area. For example, the processing unit 114 may be configured for dividing a sensor region of the CMOS sensor into at least two sub-regions, wherein the processing unit may be configured for dividing the sensor region of the CMOS sensor into at least one left part and at least one right part and/or at least one upper part and at least one lower part and/or at least one inner and at least one outer part. Additionally or alternatively, the camera 112 may comprise at least two optical sensors 126, wherein the light-sensitive areas of a first optical sensor 126 and of a second optical sensor 126 may be arranged such that the first optical sensor 126 is adapted to determine the first area of the beam profile of the reflection feature and that the second optical sensor 126 is adapted to determine the second area of the beam profile of the reflection feature. The processing unit 114 may be adapted to integrate the first area and the second area. The processing unit 114 may be configured for using at least one predetermined relationship between the quotient Q and the longitudinal coordinate for determining the longitudinal coordinate. The predetermined relationship may be one or more of an empiric relationship, a semi-empiric relationship and an analytically derived relationship. The processing unit 114 may comprise at least one data storage device for storing the predetermined relationship, such as a lookup list or a lookup table.

The 3D detection step may comprise determining the at least one depth level from the second beam profile information of said reflection features by using the processing unit.

The processing unit 114 may be configured for determining the depth map of at least parts of the scene by determining at least one depth information of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified geometrical feature. The processing unit 114 may be configured for determining the depth information of the reflection features by one or more of the following techniques: depth-from-photon-ratio, structured light, beam profile analysis, time-of-flight,

shape-from-motion, depth-from-focus, triangulation, depth-from-defocus, stereo sensors. The depth map may be a thinly filled depth map comprising a few entries. Alternatively, the depth may be crowded comprising a large amount of entries.

- 5 The detected face is characterized as 3D object if the depth level deviates from a pre-determined or pre-defined depth level of plane objects. Step c) 120 may comprise using 3D topology data of the face in front of the camera. The method may comprise determining a curvature from the at least four of the reflection features located inside the image region of the second image corresponding to the image region of the first image comprising the identified
- 10 geometrical feature. The method may comprise comparing the curvature determined from the at least four of the reflection features to the pre-determined or pre-defined depth level of plane objects. If the curvature exceeds an assumed curvature of plane object the detected face may be characterized as 3D object, otherwise as plane object. The pre-determined or pre-defined depth level of plane objects may be stored in at least one data storage of the processing unit
- 15 such as a lookup list or a lookup table. The pre-determined or pre-defined level of plane objects may be experimentally determined and/or may be a theoretical level of plane objects. The pre-determined or pre-defined depth level of plane objects may be at least one limit for at least one curvature and/or a range for at least one curvature.
- 20 3D features determined step c) 120 may allow differentiating between high quality photographs and a 3D face-like structure. The combination of steps b) 116 and c) 120 may allow strengthening reliability of the authentication with respect to attacks. 3D features can be combined with material features to increase the security level. Since the same computational pipeline can be used to generate the input data for the skin classification and the generation of
- 25 the 3D point cloud, both properties can be calculated from the same frame with low computational effort.

- Preferably subsequent to steps a) 110, b) 116 and c) 120 the authentication step 122 may be performed. The authentication step 122 may be performed partially after each of steps a) to c).
- 30 The authentication may be aborted in case in step a) 110 no face is detected and/or in step b) 116 the reflection features is determined not to be generated by skin and/or in step c) 120 the depth map refers to a plane object. The authentication step comprises authenticating the detected face by using at least one authentication unit if in step b) 116 the detected face is characterized as skin and in step c) 122 the detected face is characterized as 3D object.

- 35 Steps a) to d) may be performed by using at least one device, for example the at least one mobile device¹²⁴ such as a mobile phone, smartphone and the like, wherein access of the device is secured by using face authentication. Other devices may be possible, too such as an access control device controlling access to buildings, machines, automobiles and the like. The
- 40 method may comprise permitting access to the device if the detected face is authenticated.

The method may comprise at least one enrollment step. In the enrollment step a user of the device may be enrolled. The enrolling may comprise a process of registering and/or signing up

and/or teach in of a user for subsequent usage of the device. Usually, enrolling may be performed at first use of the device and/or for initiating the device. However, embodiments are feasible in which a plurality of users may be enrolled, e.g. successively, such that the enrolling may be performed and/or repeated at an arbitrary time during usage of the device. The enrolling may comprise generating a user account and/or user profile. The enrolling may comprise entering and storing user data, in particular image data, via at least one user interface. Specifically, at least one 2D image of the user is stored in at least one database. The enrollment step may comprise imaging at least one image of the user, in particular a plurality of images. The images may be recorded from different direction and/or the user may change his orientation. Additionally, the enrollment step may comprise generating at least one 3D image and/or a depth map for the user which may be used in step d) for comparison. The database may be a database of the device, e.g. of the processing unit 114, and/or may be an external database such as a cloud. The method comprises identifying the user by comparing the 2D image of the user with the first image. The method according to the present invention may allow significantly improving the presentation attack detection capabilities of biometric authentication methods. In order to improve the overall authentication, person specific material fingerprints as well as 3D topological features may be stored during the enrollment process in addition to the 2D image of the user. This may allow a multifactor authentication within one device by using 2D, 3D and material-derived features.

The method according to the present invention using beam profile analysis technology may provide a concept to reliably detect human skin by analyzing reflections of laser spots, in particular in the NIR regime, on a face and distinguish it from reflections coming from attack materials that were produced to mimic a face. Additionally, beam profile analysis simultaneously provides depth information by analyzing the same camera frame. Therefore, 3D as well as skin security features may be provided by the exact same technology.

Since also the 2D image of the face can be recorded by simply switching off the laser illumination, a fully secure face recognition pipeline can be established solving the above-stated problem.

The reflection properties of human skin with respect to ethnic origin become more similar when the laser wavelength is shifted towards the NIR regime. At a wavelength of 940 nm differences are at a minimum. Accordingly, different ethnic origins do not play a role for skin authentication.

No time-consuming analysis of a series of frames may be necessary since presentation attack detection (via skin classification) is provided by just one frame. A time frame for performing the complete method may be ≤ 500 ms, preferably ≤ 250 ms. However, embodiments may be feasible in which the skin detection may be performed using a plurality of frames. Depending on confidence for identifying reflection features in the second image, and speed of the method, the method may comprise sampling reflection features over several frames in order to reach a more stable classification.

Figure 3 shows experimental results, in particular density as a function of skin score. On the x-axis the score is shown and on the y-axis the frequency. The score is a measure for the classification quality and has a value range between 0 and 1, wherein 1 indicates very high skin-similarity and 0 very low skin-similarity. A threshold for decision may be around 0.5. A reference distribution of skin-scores for bona fide presentations has been generated using 10 subjects. Skin-scores have also been recorded for presentation attacks (PA) of Level A, Level B and Level C (as defined in the relevant ISO standards). The experimental setup (target of evaluation, ToE) included a proprietary hardware device, e.g. as described in Figure 2, which includes the necessary sensors as well as the computational platform on which the PAD software is executed. The ToE was tested using six species of target of Level A PAIs (presentation attack instruments), five PAI species of Level B, and one PAI species of Level C. For each PAI species, 10 PAIs have been used. The PAI species used in this study are listed in the table below. In the table, APCER is the Attack Presentation Classification Error Rate, which refers to the number of successful attacks/all attacks*100. In the table, BPCER is the Bona Fide Presentation Classification Error Rate, which refers to the number of rejected unlock-attempts/all unlock-attempts*100 Level A and Level B attacks are all based on 2D PAIs, whereas for Level C attacks, based on 3D masks. For Level C attacks, custom rigid masks constructed using a 3D printer were used. A test crew of 10 subjects has been used to obtain the reference distribution of skin-scores for bona fide presentations.

Level	#PAI species	PAI species	#objects	#presentations	Results
A	6	4 printed, 2 digital replay (normal quality)	60	300 (60x5)	APCER=0%
B	5	4 printed, 1 digital replay (optimized for spoof)	50	250 (50x5)	APCER=0%
C	1	3D printed mask	10	50 (10x5)	APCER=0%
Bona Fide	-	-	10	50 (10x5)	BPCER=0%

The experiments with these PAIs show that the two classes of presentations (bona fide or PA) are clearly distinguishable based on the skin-score. Clear distinction between paper, 3D print and skin is possible using the method according to the present invention.

List of reference numbers

110	face detection step
112	camera
114	processing unit
116	skin detection step
118	illumination unit
120	3D detection step
122	authentication step
124	mobile device
126	optical sensor

Claims

1. A system, comprising:

a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising

5 a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area;

a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams;

a second VCSEL comprising a periodic arrangement and configured to output a second plurality of NIR light beams, wherein the second VCSEL array is offset from the first VCSEL

10 array;

a diffractive optical element (DOE);

at least one lens disposed between the DOE and the first and second VCSELs,

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens,

15 wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern,

wherein the CMOS sensor is configured to receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and output a first image

20 corresponding to the first reflective light pattern and a second image corresponding to the second reflective light pattern,

wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams; and

25 a processor configured to:

detect a face in a first region in the first image;

determine a distribution of light intensity of a second region in the second image, wherein the second region corresponds to the first region;

compare the distribution of light intensity of the second region of the second image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the second image and the corresponding region;

determine a depth map of the face in the scene in the second image based on a distortion in the pseudo-random structured light pattern; and

authenticate the face based on the detection of the face, the classification of the object, and the depth map.

2. The system of claim 1, wherein the first VCSEL and the second VCSEL are disposed at a different distance from the at least one lens.

3. The system of claim 1, wherein the processor is configured to detect the face by identifying at least one pre-defined facial characteristic in the region of the first image.

4. The system of claim 1, wherein the distribution of light intensity comprises a shape and a brightness of the light beam.

5. The system of claim 1, wherein the NIR light is in a range of 700 nm to 1100 nm.

6. The system of claim 1, wherein the CMOS sensor comprises a matrix of pixels, each pixel forming a NIR light sensitive area.

7. The system of claim 6, wherein the image comprises a plurality of pixels corresponding to the
5 pixels of the matrix of the CMOS sensor and wherein the distribution of light intensity is determined in a region of the matrix.

8. The system of claim 1, wherein in response to detecting the face, the processor is configured to authenticate the face based on the classification and the depth map.
10

9. The system of claim 1, wherein the depth map is a spatial distribution of a distance between the surfaces and the CMOS sensor.

10. The system of claim 1, wherein the processor is configured to:
15 compare the depth map of the object with a pre-defined depth map; and
classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

11. The system of claim 1, wherein the processor is configured to authenticate the object when
20 the object is classified as three-dimensional.

12. The system of claim 1, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.

13. The system of claim 12, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.

14. The system of claim 1, wherein the system includes a mobile device.

5

15. A system, comprising:

a near infrared (NIR) camera comprising a CMOS sensor and configured to capture an object, wherein the CMOS sensor comprises an NIR light sensitive area;

a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random
10 arrangement and configured to output a plurality of light beams in a near-infrared range of 700 nm to 1100 nm;

a second VCSEL comprising a periodic arrangement and configured to output a plurality of beams of NIR light of 700 nm to 1100 nm, wherein the second VCSEL array is offset from the first VCSEL;

15 a diffractive optical element (DOE);

at least one lens disposed between the DOE and the first and second VCSELs,

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens,

wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in
20 a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern,

wherein the CMOS sensor is configured to receive at least one reflective light pattern and output one or more images corresponding to the at least one reflective light pattern; and

a processor configured to:

25 detect a face of the object in a first region of the one or more images;

determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap;

compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image

5 data derived from a reference image;

determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region;

10 determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

16. The system of claim 15, wherein the first VCSEL and the second VCSEL are disposed at a
15 different distance from the at least one lens.

17. The system of claim 15, wherein the processor is configured to detect the face of the object by identifying at least one pre-defined facial characteristic in the first region of the one or more images.

20

18. The system of claim 15, wherein the distribution of light intensity comprises a shape and a brightness of the light beam.

19. The system of claim 15, wherein the system includes a mobile device.

25

20. The system of claim 15, wherein in response to detecting the face, the processor is configured to authenticate the object based on the classification of the object and the depth map.

5 21. The system of claim 15, wherein the depth map is a spatial distribution of a distance between the object and the CMOS sensor.

22. The system of claim 15, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

10 classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

23. The system of claim 22, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.

15

24. The system of claim 22, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.

25. The system of claim 24, wherein the processor is configured to authenticate the object when
20 the object is classified as skin and three-dimensional.

26. The system of claim 15, further comprising a neural network coupled to the processor, the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained to detect features related to classifications of the
25 objects and output the classification of the object based on the processed images.

27. A system, comprising:

at least one vertical cavity surface emitting laser (VCSEL) configured to:

emit a plurality of light beams in a near-infrared range (NIR), arranged in a first
5 spatial arrangement relative to each other, towards an object;

emit at least one substantially uniform light beam in the near-infrared range
towards the object;

a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area,
wherein the CMOS sensor is configured to receive light reflected from the object and

10 output one or more images characterizing at least an intensity of the reflected light; and

a processor configured to:

detect a face of the object in a first region of the one or more images;

determine a distribution of light intensity of a second region of the one or more
images, wherein the first region and the second region at least partially overlap;

15 compare the distribution of light intensity of the second region of the one or more
images to a distribution of light intensity of a corresponding region in reference image
data derived from a reference image;

determine a classification of the object based on the distribution of light intensity
comparison of the second region of the one or more images and the corresponding
20 region;

determine a depth map of the object in at least one image based on a distortion
in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of
the object, and the depth map.

28. The system of claim 27, wherein the system includes a mobile device.

29. The system of claim 27, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

5 classify the object as three-dimensional in response to determining a deviation of the
depth map from the pre-defined depth map.

30. The system of claim 29, wherein the processor is configured to differentiate skin from non-
living surfaces in the classification of the object and authenticate the object when the object is

10 classified as skin and three-dimensional.

15

Optical Skin Detection for Face Unlock

Abstract

Proposed herein are systems for face authentication. The systems are configured to classify an object based on a distribution of light intensity.



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ELECTRONIC PAYMENT RECEIPT

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Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

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CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	69707303	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	-
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

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1311	PATENT APPL. EXAMINATION FEE	880.00	1	880.00
1817	REQUEST FOR PRIORITIZED EXAMINATION	4515.00	1	4515.00
1830	PROCESSING FEE, EXCEPT IN PROVISIONAL APPLICATIONS	150.00	1	150.00
1202	EACH CLAIM IN EXCESS OF 20	200.00	10	2000.00

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Armstrong Teasdale LLP
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FILING RECEIPT



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Foreign Applications for which priority is claimed (You may be eligible to benefit from the **Patent Prosecution Highway** program at the USPTO. Please see <http://www.uspto.gov> for more information.)
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Title

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Preliminary Class

Statement under 37 CFR 1.55 or 1.78 for AIA (First Inventor to File) Transition Applications: No

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PATENT APPLICATION FEE DETERMINATION RECORD

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Application or Docket Number
19/088,110

APPLICATION AS FILED - PART I

(Column 1)		(Column 2)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
FOR	NUMBER FILED	NUMBER EXTRA	RATE(\$)	FEE(\$)		RATE(\$)	FEE(\$)
BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A			N/A	350
SEARCH FEE (37 CFR 1.16(k), (l), or (m))	N/A	N/A	N/A			N/A	770
EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A			N/A	880
TOTAL CLAIMS (37 CFR 1.16(i))	30 minus 20 =	* 10			OR	x 200 =	2000
INDEPENDENT CLAIMS (37 CFR 1.16(h))	3 minus 3 =	*				x =	0
APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).						0
MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))							0
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL			TOTAL	4000

APPLICATION AS AMENDED - PART II

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT A	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	* Minus **	=		x =		OR	x =	
Independent (37 CFR 1.16(h))	* Minus ***	=		x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

(Column 1)		(Column 2)	(Column 3)	SMALL ENTITY		OR	OTHER THAN SMALL ENTITY	
AMENDMENT B	CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE(\$)	ADDITIONAL FEE(\$)		RATE(\$)	ADDITIONAL FEE(\$)
Total (37 CFR 1.16(i))	* Minus **	=		x =		OR	x =	
Independent (37 CFR 1.16(h))	* Minus ***	=		x =		OR	x =	
Application Size Fee (37 CFR 1.16(s))						OR		
FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						OR		
				TOTAL ADD'L FEE		OR	TOTAL ADD'L FEE	

* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	APP.FILE.REC	04/28/2025	27843-2771 202595US04
19088110	WELCOME.LET	04/28/2025	27843-2771 202595US04

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APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	05/20/2025 09:59:35 AM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	70500750	FILING DATE	03/24/2025
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 3

DOCUMENT		PAGES	DESCRIPTION	SIZE (KB)
2025-05-20 Preliminary Amendment 27843-2771.pdf		11	-	129 KB
2025-05-20 Preliminary Amendment 27843-2771- A.PE.pdf	(1-1)	1	Preliminary Amendment	103 KB
2025-05-20 Preliminary Amendment 27843-2771- CLM.pdf	(2-10)	9	Claims	125 KB
2025-05-20 Preliminary Amendment 27843-2771- REM.pdf	(11-11)	1	Applicant Arguments/Remarks Made in an Amendment	103 KB

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2025-05-20 Preliminary
Amendment 27843-2771.pdf

F9431166A3E17B2D8CC36C377581B3F7C8BC40866F7C9718F
FBF63039FA5F520321162B821B0E6890EA87368607D69D1FA4
D04950F8FA062720E284B73A6CE1D

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A.PE.pdf

1EC36854623DF3A97CA29B869A8246BE215584B2C903666F96
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DB15E4D8BB605D97646494E0DAF765

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REM.pdf

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378FAA17792F5BF0D0CA694BF35C64CF5214CA94C027A86F5
54B9F70EC15720D8C1F3A77643BCB7C

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IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Named Inventor: Peter Schillen

Serial No.: 19/088,110

Filed: March 24, 2025

Confirmation No.: 4090

For: OPTICAL SKIN DETECTION FOR FACE UNLOCK

May 20, 2025

PRELIMINARY AMENDMENT

TO THE COMMISSIONER FOR PATENTS:

Please enter the following amendments prior to calculation of fees due in the present application.

Amendments to the Claims begin on page 2.

Remarks are on page 11.

AMENDMENTS TO THE CLAIMS

1. (Currently Amended) A system, comprising:

a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength ~~wherein the CMOS sensor comprises an NIR light sensitive area;~~

a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement ~~comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams;~~

a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement ~~comprising a periodic arrangement and configured to output a second plurality of NIR light beams~~, wherein the second VCSEL ~~array~~ is offset from the first VCSEL ~~array~~;

a diffractive optical element (DOE);
~~at least one lens disposed between the DOE and the first and second VCSELs;~~

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through ~~the~~ at least one lens toward the DOE,

wherein the DOE is configured to diffract the first plurality of NIR light beams, ~~resulting~~ in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern ~~and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern,~~

wherein the CMOS sensor is configured to ~~receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and~~ output ~~a first~~ an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern ~~and a second image corresponding to the second reflective light pattern, and~~

wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the range ~~wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams;~~ and

a processor configured to:

detect a face in a first region in the **first** image;

determine a distribution of light intensity of a second region in the **second** image, wherein the second region corresponds to the first region;

compare the distribution of light intensity of the second region of the **second** image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the **second** image and the corresponding region;

determine a depth map of the face in the scene in the **second** image based on a distortion in the pseudo-random structured light pattern; and

authenticate the face based on the detection of the face, the classification of the object, and the depth map.

2. (Cancelled)

3. (Currently Amended) The system of claim 1, wherein the processor is configured to detect the face by identifying at least one pre-defined facial characteristic in the region of the ~~first~~ image.
4. (Original) The system of claim 1, wherein the distribution of light intensity comprises a shape and a brightness of the light beam.
5. (Cancelled)
6. (Cancelled)
7. (Cancelled)
8. (Original) The system of claim 1, wherein in response to detecting the face, the processor is configured to authenticate the face based on the classification and the depth map.
9. (Original) The system of claim 1, wherein the depth map is a spatial distribution of a distance between the surfaces and the CMOS sensor.
10. (Original) The system of claim 1, wherein the processor is configured to:
 - compare the depth map of the object with a pre-defined depth map; and
 - classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.
11. (Original) The system of claim 1, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.
12. (Original) The system of claim 1, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.
13. (Original) The system of claim 12, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.
14. (Original) The system of claim 1, wherein the system includes a mobile device.

15. (Currently Amended) A system, comprising:

a near infrared (NIR) camera ~~comprising a CMOS sensor and~~ configured to capture an object, the NIR camera comprising a CMOS sensor, wherein the ~~CMOS sensor comprises an NIR light sensitive area~~ NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength;

a first vertical cavity surface emitting laser (VCSEL) ~~comprising a pseudo-random arrangement and~~ configured to output a first plurality of light beams ~~in a near-infrared range of 700 nm to 1100 nm toward the object~~, the first plurality of NIR light beams comprising a pseudo-random arrangement;

a second VCSEL ~~comprising a periodic arrangement and~~ configured to output a second plurality of NIR light beams ~~of NIR light of 700 nm to 1100 nm toward the scene~~, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL array is offset from the first VCSEL;

~~a diffractive optical element (DOE)~~ an optical element comprising at least one of a diffractive optical element (DOE) and a lens;

~~at least one lens disposed between the DOE and the first and second VCSELs~~;

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the ~~at least one lens~~ optical element toward the scene,

wherein the ~~DOE optical element~~ is configured to ~~diffract~~ modify the first plurality of NIR light beams, ~~resulting~~ in a pseudo-random structured light pattern and ~~configured to diffract~~ the second plurality of NIR light beams ~~resulting~~ in a substantially uniform light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern and the periodic light pattern into a second reflective light pattern,

wherein the CMOS sensor is configured to ~~receive at least one reflective light pattern and~~ output one or more images corresponding to the ~~at least one~~ first reflective

light pattern and the second reflective light pattern in response to receiving the first reflective light pattern and the second reflective light pattern,[[;]] and

wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the NIR range; and

a processor configured to:

detect a face of the object in a first region of the one or more images;

determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap;

compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region;

determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

16. (Cancelled)

17. (Original) The system of claim 15, wherein the processor is configured to detect the face of the object by identifying at least one pre-defined facial characteristic in the first region of the one or more images.

18. (Original) The system of claim 15, wherein the distribution of light intensity comprises a shape and a brightness of the light beam.

19. (Original) The system of claim 15, wherein the system includes a mobile device.
20. (Original) The system of claim 15, wherein in response to detecting the face, the processor is configured to authenticate the object based on the classification of the object and the depth map.
21. (Original) The system of claim 15, wherein the depth map is a spatial distribution of a distance between the object and the CMOS sensor.
22. (Original) The system of claim 15, wherein the processor is configured to:
- compare the depth map of the object with a pre-defined depth map; and
 - classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.
23. (Original) The system of claim 22, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.
24. (Original) The system of claim 22, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.
25. (Original) The system of claim 24, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.
26. (Original) The system of claim 15, further comprising a neural network coupled to the processor, the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained to detect features related to classifications of the objects and output the classification of the object based on the processed images.
27. (Currently Amended) A system, comprising:
- a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;**

~~at least one~~ a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement;[::]

a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement;

~~emit a plurality of light beams in a near-infrared range (NIR), arranged in a first spatial arrangement relative to each other, towards an object;~~

~~emit at least one substantially uniform light beam in the near-infrared range towards the object;~~

~~a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area;~~

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the scene,

wherein the at least one lens is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern,

~~wherein the CMOS sensor is configured to receive light reflected from the object and output one or more images characterizing at least an intensity of the reflected light~~ an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern; and

wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in an NIR light range of 700 nm to 1100 nm in wavelength; and

a processor configured to:

detect a face of ~~the~~ an object in the scene in a ~~first~~ region of the ~~one or more images~~ image;

determine a distribution of light intensity of ~~a second~~ the region of the ~~one or more images, wherein the first region and the second region at least partially overlap~~ image;

compare the distribution of light intensity of the ~~second~~ region of the ~~one or more images~~ image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the object in the scene based on the distribution of light intensity comparison of the ~~second~~ region of the ~~one or more images~~ image and the corresponding region;

determine a depth map of the object in ~~at least one~~ the image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

28. (Original) The system of claim 27, wherein the system includes a mobile device.

29. (Original) The system of claim 27, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

30. (Original) The system of claim 29, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object and authenticate the object when the object is classified as skin and three-dimensional.

31. (New) The system of claim 27, further comprising:

a diffractive optical element (DOE), wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens toward the DOE.

32. (New) The system of claim 27, wherein the light projecting device is a light-emitting diode (LED).

33. (New) The system of claim 27, wherein the light projecting device is a VCSEL.

34. (New) The system of claim 27, wherein the system includes a computing device.

35. (New) The system of claim 15, wherein the system includes a computing device.

REMARKS

Claims 2, 5-7, and 16 are cancelled. Claims 1, 3, 15, and 27 are amended. Claims 31-35 are new. No new matter has been added in this amendment.

The Commissioner is hereby authorized to charge any fees under 37 CFR 1.16 and 1.17 which may be required during the entire pendency of this application to Account No. 012384.

Respectfully submitted,

/Christopher M. Goff/

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090
76656	7590	05/27/2025	EXAMINER	
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	Examiner JONYA SMALLS	Art Unit OPET	AIA (FITF) Status Yes
1. THE REQUEST FILED <u>24 March 2025</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☒ for an original nonprovisional application (Track I). B. ☐ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to JONYA SMALLS at (571)272-1619. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
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Application	Document	Mailroom Date	Attorney Docket No.
19088110	TRACK1.GRANT	05/27/2025	27843-2771 202595US04

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090

76656 7590 06/03/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

TRAN, PHUOC

ART UNIT	PAPER NUMBER
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2668

NOTIFICATION DATE	DELIVERY MODE
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06/03/2025

ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com

Office Action Summary

Application No.

19/088,110

Applicant(s)

SCHILLEN et al.

Examiner

PHUOC TRAN

Art Unit

2668

AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1) ☒ Responsive to communication(s) filed on 20 May 2025.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on ____.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on ____; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☒ Claim(s) 1,3-4,8-15 and 17-35 is/are pending in the application.

5a) Of the above claim(s) ____ is/are withdrawn from consideration.

6) ☐ Claim(s) ____ is/are allowed.

7) ☒ Claim(s) 1,3-4,8-15 and 17-35 is/are rejected.

8) ☐ Claim(s) ____ is/are objected to.

9) ☐ Claim(s) ____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.

Application Papers

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 24 March 2025 is/are: a) ☐ accepted or b) ☒ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12) ☒ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a) ☒ All b) ☐ Some** c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☒ Certified copies of the priority documents have been received in Application No. 18/264,326.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date ____.

2) ☒ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: ____.

Paper No(s)/Mail Date ____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Drawings

2. The drawings are objected to under 37 CFR 1.83(a). The drawings must show every feature of the invention specified in the claims. Therefore, the following claimed subject matter must be shown or the feature(s) canceled from the claim(s):

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 1;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL is offset from the first VCSEL” recited in claim 1;

“a diffractive optical element (DOE)” recited in claim 1;

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 15;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL array is offset from the first VCSEL” recited in claim 15;

“an optical element comprising at least one of a diffractive optical element (DOE) and a lens” recited in claim 15;

“a neural network coupled to the processor, the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained

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to detect features related to classifications of the objects and output the classification of the object based on the processed images” recited in claim 26.

“a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 27;

“a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 27;

“at least one lens is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern” recited in claim 27;

“a diffractive optical element (DOE), wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens toward the DOE” recited in claim 31.

No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as “amended.” If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either “Replacement Sheet” or “New Sheet” pursuant to 37 CFR 1.121(d). If the changes are not accepted by

the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 112

3. The following is a quotation of 35 U.S.C. 112(b):
(b) CONCLUSION.—The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the inventor or a joint inventor regards as the invention.

The following is a quotation of 35 U.S.C. 112 (pre-AIA), second paragraph:
The specification shall conclude with one or more claims particularly pointing out and distinctly claiming the subject matter which the applicant regards as his invention.

4. Claims 4, 15, 17-26, 35 are rejected under 35 U.S.C. 112(b) or 35 U.S.C. 112 (pre-AIA), second paragraph, as being indefinite for failing to particularly point out and distinctly claim the subject matter which the inventor or a joint inventor (or for applications subject to pre-AIA 35 U.S.C. 112, the applicant), regards as the invention.

Claim 4 recites the limitation "the light beam" in line 2. There is insufficient antecedent basis for this limitation in the claim.

Claim 15 recites the limitation "the second VCEL array" in line 15. There is insufficient antecedent basis for this limitation in the claim.

Claim 18 recites the limitation "the light beam" in line 2. There is insufficient antecedent basis for this limitation in the claim.

Claims 17-26, 35 are rejected as being dependent upon rejected claims.

5. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

6. Claims 1, 3-4, 8-15, 17-35 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

The original disclosure fails to provide support for the following claimed subject matter:

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 1;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL is offset from the first VCSEL” recited in claim 1;

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 15;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL array is offset from the first VCSEL” recited in claim 15;

“a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 27;

“a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 27.

The specification discloses the term “vertical cavity surface emitting lasers” twice, at page 14, line 27 and page 49, line 11. However, it fails to provide support for the above claimed subject matter.

Double Patenting

7. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

8. Claim 1, 3-4, 8-15, 17-35 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claim 21-50 of copending Application No. 18/468,352. Although the claims at issue are not identical, they are not patentably distinct from each other because the present claims are either anticipated by or obvious variants of the copending application claims. The following table shows the corresponding limitations between representative claims and representative copending application claims.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Present Application Claims	Copending Application Claims
1. (Currently Amended) A system, comprising: a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, <u>wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength wherein the CMOS sensor comprises an NIR light sensitive area</u> ; a first vertical cavity surface emitting laser (VCSEL) <u>configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams</u> ; a second VCSEL configured to output a second	21. A system, comprising: a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area; a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams; a second VCSEL comprising a periodic arrangement and configured to output a second plurality of NIR light beams, wherein the second VCSEL array is offset from the first VCSEL array; a diffractive optical element (DOE); at least one lens disposed between the DOE and the first and second VCSELs, wherein

<u>plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement comprising a periodic arrangement and configured to output a second plurality of NIR light beams, wherein the second VCSEL array is offset from the first VCSEL array; a diffractive optical element (DOE); at least one lens disposed between the DOE and the first and second VCSELs, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens toward the DOE, wherein the DOE is configured to diffract the first plurality of NIR light beams resulting in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein the CMOS sensor is configured to receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and output a first an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern and a second image corresponding to the second reflective light pattern, and wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the range wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams; and a processor configured to: detect a face in a first region in the first image; determine a distribution of light intensity of a second region in the second image, wherein the second region corresponds to the first region; compare the distribution of light intensity of the second region of the second image to a distribution of light intensity of a</u>	the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens, wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein the CMOS sensor is configured to receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and output a first image corresponding to the first reflective light pattern and a second image corresponding to the second reflective light pattern, wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams; and a processor configured to: detect a face in a first region in the first image; determine a distribution of light intensity of a second region in the second image, wherein the second region corresponds to the first region; compare the distribution of light intensity of the second region of the second image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the second image and the corresponding region; determine a depth map of the face in the scene in the second image based on a distortion in the pseudo-random structured light pattern; and authenticate the face based on the detection of the face, the classification of the object, and the depth map.
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corresponding region in reference image data derived from a reference image; determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the ~~second~~ image and the corresponding region; determine a depth map of the face in the scene in the ~~second~~ image based on a distortion in the pseudo-random structured light pattern; and authenticate the face based on the detection of the face, the classification of the object, and the depth map.

15. (Currently Amended) A system, comprising: a near infrared (NIR) camera ~~comprising a CMOS sensor and~~ configured to capture an object, the NIR camera comprising a CMOS sensor, wherein the ~~CMOS sensor comprises an NIR light sensitive area~~ NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength; a first vertical cavity surface emitting laser (VCSEL) ~~comprising a pseudo-random arrangement and~~ configured to output a first plurality of light beams in a near-infrared range of 700 nm to 1100 nm toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement; a second VCSEL ~~comprising a periodic arrangement and~~ configured to output a second plurality of NIR light beams of NIR light of 700 nm to 1100 nm toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL array is offset from the first VCSEL; ~~a diffractive optical element (DOE)~~ an optical element comprising at least one of a diffractive optical element (DOE) and a lens; ~~at least one lens disposed between the DOE and the first and second VCSELs~~, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens optical element toward the scene, wherein the DOE optical element is configured to diffract

35. A system, comprising: a near infrared (NIR) camera comprising a CMOS sensor and configured to capture an object, wherein the CMOS sensor comprises an NIR light sensitive area; a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a plurality of light beams in a near-infrared range of 700 nm to 1100 nm; a second VCSEL comprising a periodic arrangement and configured to output a plurality of beams of NIR light of 700 nm to 1100 nm, wherein the second VCSEL array is offset from the first VCSEL; a diffractive optical element (DOE); at least one lens disposed between the DOE and the first and second VCSELs, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens, wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein the CMOS sensor is configured to receive at least one reflective light pattern and output one or more images corresponding to the at least one reflective light pattern; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light

<u>modify</u> the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern and the periodic light pattern into a second reflective light pattern, wherein the CMOS sensor is configured to receive at least one reflective light pattern and output one or more images corresponding to the at least one first reflective light pattern and the second reflective light pattern in response to receiving the first reflective light pattern and the second reflective light pattern,[[;]] and wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the NIR range; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.	intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.
27. (Currently Amended) A system, comprising: <u>a near infrared (NIR) camera</u>	47. A system, comprising: at least one vertical cavity surface emitting laser (VCSEL)

<u>configured to capture a scene, the NIR camera comprising a CMOS sensor; at least one a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement;[::]a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising aperiodic arrangement; emit a plurality of light beams in a near-infrared range (NIR), arranged in a first spatial arrangement relative to each other, towards an object emit at least one substantially uniform light beam in the near-infrared range towards the object; a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the scene, wherein the at least one lens is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern, wherein the CMOS sensor is configured to receive light reflected from the object and output one or more images characterizing at least an intensity of the reflected light an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern; and wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in an NIR light range of 700 nm to 1100 nm in wavelength; and a processor configured to: detect a face of the an object in the scene in a first region of the one or more images image; determine a distribution of light intensity of a second the region of the one or more images, wherein the first region and the second region at least partially overlap image; compare the distribution of light intensity of</u>	configured to: emit a plurality of light beams in a near-infrared range (NIR), arranged in a first spatial arrangement relative to each other, towards an object; emit at least one substantially uniform light beam in the near-infrared range towards the object; a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area, wherein the CMOS sensor is configured to receive light reflected from the object and output one or more images characterizing at least an intensity of the reflected light; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.
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the second region of the one or more images <u>image</u> to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object <u>in the scene</u> based on the distribution of light intensity comparison of the second region of the one or more images <u>image</u> and the corresponding region; determine a depth map of the object in at least one the image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map. Claims 3-4, 8-14, 17-26, 28-35 recite features similar to features recited in claims 22-34, 36-46, 48-50. Therefore, they are rejected for reasons similar to those discussed above.	
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Conclusion

9. The prior art made of record and not relied upon is considered pertinent to applicant's disclosure. See PTO-892 Form.

10. Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHUOC TRAN whose telephone number is (571)272-7399. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vu Le can be reached at 571-272-7332. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/PHUOC TRAN/

Primary Examiner, Art Unit 2668

Notice of References Cited	Application/Control No. 19/088,110		Applicant(s)/Patent Under Reexamination SCHILLEN et al.	
	Examiner PHUOC TRAN		Art Unit 2668	Page 1 of 1

U.S. PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Name	CPC Classification	US Classification
*	A	US-20190042718-A1	2019-02-07	Prakash; Deepti S.	H04L9/3247	1/1
*	B	US-20200364489-A1	2020-11-19	YE; Jianhui	G06V40/166	1/1
*	C	US-20200084439-A1	2020-03-12	TANG; Cheng	G01S17/86	1/1
	D					
	E					
	F					
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	I					
	J					
	K					
	L					
	M					

FOREIGN PATENT DOCUMENTS

*		Document Number Country Code-Number-Kind Code	Date YYYY-MM-DD	Country	Name	CPC Classification
	N					
	O					
	P					
	Q					
	R					
	S					
	T					

NON-PATENT DOCUMENTS

*		Include as applicable: Author, Title, Date, Publisher, Edition or Volume, Pertinent Pages
	U	
	V	
	W	
	X	

*A copy of this reference is not being furnished with this Office action. (See MPEP § 707.05(a).)
Dates in YYYY-MM-DD format are publication dates. Classifications may be US or foreign.

<i>Search Notes</i> 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC - Searched*		
Symbol	Date	Examiner
G06V40/40, G01S17/894, G06F21/32, G06T7/50, G06V10/141, G06V10/145, G06V10/25, G06V10/54, G06V10/60, G06V10/764, G06V10/82, G06V40/165, G06V40/168, G06V40/172, G06T2207/30201	05/29/2025	PT

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search	05/29/2025	PT
S.N. 18/264,326 refs. checked	05/29/2025	PT
S.N. 18/468,352 refs. checked	05/29/2025	PT
See search history printout(s)	05/29/2025	PT

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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Bibliographic Data

Application No: 19/088,110

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/PHUOC TRAN/

Examiner's Signature

Initials

Title:

OPTICAL SKIN DETECTION FOR FACE UNLOCK

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
03/24/2025	382	2668	27843-2771 202595US04
RULE			

APPLICANTS

trinamiX GmbH, Ludwigshafen Am Rhein, GERMANY

INVENTORS

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY

Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY

Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY

Manuel GUENTHER, Zuerich, SWITZERLAND

Lars DIESSELBERG, Karlsruhe, GERMANY

Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY

CONTINUING DATA

This application is a CON of 18468352 09/15/2023

18468352 is a CON of 18264326 08/04/2023

18264326 is a 371 of PCT/EP22/53941 02/17/2022

FOREIGN APPLICATIONS

EPO EP21157800.0 02/18/2021

IF REQUIRED, FOREIGN LICENSE GRANTED**

04/25/2025

STATE OR COUNTRY

GERMANY

ADDRESS

Patent Docket Department

Armstrong Teasdale LLP

7700 Forsyth Boulevard

Suite 1800

St. Louis, MO 63105

UNITED STATES

FILING FEE RECEIVED

\$8,665

Doc code: IDS

PTO/SB/08a (01-25)

Doc description: Information Disclosure Statement (IDS) Filed

Approved for use through 11/30/2027. OMB 0651-0031

U.S. Patent and Trademark Office; U.S. DEPARTMENT OF COMMERCE

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INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number		
	Filing Date		2025-03-24
	First Named Inventor	Peter SCHILLEN	
	Art Unit		
	Examiner Name		
	Attorney Docket Number		27843-2771 202595US04

U.S. PATENTS Remove						
Examiner Initial*	Cite No	Patent Number	Kind Code ¹	Issue Date	Name of Inventor of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1					
If you wish to add additional U.S. Patent citation information, please click the Add button. Add						
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Examiner Initial*	Cite No	Publication Number	Kind Code ¹	Publication Date	Name of Inventor or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear
	1	20150374277	A1	2015-12-31	PATWARDHAN	
	2	20160242853	A1	2016-08-25	BANGERAL et al.	
	3	20190213309	A1	2019-07-11	MORESTIN et al.	
	4	20200082155	A1	2020-03-12	KALSCHEUR et al.	
	5	20200211180	A1	2020-07-02	GILLIES et al.	

INFORMATION DISCLOSURE STATEMENT BY APPLICANT	Application Number			
	Filing Date		2025-03-24	
	First Named Inventor	Peter SCHILLEN		
	Art Unit			
	Examiner Name			
	Attorney Docket Number		27843-2771 202595US04	

	6	20210049391	A1	2021-02-18	ZOU et al.	
	7	20220222464	A1	2022-07-14	DIEDRICH et al.	
	8	20240119130	A1	2024-04-11	HU et al.	
	9	20160335483	A1	2016-11-17	PFURISCH et al.	

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Examiner Initial*	Cite No	Foreign Document Number ³	Country Code ²	Kind Code ⁴	Publication Date	Name of Patentee or Applicant of cited Document	Pages, Columns, Lines where Relevant Passages or Relevant Figures Appear	T ⁵
	1	111164610	CN	A	2020-05-15	SHENZHEN GOODIX TECH CO LTD.	English-language translation attached	☒

If you wish to add additional Foreign Patent Document citation information, please click the Add button [Add](#)

NON-PATENT LITERATURE DOCUMENTS

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Examiner Initials*	Cite No	Include name of the author (in CAPITAL LETTERS), title of the article (when appropriate), title of the item (book, magazine, journal, serial, symposium, catalog, etc), date, pages(s), volume-issue number(s), publisher, city and/or country where published.	T ⁵
	1	International Search Report and Written Opinion for PCT/EP2022/053941 dated 2022-06-09, 12 pages	☐

If you wish to add additional non-patent literature document citation information, please click the Add button [Add](#)

EXAMINER SIGNATURE

Examiner Signature	/PHUOC TRAN/	Date Considered	05/29/2025
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*EXAMINER: Initial if reference considered and whether or not citation is in conformance with MPEP 609. Draw line through a citation if not in conformance and not considered. Include copy of this form with next communication to applicant.

INFORMATION DISCLOSURE
STATEMENT BY APPLICANT

Application Number		
Filing Date		2025-03-24
First Named Inventor	Peter SCHILLEN	
Art Unit		
Examiner Name		
Attorney Docket Number		27843-2771 202595US04

¹ See Kind Codes of USPTO Patent Documents at www.USPTO.GOV or MPEP 901.04. ² Enter office that issued the document, by the two-letter code (WIPO Standard ST.3). ³ For Japanese patent documents, the indication of the year of the reign of the Emperor must precede the serial number of the patent document. ⁴ Kind of document by the appropriate symbols as indicated on the document under WIPO Standard ST.16 if possible. ⁵ Applicant is to place a check mark here if English language translation is attached.

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	
Filing Date	2025-03-24
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2771 202595US04

TIMING STATEMENT

Please see 37 CFR 1.97(e) to make the appropriate selection(s):

☐ That each item of information contained in the information disclosure statement was first cited in any communication from a foreign patent office in a counterpart foreign application not more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(1).

OR

☐ That no item of information contained in the information disclosure statement was cited in a communication from a foreign patent office in a counterpart foreign application, and, to the knowledge of the person signing the certification after making reasonable inquiry, no item of information contained in the information disclosure statement was known to any individual designated in 37 CFR 1.56(c) more than three months prior to the filing of the information disclosure statement. See 37 CFR 1.97(e)(2).

☐ See attached timing statement under 37 CFR 1.97(e).

☒ A timing statement under 37 CFR 1.97(e) is not submitted herewith.

TIMING FEE

☐ The fee set forth in 37 CFR 1.17(p) has been submitted herewith.

COPIES

☒ An identification of an earlier application pursuant to 37 CFR 1.98(d)(1) is attached.

SIZE FEE ASSERTION

Please see 37 CFR 1.17(v) and the [IDS Auto-Load Instructions](#) for completing this form to make the appropriate selection of an assertion under 37 CFR 1.98. For the information disclosure statement (IDS) submitted herewith, the applicant or patent owner certifies the following with respect to the cumulative number of applicant-provided or patent-owner provided items of information submitted to date including those in the accompanying IDS (select only one):

☒ No IDS size fee is required under 37 CFR 1.17(v) at this time.

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(1).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(2).

☐ The IDS is accompanied by the IDS size fee under 37 CFR 1.17(v)(3).

SIGNATURE

A signature of the applicant or representative is required in accordance with 37 CFR 1.33 and 11.18. Please see 37 CFR 1.4(d) for the form of the signature.

Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2025-03-24
Name/Print	Christopher M. Goff	Registration Number	41785

**INFORMATION DISCLOSURE
STATEMENT BY APPLICANT**

Application Number	
Filing Date	2025-03-24
First Named Inventor	Peter SCHILLEN
Art Unit	
Examiner Name	
Attorney Docket Number	27843-2771 202595US04

This collection of information is required by 37 CFR 1.97 and 1.98. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 1 hour to complete, including gathering, preparing and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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The information provided by you in this form will be subject to the following routine uses:

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2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	282	((("SCHILLEN") near3 ("Peter")) OR ("GUTHIER") near3 ("Benjamin")) OR ("SCHICK") near3 ("Friedrich")) OR ("GUENTHER") near3 ("Manuel")) OR ("DIESELBERG") near3 ("Lars")) OR ("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/09/07 03:49 PM
L2	537832	(G06V40/40 or G01S17/894 or G06F21/32 or G06T7/50 or G06V10/141 or G06V10/145 or G06V10/25 or G06V10/54 or G06V10/60 or G06V10/764 or G06V10/82 or G06V40/165 or G06V40/168 or G06V40/172 or G06T2207/30201).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/09/07 03:50 PM
L3	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authentikat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:03 PM
L4	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authentikat\$4 OR identif\$8 OR	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:04 PM

L5	2	verifi\$8)).clm. 2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:05 PM
L6	0	2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:08 PM
L7	0	(image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:09 PM
L10	329	(image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:14 PM

L11	120	OR skin)) 2 AND (image SAME (face OR facial OR skin) SAME (three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial OR skin))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:15 PM
L12	14	11 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:18 PM
L13	24	10 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:19 PM
L14	10	13 NOT 12	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:26 PM
L15	106	11 NOT 13	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:29 PM
L16	2	("20190213309" OR "2020082155").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2024/09/07 05:27 PM
L17	305	((("SCHILLEN") near3 ("Peter")) OR (("GUTHIER") near3 ("Benjamin")) OR (("SCHICK") near3 ("Friedrich")) OR (("GUENTHER") near3 ("Manuel")) OR (("DIESELBERG") near3 ("Lars")) OR (("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/05/29 11:50 AM
L18	3	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 11:55 AM

L19	8	OR light) SAME (structur\$4 OR pattern) SAME (classif\$8 OR authentikat\$4 OR identif\$8 OR verifi\$8)).clm. ("20150374277" "20160242853" "20190213309" "20200082155" "20200211180" "20210049391" "20220222464" "20240119130").pn. OR ("12288421").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:00 PM
L20	3	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern)) .clm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:02 PM
L21	11	17 AND (infrar\$4 SAME laser) AND ((face OR facial) SAME (depth OR three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:06 PM
L22	11	17 AND (infrar\$4 SAME (camera OR sensor)) AND ((vertical SAME laser) OR VCSEL) AND ((face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:12 PM
L23	11	21 AND 22	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:16 PM
L24	5	17 AND (infrar\$4 SAME	(US-PGPUB; USPAT;	ADJ	ON	ON	2025/05/29

		(camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	USOCR)				12:20 PM
L25	0	20 AND (infrar\$4 SAME (camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:26 PM
L26	3	20 AND ((vertical SAME laser) OR VCSEL) AND (infrar\$4 OR NIF) AND (pseudo\$11 OR random\$4 OR periodic)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:27 PM
L27	0	("20240037201" OR "20240005703" OR "12288421").pn. AND (((vertical SAME laser) OR VCSEL) SAME ((infrar\$4 OR NIF OR pseudo\$11 OR random\$4 OR periodic)))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:39 PM
L28	0	("20240037201" OR "20240005703" OR "12288421").pn. AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL)) AND ((infrar\$4 OR NIF) SAME laser) AND ((infrar\$4 OR NIF OR laser) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L29	3	("20240037201" OR "20240005703" OR "12288421").pn.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L30	0	29 AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:44 PM
L31	3	29 AND ((vertical SAME laser) OR VCSEL)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:45 PM
L33	3	29 AND ((infrar\$4 OR NIR) SAME laser) AND ((infrar\$4 OR NIR OR laser) SAME (pseudo\$11 OR	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:56 PM

L34	9	random\$4 OR periodic)) ("20150374277" OR "20160242853" OR "20160335483" OR "20190213309" OR "20200082155" OR "20200211180" OR "20210049391" OR "20220222464" OR "20240119130").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/05/29 12:59 PM
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
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UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	CTNF	06/03/2025	27843-2771 202595US04
19088110	892	06/03/2025	27843-2771 202595US04
19088110	1449	06/03/2025	27843-2771 202595US04

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 19/088,110	Filing Date 03/24/2025	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$200 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$ 600 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

		(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	05/20/2025	CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA		
	Total (37 CFR 1.16(i))	* 30	Minus	** 30	= 0	x \$ 200 =	0
	Independent (37 CFR 1.16(h))	* 3	Minus	*** 3	= 0	x \$ 600 =	0
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	0
AMENDMENT		CLAIMS REMAINING AFTER AMENDMENT		HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	RATE (\$)	ADDITIONAL FEE (\$)
	Total (37 CFR 1.16(i))	*	Minus	**	=	x \$ 0 =	
	Independent (37 CFR 1.16(h))	*	Minus	***	=	x \$ 0 =	
	☐ Application Size Fee (37 CFR 1.16(s))						
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))						
						TOTAL ADD'L FEE	
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE	
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/RENEE D HAWKINS/	
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".							
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.							

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090
76656	7590	07/22/2025	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	PAPER NUMBER
7700 Forsyth Boulevard			2668	
Suite 1800			NOTIFICATION DATE	DELIVERY MODE
St. Louis, MO 63105			07/22/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com

<i>Applicant-Initiated Interview Summary</i>	Application No. 19/088,110	Applicant(s) SCHILLEN et al.		
	Examiner PHUOC TRAN	Art Unit 2668	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
PHUOC TRAN	Primary Examiner	Telephonic
ZACHARY WIERSMA	Attorney of Record	

Date of Interview: 17 July 2025

Issues Discussed:

35 U.S.C. 112

Whether the claim features, such as “a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 1 and “a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, wherein the second VCSEL is offset from the first VCSEL” recited in claim 1, have adequate written support in the specification were discussed. No agreement was reached.

Drawings

Whether the drawings must show every feature of the invention specified in the claims was discussed. No agreement was reached.

☒ Attachment

/PHUOC TRAN/ Primary Examiner, Art Unit 2668	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

IN THE UNITED STATES PATENT AND TRADEMARK OFFICE

First Inventor:	Peter Schillen	Art Unit:	2688
Serial No.:	19/088,110	Confirmation No.:	4090
Date Filed:	March 24, 2025	Examiner:	Phuoc Tran
Title:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

To: Phuoc Tran

From: Zach Wiersma, Registration No. 77,808

EXAMINER INTERVIEW AGENDA

The following interview agenda is provided for the interview scheduled for 1:00 P.M. EST on July 17, 2025.

- The drawing objections. In particular, that before the content of the drawing is addressed, it must first be assessed whether drawings are required. To this end, as set forth in 37 CFR 1.81(a), “The applicant for a patent is required to furnish a drawing of the invention where necessary for the understanding of the subject matter sought to be patented” (emphasis added). Applicant respectfully submits that drawings are not required for understanding, and therefore the drawing objections should be withdrawn on this basis.
- The written description requirement. In particular, that the present claims comply with MPEP 2163.02 and MPEP 2163(I)(B) regarding express, implicit, or inherent disclosures (see the below table).

Claim element	Exemplary support in the application as-filed
vertical cavity surface emitting lasers	Page 14, Lines 9-20 and Page 48, Lines 15-26

Plurality of illumination units (i.e., lasers)	Page 14, Lines 31-34
pseudo-random and periodic point patterns	Page 14, Lines 20-32
NIR light beams	Page 48, Lines 7-13 and Page 57, Lines 33-38
offset or angled light beams relative to the axis	Page 46, Lines 10-18

- Further, the present claims comply with MPEP 2163.07(b) regarding incorporation by reference (see Page 18, Lines 15-20 and Page 29, Lines 43-45 of the application as-filed).
- The clarity and patentability of the present claims. In particular, that the present claims are directed towards a system for authenticating a face. The system analyzes a beam profile of at least two different areas (see Page 55, Lines 13-17 of the application as-filed). This unique approach allows face authentication to reliably detect presentation attacks with 3D face masks (see Page 3, Lines 7-21 of the application as-filed). The present claims are the first report of such a system.
- Related U.S. Application No. 18/468,352 (Attorney docket number 27843-2437). In particular, that the related application is under similar rejections and the above aspects are applicable there as well.

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 76656

Tue Jul 22 05:02:54 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MISSOURI, 63105
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 76656, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	EXIN	07/22/2025	27843-2771 202595US04
19088110	OA.APPENDIX	07/22/2025	27843-2771 202595US04

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Sincerely,

The Patent Center Team

In the United States Patent & Trademark Office

First Inventor:	Peter Schillen	Art Unit:	2688
Serial No.:	19/088,110	Confirmation No.:	4090
Date Filed:	March 24, 2025	Examiner:	Phuoc Tran
Title:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

July 28, 2025

RESPONSE TO NON-FINAL OFFICE ACTION

Commissioner for Patents
Alexandria, VA 22313

Commissioner:

In response to the Non-Final Office Action dated June 3, 2025, please consider the following remarks.

Amendments to the Claims begin on page 2.

Remarks begin on page 10.

Conclusion is on page 15.

AMENDMENTS TO THE CLAIMS

1. (Currently Amended) A system, comprising:

a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength;

a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement;

a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, ~~wherein the second VCSEL is offset from the first VCSEL;~~

a diffractive optical element (DOE);

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the DOE,

wherein the DOE is configured to diffract the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern,

wherein the CMOS sensor is configured to output an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern, and

wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the range; and

a processor configured to:

detect a face in a first region in the image;

determine a distribution of light intensity of a second region in the image, wherein the second region corresponds to the first region;

compare the distribution of light intensity of the second region of the image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the image and the corresponding region;

determine a depth map of the face in the scene in the image based on a distortion in the pseudo-random structured light pattern; and

authenticate the face based on the detection of the face, the classification of the object, and the depth map.

2. (Cancelled)

3. (Previously Presented) The system of claim 1, wherein the processor is configured to detect the face by identifying at least one pre-defined facial characteristic in the region of the image.

4. (Currently Amended) The system of claim 1, wherein the distribution of light intensity **comparison** comprises a shape and a brightness ~~of the light beam~~.

5. (Cancelled)

6. (Cancelled)

7. (Cancelled)

8. (Original) The system of claim 1, wherein in response to detecting the face, the processor is configured to authenticate the face based on the classification and the depth map.

9. (Original) The system of claim 1, wherein the depth map is a spatial distribution of a distance between the surfaces and the CMOS sensor.

10. (Original) The system of claim 1, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

11. (Original) The system of claim 1, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.

12. (Original) The system of claim 1, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.

13. (Original) The system of claim 12, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.

14. (Original) The system of claim 1, wherein the system includes a mobile device.

15. (Currently Amended) A system, comprising:

a near infrared (NIR) camera configured to capture an object, the NIR camera comprising a CMOS sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength;

a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement;

a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement, ~~wherein the second VCSEL array is offset from the first VCSEL;~~

an optical element comprising at least one of a diffractive optical element (DOE) and a lens;

wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the optical element toward the scene,

wherein the optical element is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern and the second plurality of NIR light beams in a substantially uniform light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern and the periodic light pattern into a second reflective light pattern,

wherein the CMOS sensor is configured to output one or more images corresponding to the first reflective light pattern and the second reflective light pattern in response to receiving the first reflective light pattern and the second reflective light pattern, and

wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the NIR range; and

a processor configured to:

detect a face of the object in a first region of the one or more images;

determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap;

compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region;

determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

16. (Cancelled)

17. (Original) The system of claim 15, wherein the processor is configured to detect the face of the object by identifying at least one pre-defined facial characteristic in the first region of the one or more images.

18. (Currently Amended) The system of claim 15, wherein the distribution of light intensity **comparison** comprises a shape and a brightness ~~of the light beam~~.

19. (Original) The system of claim 15, wherein the system includes a mobile device.

20. (Original) The system of claim 15, wherein in response to detecting the face, the processor is configured to authenticate the object based on the classification of the object and the depth map.

21. (Original) The system of claim 15, wherein the depth map is a spatial distribution of a distance between the object and the CMOS sensor.

22. (Original) The system of claim 15, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

23. (Original) The system of claim 22, wherein the processor is configured to authenticate the object when the object is classified as three-dimensional.

24. (Original) The system of claim 22, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object.

25. (Original) The system of claim 24, wherein the processor is configured to authenticate the object when the object is classified as skin and three-dimensional.

26. (Original) The system of claim 15, further comprising a neural network coupled to the processor, the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained to detect features related to

classifications of the objects and output the classification of the object based on the processed images.

27. (Previously Presented) A system, comprising:

- a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;

- a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement;

- a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement;

- wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the scene,

- wherein the at least one lens is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern,

- wherein the CMOS sensor is configured to output an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern;

- wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in an NIR light range of 700 nm to 1100 nm in wavelength; and

- a processor configured to:

 - detect a face of an object in the scene in a region of the image;

 - determine a distribution of light intensity of the region of the image;

compare the distribution of light intensity of the region of the image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image;

determine a classification of the object in the scene based on the distribution of light intensity comparison of the region of the image and the corresponding region;

determine a depth map of the object in the image based on a distortion in the pseudo-random structured light pattern; and

authenticate the object based on the detection of the face, the classification of the object, and the depth map.

28. (Original) The system of claim 27, wherein the system includes a mobile device.

29. (Original) The system of claim 27, wherein the processor is configured to:

compare the depth map of the object with a pre-defined depth map; and

classify the object as three-dimensional in response to determining a deviation of the depth map from the pre-defined depth map.

30. (Original) The system of claim 29, wherein the processor is configured to differentiate skin from non-living surfaces in the classification of the object and authenticate the object when the object is classified as skin and three-dimensional.

31. (Previously Presented) The system of claim 27, further comprising:

a diffractive optical element (DOE), wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens toward the DOE.

32. (Previously Presented) The system of claim 27, wherein the light projecting device is a light-emitting diode (LED).

33. (Previously Presented) The system of claim 27, wherein the light projecting device is a VCSEL.

34. (Previously Presented) The system of claim 27, wherein the system includes a computing device.

35. (Previously Presented) The system of claim 15, wherein the system includes a computing device.

REMARKS

I. Status of the Claims

Claims 2, 5-7, and 16 are cancelled.

Claims 1, 4, 15, and 18 are amended for clarity. Support can be found at least at Page 48, Lines 1-2; Page 48, Lines 15-26; Page 48, Lines 5-8; Page 48, Line 39 through Page 49, Line 7; and Figure 2 of the application as-filed.

Claims 1, 3-4, 8-15, and 17-35 are currently pending and under examination. No new matter is entered herein.

II. Examiner Interview

Applicant thanks Examiner Tran for the Examiner Interview with Applicant's representative Zach Wiersma (Reg. No. 77,808) on July 17, 2025. During the interview, the clarity rejections and drawing objections were discussed. No agreement on patentability was reached during the interview.

III. Drawing Objections

The drawings stand objected to under 37 CFR 1.83(a).

The Office based the rejections in part on, "The drawings must show every feature of the invention specified in the claims. Therefore, the following claimed subject matter must be shown or the feature(s) canceled from the claim(s)" (see Page 2 of the Office Action dated 06/03/25).

Applicant respectfully disagrees. Before the content of the drawing is addressed, it must first be assessed whether drawings are required. To this end, as set forth in 37 CFR 1.81(a), "The applicant for a patent is required to furnish a drawing of the invention where **necessary** for the understanding of the subject matter sought to be patented" (emphasis added).

Here, the Office has broadly asserted that certain claim elements are not present in the drawings. However, such claim elements are known and explicitly referenced in the present

application, and therefore a drawing showing these specific elements is not required for understanding. To this end, the Office has not expressed **confusion** about what is claimed, but only a **disapproval** that certain elements are not depicted in the figures. However, a desire to observe a certain element in a figure is inconsistent with the requirements of 37 CFR 1.81(a), which requires drawings only when they are **necessary** for the understanding of the subject matter sought to be patented.

Accordingly, Applicant respectfully submits that corrected drawing sheets are not required, and further respectfully requests withdrawal of the objections at least for these reasons.

Moreover, Applicant respectfully notes that Fig. 2 shows an embodiment of a mobile device according to the present claims. The elements of Fig. 2, including the claim elements, are described in detail with respect to reference numerals (see at least Page 48, Lines 1-2; Page 48, Lines 15-26; Page 48, Lines 5-8; and Page 48, Line 39 through Page 49, Line 7 of the application as-filed). Thus, Applicant respectfully submits that the present application already includes a sufficient drawing.

In view of the foregoing, Applicant respectfully requests withdrawal of the objections.

IV. Rejections under 35 U.S.C. § 112

i. Claims 4, 15, 17-26, and 35 stand rejected under 35 U.S.C. 112(b) as allegedly being indefinite.

Without commenting on the appropriateness of the rejections, the present claims are amended consistent with the comments of the Office. It is believed that these amendments overcome the rejections.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

ii. Claims 21-50 stand rejected under 35 U.S.C. 112(a) as allegedly failing to comply with the written description requirement.

The Office based the rejections in part on, “The specification discloses the term ‘vertical cavity surface emitting lasers’ twice, at page 14, line 27 and page 48, line 11. However, it fails to provide support for the above claimed subject matter” (see Page 6 of the Office Action dated 06/03/25).

Applicant respectfully disagrees and submits that the present claims comply with the written description requirement. For clarity, the table below recites claim elements referenced by the Office, as well as exemplary support in the present application.

Claim element	Exemplary support in the application as-filed
vertical cavity surface emitting lasers	Page 14, Lines 9-20 and Page 48, Lines 15-26
Plurality of illumination units (i.e., lasers)	Page 14, Lines 31-34
pseudo-random and periodic point patterns	Page 14, Lines 20-32
NIR light beams	Page 48, Lines 7-13 and Page 57, Lines 33-38

Applicant also respectfully notes specific support for the claim elements:

- Figure 2 of the application as-filed (depicting an embodiment of a mobile device);
- Page 48, Lines 1-2 (disclosing that the illumination unit 118 may include a plurality of light sources);
- Page 48, Lines 15-26 (disclosing that the illumination unit 118 may include vertical cavity surface-emitting lasers);
- Page 48, Lines 5-8 (disclosing NIR illumination features by the illumination unit 118);
- Page 48, Line 39 through Page 49, Line 7 (disclosing a plurality of illumination features, including periodic or non periodic illumination features and pseudo-random point patterns);

- Page 47, Lines 37-44 (disclosing reflection of the illumination pattern by surfaces of the scene);
- Page 47, Lines 38-41 (disclosing that the camera is configured to receive reflected light); and
- Page 50, Lines 16-20 (disclosing that the CMOS is configured to output an image corresponding to a reflective light pattern).

As can be seen, the present application discloses each and every element of the present claims. Thus, the present claims comply with the written description requirement at least for this reason.

Further, Applicant respectfully notes that the fundamental factual inquiry to determine compliance with the written description requirement is whether the specification conveys with reasonable clarity to those skilled in the art that, as of the filing date sought, inventor was in possession of the invention as now claimed. See, e.g., *Vas-Cath, Inc. v. Mahurkar*, 935 F.2d 1555, 1563-64, 19 USPQ2d 1111, 1117 (Fed. Cir. 1991). The subject matter of the claim need not be described literally (i.e., using the same terms or *in haec verba*) in order for the disclosure to satisfy the description requirement. See MPEP 2163.02. Moreover, newly added claims or claim limitations may be supported in the specification through express, implicit, or inherent disclosure. See MPEP 2163(I)(B). Finally, instead of repeating some information contained in another document, an application may attempt to incorporate the content of another document or part thereof by reference to the document in the text of the specification. The information incorporated is as much a part of the application as filed as if the text was repeated in the application, and should be treated as part of the text of the application as filed. See MPEP 2163.07(b).

Here, a person having ordinary skill in the art would have understood that the present claims are conveyed by the specification with reasonable clarity. The present application discloses a system and the individual elements of the system. The present application also discloses interactions between the elements. The present application also incorporates by reference several documents, including EP application 20159984.2, EP application 20154961.5, WO 2020/187719, WO 2018/091649, WO 2018/091638, WO 2018/091640, and EP application 20170905.2 (see Page 18, Lines 15-20 and Page 29, Lines 43-45 of the application as-filed). These incorporated

documents serve as further support. Thus, altogether, the present application provides sufficient written description for the present claims.

For further clarity, Applicant respectfully notes that the present claims are directed towards a system for authenticating a face. The system analyzes a beam profile of at least two different areas (see Page 55, Lines 13-17 of the application as-filed). This unique approach allows face authentication to reliably detect presentation attacks with 3D face masks (see Page 3, Lines 7-21 of the application as-filed). The present claims are the first report of such a system.

In view of the foregoing, Applicant respectfully requests withdrawal of the rejections and allowance of the claims.

V. Double Patenting Rejections

Claims 1, 3-4, 8-15, 17-35 stand rejected on the ground of nonstatutory double patenting as allegedly being unpatentable over claims 21-50 of copending Application No. 18/468,352.

Applicant respectfully requests reconsideration of these rejections in view of the present response.

CONCLUSION

Applicant requests reconsideration and allowance of all pending claims. The Commissioner is hereby authorized to charge Deposit Account No. 01-2384 for any outstanding fees incurred in connection with this Response. Please reference docket number 27843-2771.

Respectfully submitted,

/Christopher M. Goff/

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CMG/EMF/ZSW

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UNITED STATES
PATENT AND TRADEMARK OFFICEP.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov**ELECTRONIC ACKNOWLEDGEMENT RECEIPT**

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	07/28/2025 04:24:01 PM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	71574275	FILING DATE	03/24/2025
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents**TOTAL DOCUMENTS: 3**

DOCUMENT		PAGES	DESCRIPTION	SIZE (KB)
2025-07-28 Response to Non-Final Office Action 27843-2771.pdf		15	-	194 KB
2025-07-28 Response to Non-Final Office Action 27843-2771-A....pdf	(1-1)	1	Amendment/Request for Reconsideration-After Non- Final Rejection	107 KB
2025-07-28 Response to Non-Final Office Action 27843-2771-CLM.pdf	(2-9)	8	Claims	122 KB
2025-07-28 Response to Non-Final Office Action 27843-2771-REM.pdf	(10-15)	6	Applicant Arguments/Remarks Made in an Amendment	173 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-07-28 Response to Non-Final Office Action 27843-2771.pdf	8B5100F36DDC3C3E487B1376D47EEE1A5491AAE67169D85C1F86E3F078DB102400420D233264FBA7E1EBD8501CD8524AACD9BAA6C439935CA13B4824F00405E3
2025-07-28 Response to Non-Final Office Action 27843-2771-A....pdf	F4CB435AC3742AE73A512DE2BC7405E9A67FD0781A5312DBF79F60B129748DF0B7C0018805C1B987033CCD40DF4F6E32C70E6CA9B11695A6B1C0F997F7C2AD31
2025-07-28 Response to Non-Final Office Action 27843-2771-CLM.pdf	B38A9E27DFDEB054B58EEB7E7F3A48D3BDD84B81BB3A220B86828B0445AC3FB8F9367AEE647E255FF277EB7474C7D58B24E5BD3260DC5DF4E76197993D0F6AD8
2025-07-28 Response to Non-Final Office Action 27843-2771-REM.pdf	8F52AA27E282D0C194456A642FFE799CE5E7288F4A53DFD7A33E103E02D228DA1CC6F87086B5B7E881D687ACC7498EB487228B5058E742222456C5A254173EA2

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If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

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PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 19/088,110	Filing Date 03/24/2025	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 =	*	x \$200 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 =	*	x \$ 600 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	07/28/2025		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	29	Minus	** 30	= 0
	Independent (37 CFR 1.16(h))	*	3	Minus	*** 3	= 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*		Minus	**	=
	Independent (37 CFR 1.16(h))	*		Minus	***	=
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						SLIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/AMANDA CHRISTINE LEMUS/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

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APPLICATION NUMBER	FILING OR 371(C) DATE	FIRST NAMED APPLICANT	ATTY. DOCKET NO./TITLE
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04

CONFIRMATION NO. 4090

76656
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

PUBLICATION NOTICE



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Date Mailed: 08/07/2025

Title:OPTICAL SKIN DETECTION FOR FACE UNLOCK

Publication No.US-2025-0252783-A1

Publication Date:08/07/2025

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Thu Aug 07 05:05:53 EDT 2025

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UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	NTC.PUB	08/07/2025	27843-2771 202595US04

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090
76656	7590	08/12/2025	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	PAPER NUMBER
7700 Forsyth Boulevard			2668	
Suite 1800			NOTIFICATION DATE	DELIVERY MODE
St. Louis, MO 63105			08/12/2025	ELECTRONIC

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Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Response to Arguments

2. Applicant's arguments filed 07/28/2025 have been fully considered but they are not persuasive.

In response to Applicant argument that “the Office has not expressed confusion about what is claimed, but only a disapproval that certain elements are not depicted in the figures”, the drawings must show every feature of the invention specified in the claims as required by under 37 CFR 1.83(a). For example, the drawings must show “a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement”; “a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement”; “a diffractive optical element (DOE)” as recited in claim 1.

In response to Applicant argument that the specification provides support for the claim elements, the specification discloses the term “vertical cavity surface emitting lasers” only twice, at page 14, line 27 and page 49, line 11, and fails to describe the claimed subject matter in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention. For example, the specification fails to describe “a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement”; “a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” as recited in claim 1.

In response to Applicant argument that application, on Page 18. Lines 15-20 and Page 29. Lines 43-45 of the specification, incorporates by reference several documents, it is noted that an incorporation by reference that does not comply with 37 CFR 1.57(c), (d), or (e) is not effective.

The nonstatutory double patenting rejection is maintained because applicant does not make any argument regarding the nonstatutory double patenting rejection.

Drawings

3. The drawings are objected to under 37 CFR 1.83(a). The drawings must show every feature of the invention specified in the claims. Therefore, the following claimed subject matter must be shown or the feature(s) canceled from the claim(s):

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 1;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 1;

“a diffractive optical element (DOE)” recited in claim 1;

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 15;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 15;

“an optical element comprising at least one of a diffractive optical element (DOE) and a lens” recited in claim 15;

“a neural network coupled to the processor, the neural network comprising one or more convolutional layers configured to process a plurality of images, wherein the neural network is trained

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to detect features related to classifications of the objects and output the classification of the object based on the processed images” recited in claim 26.

“a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 27;

“a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 27;

“at least one lens is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern” recited in claim 27;

“a diffractive optical element (DOE), wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens toward the DOE” recited in claim 31.

No new matter should be entered.

Corrected drawing sheets in compliance with 37 CFR 1.121(d) are required in reply to the Office action to avoid abandonment of the application. Any amended replacement drawing sheet should include all of the figures appearing on the immediate prior version of the sheet, even if only one figure is being amended. The figure or figure number of an amended drawing should not be labeled as “amended.” If a drawing figure is to be canceled, the appropriate figure must be removed from the replacement sheet, and where necessary, the remaining figures must be renumbered and appropriate changes made to the brief description of the several views of the drawings for consistency. Additional replacement sheets may be necessary to show the renumbering of the remaining figures. Each drawing sheet submitted after the filing date of an application must be labeled in the top margin as either “Replacement Sheet” or “New Sheet” pursuant to 37 CFR 1.121(d). If the changes are not accepted by

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the examiner, the applicant will be notified and informed of any required corrective action in the next Office action. The objection to the drawings will not be held in abeyance.

Claim Rejections - 35 USC § 112

4. The following is a quotation of the first paragraph of 35 U.S.C. 112(a):

(a) IN GENERAL.—The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor or joint inventor of carrying out the invention.

The following is a quotation of the first paragraph of pre-AIA 35 U.S.C. 112:

The specification shall contain a written description of the invention, and of the manner and process of making and using it, in such full, clear, concise, and exact terms as to enable any person skilled in the art to which it pertains, or with which it is most nearly connected, to make and use the same, and shall set forth the best mode contemplated by the inventor of carrying out his invention.

5. Claims 1, 3-4, 8-15, 17-35 are rejected under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as failing to comply with the written description requirement. The claim(s) contains subject matter which was not described in the specification in such a way as to reasonably convey to one skilled in the relevant art that the inventor or a joint inventor, or for applications subject to pre-AIA 35 U.S.C. 112, the inventor(s), at the time the application was filed, had possession of the claimed invention.

The original disclosure fails to provide support for the following claimed subject matter:

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 1;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 1;

“a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 15;

“a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 15;

“a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement” recited in claim 27;

“a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement” recited in claim 27.

The specification discloses the term “vertical cavity surface emitting lasers” twice, at page 14, line 27 and page 49, line 11. However, it fails to provide support for the above claimed subject matter.

Double Patenting

6. The nonstatutory double patenting rejection is based on a judicially created doctrine grounded in public policy (a policy reflected in the statute) so as to prevent the unjustified or improper timewise extension of the “right to exclude” granted by a patent and to prevent possible harassment by multiple assignees. A nonstatutory double patenting rejection is appropriate where the conflicting claims are not identical, but at least one examined application claim is not patentably distinct from the reference claim(s) because the examined application claim is either anticipated by, or would have been obvious over, the reference claim(s). See, e.g., *In re Berg*, 140 F.3d 1428, 46 USPQ2d 1226 (Fed. Cir. 1998); *In re Goodman*, 11 F.3d 1046, 29 USPQ2d 2010 (Fed. Cir. 1993); *In re Longi*, 759 F.2d 887, 225 USPQ 645 (Fed. Cir. 1985); *In re Van Ornum*, 686 F.2d 937, 214 USPQ 761 (CCPA 1982); *In re Vogel*, 422 F.2d 438, 164 USPQ 619 (CCPA 1970); *In re Thorington*, 418 F.2d 528, 163 USPQ 644 (CCPA 1969).

A timely filed terminal disclaimer in compliance with 37 CFR 1.321(c) or 1.321(d) may be used to overcome an actual or provisional rejection based on nonstatutory double patenting provided the reference application or patent either is shown to be commonly owned with the examined application, or claims an invention made as a result of activities undertaken within the scope of a joint research agreement. See MPEP § 717.02 for applications subject to examination under the first inventor to file provisions of the AIA as explained in MPEP § 2159. See MPEP § 2146 *et seq.* for applications not subject to examination under the first inventor to file provisions of the AIA. A terminal disclaimer must be signed in compliance with 37 CFR 1.321(b).

The USPTO Internet website contains terminal disclaimer forms which may be used. Please visit www.uspto.gov/patent/patents-forms. The filing date of the application in which the form is filed determines what form (e.g., PTO/SB/25, PTO/SB/26, PTO/AIA/25, or PTO/AIA/26) should be used. A web-based eTerminal Disclaimer may be filled out completely online using web-screens. An eTerminal Disclaimer that meets all requirements is auto-processed and approved immediately upon submission. For more information about eTerminal Disclaimers, refer to www.uspto.gov/patents/process/file/efs/guidance/eTD-info-I.jsp.

7. Claim 1, 3-4, 8-15, 17-35 are provisionally rejected on the ground of nonstatutory double patenting as being unpatentable over claim 21-50 of copending Application No. 18/468,352. Although the claims at issue are not identical, they are not patentably distinct from each other because the present claims are either anticipated by or obvious variants of the copending application claims. The following table shows the corresponding limitations between representative claims and representative copending application claims.

This is a provisional nonstatutory double patenting rejection because the patentably indistinct claims have not in fact been patented.

Present Application Claims	Copen ding Application Claims
1. A system, comprising: a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength; a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement; a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement; a diffractive optical element (DOE); wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the DOE, wherein the DOE is configured to diffract the first plurality of NIR light beams in a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern, wherein the CMOS sensor is configured to output an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern, and wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the range; and a processor configured to: detect a face in a first region in the image; determine a distribution of light intensity of a second region in the image, wherein the second region corresponds to the first region; compare the distribution of light intensity of the second region of the image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the image and the corresponding region; determine a depth map	21. A system, comprising: a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area; a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a first plurality of NIR light beams; a second VCSEL comprising a periodic arrangement and configured to output a second plurality of NIR light beams, wherein the second VCSEL array is offset from the first VCSEL array; a diffractive optical element (DOE); at least one lens disposed between the DOE and the first and second VCSELs, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens, wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein the CMOS sensor is configured to receive a first reflective light pattern and a second reflective light pattern reflected by surfaces of the scene and output a first image corresponding to the first reflective light pattern and a second image corresponding to the second reflective light pattern, wherein the first reflective light pattern originates from the first plurality of NIR light beams and the second reflective light pattern originates from the second plurality of NIR light beams; and a processor configured to: detect a face in a first region in the first image; determine a distribution of light intensity of a second region in the second image, wherein the second region corresponds to the first region; compare the distribution of light intensity of the second region of the second image to a distribution of light intensity of a corresponding region in

of the face in the scene in the image based on a distortion in the pseudo-random structured light pattern; and authenticate the face based on the detection of the face, the classification of the object, and the depth map.

15. A system, comprising: a near infrared (NIR) camera configured to capture an object, the NIR camera comprising a CMOS sensor, wherein the NIR camera is sensitive to NIR light in a range of 700 nm to 1100 nm in wavelength; a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of light beams toward the object, the first plurality of NIR light beams comprising a pseudo-random arrangement; a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement; an optical element comprising at least one of a diffractive optical element (DOE) and a lens; wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the optical element toward the scene, wherein the optical element is configured to modify the first plurality of NIR light beams in a pseudo-random structured light pattern and the second plurality of NIR light beams in a substantially uniform light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern and the periodic light pattern into a second reflective light pattern, wherein the CMOS sensor is configured to output one or more images corresponding to the first reflective light pattern and the second reflective light pattern in response to receiving

reference image data derived from a reference image; determine a classification of the face in the scene based on the distribution of light intensity comparison of the second region of the second image and the corresponding region; determine a depth map of the face in the scene in the second image based on a distortion in the pseudo-random structured light pattern; and authenticate the face based on the detection of the face, the classification of the object, and the depth map.

35. A system, comprising: a near infrared (NIR) camera comprising a CMOS sensor and configured to capture an object, wherein the CMOS sensor comprises an NIR light sensitive area; a first vertical cavity surface emitting laser (VCSEL) comprising a pseudo-random arrangement and configured to output a plurality of light beams in a near-infrared range of 700 nm to 1100 nm; a second VCSEL comprising a periodic arrangement and configured to output a plurality of beams of NIR light of 700 nm to 1100 nm, wherein the second VCSEL array is offset from the first VCSEL; a diffractive optical element (DOE); at least one lens disposed between the DOE and the first and second VCSELs, wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through the at least one lens, wherein the DOE is configured to diffract the first plurality of NIR light beams, resulting in a pseudo-random structured light pattern and configured to diffract the second plurality of NIR light beams resulting in a substantially uniform light pattern, wherein the CMOS sensor is configured to receive at least one reflective light pattern and output one or more images corresponding to the at least one reflective light pattern; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light intensity of a second region of the one or more

the first reflective light pattern and the second reflective light pattern, and wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in the NIR range; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light intensity of a second region of the one or more images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.	images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.
27. A system, comprising: a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor; a vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement; a light projecting device configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement; wherein the first plurality of NIR light beams and the second plurality of NIR light beams pass through at least one lens toward the scene, wherein the at least one lens is configured to modify the first plurality of NIR light beams in	47. A system, comprising: at least one vertical cavity surface emitting laser (VCSEL) configured to: emit a plurality of light beams in a near-infrared range (NIR), arranged in a first spatial arrangement relative to each other, towards an object; emit at least one substantially uniform light beam in the near-infrared range towards the object; a CMOS sensor, wherein the CMOS sensor comprises an NIR light sensitive area, wherein the CMOS sensor is configured to receive light reflected from the object and output one or more images characterizing at least an intensity of the reflected light; and a processor configured to: detect a face of the object in a first region of the one or more images; determine a distribution of light intensity of a second

a pseudo-random structured light pattern, wherein surfaces in the scene reflect the pseudo-random structured light pattern into a first reflective light pattern, wherein the CMOS sensor is configured to output an image corresponding to the first reflective light pattern in response to receiving the first reflective light pattern; wherein light of each of the first plurality of NIR light beams and the second plurality of NIR light beams is in an NIR light range of 700 nm to 1100 nm in wavelength; and a processor configured to: detect a face of an object in the scene in a region of the image; determine a distribution of light intensity of the region of the image; compare the distribution of light intensity of the region of the image to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object in the scene based on the distribution of light intensity comparison of the region of the image and the corresponding region; determine a depth map of the object in the image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map. Claims 3-4, 8-14, 17-26, 28-35 recite features similar to features recited in claims 22-34, 36-46, 48-50. Therefore, they are rejected for reasons similar to those discussed above.	region of the one or more images, wherein the first region and the second region at least partially overlap; compare the distribution of light intensity of the second region of the one or more images to a distribution of light intensity of a corresponding region in reference image data derived from a reference image; determine a classification of the object based on the distribution of light intensity comparison of the second region of the one or more images and the corresponding region; determine a depth map of the object in at least one image based on a distortion in the pseudo-random structured light pattern; and authenticate the object based on the detection of the face, the classification of the object, and the depth map.
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Conclusion

8. **THIS ACTION IS MADE FINAL.** Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date

of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any nonprovisional extension fee (37 CFR 1.17(a)) pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the mailing date of this final action.

9. Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHUOC TRAN whose telephone number is (571)272-7399. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vu Le can be reached at 571-272-7332. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

Application/Control Number: 19/088,110

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Art Unit: 2668

/PHUOC TRAN/

Primary Examiner, Art Unit 2668

<i>Applicant-Initiated Interview Summary</i>	Application No. 19/088,110	Applicant(s) SCHILLEN et al.		
	Examiner PHUOC TRAN	Art Unit 2668	AIA (First Inventor to File) Status Yes	Page 1 of 1

All Participants (applicant, applicants representative, PTO personnel)	Title	Type
PHUOC TRAN	Primary Examiner	Telephonic
ZACHARY WIERSMA	Attorney of Record	

Date of Interview: 07 August 2025

Issues Discussed:

35 U.S.C. 112

Portions of the specification provided by Application were discussed. The Examiner pointed out that the specification discloses the term “vertical cavity surface emitting lasers” only twice, at page 14, line 27 and page 49, line 11. No agreement was reached.

Drawings

Illumination Unit 118 in Fig. 2 was discussed. Whether the drawings must show claim features “a first vertical cavity surface emitting laser (VCSEL) configured to output a first plurality of NIR light beams toward the scene, the first plurality of NIR light beams comprising a pseudo-random arrangement”; “a second VCSEL configured to output a second plurality of NIR light beams toward the scene, the second plurality of NIR light beams comprising a periodic arrangement”; “a diffractive optical element (DOE)” as recited in claim 1 was discussed. No agreement was reached.

/PHUOC TRAN/ Primary Examiner, Art Unit 2668	
Applicant is reminded that a complete written statement as to the substance of the interview must be made of record in the application file. It is the applicants responsibility to provide the written statement, unless the interview was initiated by the Examiner and the Examiner has indicated that a written summary will be provided. See MPEP 713.04 Please further see: MPEP 713.04 Title 37 Code of Federal Regulations (CFR) § 1.133 Interviews, paragraph (b) 37 CFR § 1.2 Business to be transacted in writing	

Applicant recordation instructions: The formal written reply to the last Office action must include the substance of the interview. (See MPEP section 713.04). If a reply to the last Office action has already been filed, applicant is given a non-extendable period of the longer of one month or thirty days from this interview date, or the mailing date of this interview summary form, whichever is later, to file a statement of the substance of the interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

<i>Search Notes</i> 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC - Searched*		
Symbol	Date	Examiner
G06V40/40, G01S17/894, G06F21/32, G06T7/50, G06V10/141, G06V10/145, G06V10/25, G06V10/54, G06V10/60, G06V10/764, G06V10/82, G06V40/165, G06V40/168, G06V40/172, G06T2207/30201	05/29/2025	PT

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search	05/29/2025	PT
S.N. 18/264,326 refs. checked	05/29/2025	PT
S.N. 18/468,352 refs. checked	05/29/2025	PT
See search history printout(s)	05/29/2025	PT
See search history printout(s)	08/08/2025	PT

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner

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<i>Search Notes</i> 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

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PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	282	((("SCHILLEN") near3 ("Peter")) OR ("GUTHIER") near3 ("Benjamin")) OR ("SCHICK") near3 ("Friedrich")) OR ("GUENTHER") near3 ("Manuel")) OR ("DIESELBERG") near3 ("Lars")) OR ("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/09/07 03:49 PM
L2	537832	(G06V40/40 or G01S17/894 or G06F21/32 or G06T7/50 or G06V10/141 or G06V10/145 or G06V10/25 or G06V10/54 or G06V10/60 or G06V10/764 or G06V10/82 or G06V40/165 or G06V40/168 or G06V40/172 or G06T2207/30201).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/09/07 03:50 PM
L3	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authentikat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:03 PM
L4	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authentikat\$4 OR identif\$8 OR	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:04 PM

L5	2	verifi\$8)).clm. 2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:05 PM
L6	0	2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:08 PM
L7	0	(image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:09 PM
L10	329	(image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:14 PM

L11	120	OR skin)) 2 AND (image SAME (face OR facial OR skin) SAME (three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial OR skin))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:15 PM
L12	14	11 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:18 PM
L13	24	10 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:19 PM
L14	10	13 NOT 12	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:26 PM
L15	106	11 NOT 13	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:29 PM
L16	2	("20190213309" OR "2020082155").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2024/09/07 05:27 PM
L17	305	((("SCHILLEN") near3 ("Peter")) OR (("GUTHIER") near3 ("Benjamin")) OR (("SCHICK") near3 ("Friedrich")) OR (("GUENTHER") near3 ("Manuel")) OR (("DIESELBERG") near3 ("Lars")) OR (("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/05/29 11:50 AM
L18	3	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 11:55 AM

L19	8	OR light) SAME (structur\$4 OR pattern) SAME (classif\$8 OR authentikat\$4 OR identif\$8 OR verifi\$8)).clm. ("20150374277" "20160242853" "20190213309" "20200082155" "20200211180" "20210049391" "20220222464" "20240119130").pn. OR ("12288421").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:00 PM
L20	3	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern)) .clm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:02 PM
L21	11	17 AND (infrar\$4 SAME laser) AND ((face OR facial) SAME (depth OR three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:06 PM
L22	11	17 AND (infrar\$4 SAME (camera OR sensor)) AND ((vertical SAME laser) OR VCSEL) AND ((face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:12 PM
L23	11	21 AND 22	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:16 PM
L24	5	17 AND (infrar\$4 SAME	(US-PGPUB; USPAT;	ADJ	ON	ON	2025/05/29

		(camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	USOCR)				12:20 PM
L25	0	20 AND (infrar\$4 SAME (camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:26 PM
L26	3	20 AND ((vertical SAME laser) OR VCSEL) AND (infrar\$4 OR NIF) AND (pseudo\$11 OR random\$4 OR periodic)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:27 PM
L27	0	("20240037201" OR "20240005703" OR "12288421").pn. AND (((vertical SAME laser) OR VCSEL) SAME ((infrar\$4 OR NIF OR pseudo\$11 OR random\$4 OR periodic)))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:39 PM
L28	0	("20240037201" OR "20240005703" OR "12288421").pn. AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL)) AND ((infrar\$4 OR NIF) SAME laser) AND ((infrar\$4 OR NIF OR laser) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L29	3	("20240037201" OR "20240005703" OR "12288421").pn.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L30	0	29 AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:44 PM
L31	3	29 AND ((vertical SAME laser) OR VCSEL)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:45 PM
L33	3	29 AND ((infrar\$4 OR NIR) SAME laser) AND ((infrar\$4 OR NIR OR laser) SAME (pseudo\$11 OR	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:56 PM

L34	9	random\$4 OR periodic)) ("20150374277" OR "20160242853" OR "20160335483" OR "20190213309" OR "20200082155" OR "20200211180" OR "20210049391" OR "20220222464" OR "20240119130").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/05/29 12:59 PM
L35	14	1 AND ((vertical WITH laser) OR vcsel).clm. AND ((vertical WITH laser) OR vcsel)	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/08/07 01:55 PM
L36	30	(Artificial Intelligence) More like doc: US-20240331450-A1 Text: (US-9358350-B2 OR US-9940709-B2 OR US-10368795-B2 OR US-9672415-B2 OR WO-2019056310-A1 OR US-10922395-B2 OR US-10990805-B2 OR CN-111164610-B OR CN-114676403-A OR US-11741747-B2 OR US-12288421-B2 OR US-12361760-B2 OR US-20240331450- A1 OR US- 20240037201-A1 OR CN-120375419-A OR CN-116897349-A OR EP-4546296-A2 OR KR-20230146540-A OR EP-4295252-A1 OR JP- 2024512247-A).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:08 AM
L37	30	(Artificial Intelligence) More like doc: US-12288421-B2 Text: (US-9358350-B2 OR US-9940709-B2 OR US-10368795-B2 OR US-9672415-B2 OR WO-2019056310-A1 OR US-10922395-B2 OR US-10990805-B2 OR CN-111164610-B OR CN-114676403-A OR US-11741747-B2 OR US-12288421-B2 OR US-12361760-B2 OR CN-120375419-A	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:08 AM

L38	30	OR US-20240331450-A1 OR US-20240037201-A1 OR EP-4546296-A2 OR EP-4295252-A1 OR KR-20230146540-A OR CN-116897349-A OR JP-2024512247-A).did. 36 OR 37	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:13 AM
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PE2E SEARCH - Search History (Interference)

There are no Interference searches to show.

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 76656

Tue Aug 12 05:03:24 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MISSOURI, 63105
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 76656, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	CTFR	08/12/2025	27843-2771 202595US04
19088110	EXIN	08/12/2025	27843-2771 202595US04

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ELECTRONIC PAYMENT RECEIPT

APPLICATION # 19/088,110	RECEIPT DATE / TIME 10/14/2025 10:23:27 AM Z ET	ATTORNEY DOCKET # 27843-2771 202595US04
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Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	72692273	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	03/24/2025
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

Payment Information

PAYMENT METHOD DA / 012384	PAYMENT TRANSACTION ID E20250DA23525123	PAYMENT AUTHORIZED BY Barbara McGregor
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FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1817	REQUEST FOR PRIORITIZED EXAMINATION	4515.00	1	4515.00
1830	PROCESSING FEE, EXCEPT IN PROVISIONAL APPLICATIONS	150.00	1	150.00
1801	REQUEST FOR CONTINUED EXAMINATION (RCE) - 1ST REQUEST (SEE 37 CFR 1.114)	1500.00	1	1500.00
TOTAL AMOUNT:				\$6,165.00

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

**REQUEST FOR CONTINUED EXAMINATION(RCE)TRANSMITTAL
(Submitted Only via EFS-Web)**

Application Number	19088110	Filing Date	2025-03-24	Docket Number (if applicable)	27843-2771 202595US04	Art Unit	2668
First Named Inventor	Peter SCHILLEN			Examiner Name	Phuoc Tran		

This is a Request for Continued Examination (RCE) under 37 CFR 1.114 of the above-identified application.

Request for Continued Examination (RCE) practice under 37 CFR 1.114 does not apply to any utility or plant application filed prior to June 8, 1995, or to any design application. The Instruction Sheet for this form is located at WWW.USPTO.GOV

SUBMISSION REQUIRED UNDER 37 CFR 1.114

Note: If the RCE is proper, any previously filed unentered amendments and amendments enclosed with the RCE will be entered in the order in which they were filed unless applicant instructs otherwise. If applicant does not wish to have any previously filed unentered amendment(s) entered, applicant must request non-entry of such amendment(s).

☐ Previously submitted. If a final Office action is outstanding, any amendments filed after the final Office action may be considered as a submission even if this box is not checked.☐ Consider the arguments in the Appeal Brief or Reply Brief previously filed on _____☐ Other _____☒ Enclosed☒ Amendment/Reply☐ Information Disclosure Statement (IDS)☐ Affidavit(s)/ Declaration(s)☐ Other _____**MISCELLANEOUS**☐ Suspension of action on the above-identified application is requested under 37 CFR 1.103(c) for a period of months _____
(Period of suspension shall not exceed 3 months; Fee under 37 CFR 1.17(i) required)☐ Other _____**FEES****The RCE fee under 37 CFR 1.17(e) is required by 37 CFR 1.114 when the RCE is filed.**☒ The Director is hereby authorized to charge any underpayment of fees, or credit any overpayments, to
Deposit Account No 012384**SIGNATURE OF APPLICANT, ATTORNEY, OR AGENT REQUIRED**☒ Patent Practitioner Signature☐ Applicant Signature

Signature of Registered U.S. Patent Practitioner			
Signature	/Christopher M. Goff/	Date (YYYY-MM-DD)	2025-10-14
Name	Christopher M. Goff	Registration Number	41785

This collection of information is required by 37 CFR 1.114. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.11 and 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450.

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether the Freedom of Information Act requires disclosure of these records.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspections or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

**CERTIFICATION AND REQUEST FOR PRIORITIZED EXAMINATION
UNDER 37 CFR 1.102(e) (Page 1 of 1)**

First Named Inventor:	Peter SCHILLEN	Nonprovisional Application Number (if known):	19/088,110
Title of Invention:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

APPLICANT HEREBY CERTIFIES THE FOLLOWING AND REQUESTS PRIORITIZED EXAMINATION FOR THE ABOVE-IDENTIFIED APPLICATION.

1. The processing fee set forth in 37 CFR 1.17(i)(1) and the prioritized examination fee set forth in 37 CFR 1.17(c) have been filed with the request. The publication fee requirement is met because that fee, set forth in 37 CFR 1.18(d), is currently \$0. The basic filing fee, search fee, and examination fee are filed with the request or have been already been paid. I understand that any required excess claims fees or application size fee must be paid for the application.
2. I understand that the application may not contain, or be amended to contain, more than four independent claims, more than thirty total claims, or any multiple dependent claims, and that any request for an extension of time will cause an outstanding Track I request to be dismissed.
3. The applicable box is checked below:

I. ☐ Original Application (Track One) - Prioritized Examination under § 1.102(e)(1)

i. (a) The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a). This certification and request is being filed with the utility application via USPTO patent electronic filing system.

---OR---

(b) The application is an original nonprovisional plant application filed under 35 U.S.C. 111(a). This certification and request is being filed with the plant application in paper.

ii. An executed inventor's oath or declaration under 37 CFR 1.63 or 37 CFR 1.64 for each inventor, or the application data sheet meeting the conditions specified in 37 CFR 1.53(f)(3)(i) is filed with the application.

II. ☒ Request for Continued Examination - Prioritized Examination under § 1.102(e)(2)

- i. A request for continued examination has been filed with, or prior to, this form.
- ii. If the application is a utility application, this certification and request is being filed via USPTO patent electronic filing system.
- iii. The application is an original nonprovisional utility application filed under 35 U.S.C. 111(a), or is a national stage entry under 35 U.S.C. 371.
- iv. This certification and request is being filed prior to the mailing of a first Office action responsive to the request for continued examination.
- v. No prior request for continued examination has been granted prioritized examination status under 37 CFR 1.102(e)(2).

Signature	/Christopher M. Goff/	Date	October 14, 2025
Name (Print/Typed)	Christopher M. Goff	Practitioner Registration Number	41,785

Note: This form must be signed in accordance with 37 CFR 1.33. See 37 CFR 1.4(d) for signature requirements and certifications. Submit multiple forms if more than one signature is required.*

☒ *Total of 1 forms are submitted.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013). <https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

In the United States Patent & Trademark Office

First Inventor:	Peter Schillen	Art Unit:	2688
Serial No.:	19/088,110	Confirmation No.:	4090
Date Filed:	March 24, 2025	Examiner:	Phuoc Tran
Title:	OPTICAL SKIN DETECTION FOR FACE UNLOCK		

October 14, 2025

RESPONSE TO FINAL OFFICE ACTION

Commissioner for Patents
Alexandria, VA 22313

Commissioner:

In response to the Final Office Action dated August 12, 2025, please consider the following remarks.

Amendments to the Claims begin on page 2.

Remarks begin on page 8.

Conclusion is on page 9.

AMENDMENTS TO THE CLAIMS

1-35. (Cancelled)

36. (New) A system comprising:

a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;

an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene;

at least one optical element, wherein the NIR light passes through the at least one optical element;

a processor configured to:

output a first image comprising light reflected from the scene and detect a face in the first image;

output a second image comprising a reflective light pattern;

analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object;

determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image;

authenticate the detected face based on the depth information and the material classification.

37. (New) The system of claim 36, wherein the NIR camera is sensitive to light in a wavelength range of approximately 700 nm to 1100 nm.

38. (New) The system of claim 36, wherein the illumination unit is configured to project the NIR light in a light pattern, the light pattern comprising a grid, a dot matrix, a pseudo-random, a random, or a stripe configuration.
39. (New) The system of claim 36, wherein the reference image used for material classification is stored in at least one data storage device of the processor or retrieved from a remote server.
40. (New) The system of claim 36, wherein the material classification is performed using a machine learning model trained to distinguish human skin from nonbiological materials.
41. (New) The system of claim 36, wherein the processor is configured to capture multiple reflective light patterns over time and determine depth information.
42. (New) The system of claim 36, wherein the processor is configured to determine unsuccessful authentication by detecting depth information inconsistent with the three-dimensional object.
43. (New) The system of claim 40, wherein the processor is configured to determine unsuccessful authentication by determining a material classification consistent with a nonbiological material.
44. (New) The system of claim 36, wherein the NIR camera and the illumination unit are integrated into a mobile device housing.
45. (New) The system of claim 36, wherein the optical element comprises at least one of a diffractive optical element (DOE) and a lens.
46. (New) The system of claim 36, wherein the optical element comprises a plurality of lenses and a diffractive optical element (DOE).
47. (New) The system of claim 36, wherein the optical element is configured to modify the light from the illumination unit into a random light pattern, a periodic light pattern, or a combination thereof.

48. (New) The system of claim 36, wherein the illumination unit further comprises a second LED or a second laser configured to output an NIR light beam toward the scene.

49. (New) The system of claim 36, wherein the illumination unit further comprises a vertical cavity surface-emitting laser (VCSEL) configured to project an NIR light beam toward the scene.

50. (New) The system of claim 48, wherein the first image comprises light reflected from the scene in response to the projection of the NIR light beam toward the scene.

51. (New) The system of claim 36, wherein the second image comprises the reflective light pattern reflected in response to the projection of the illumination unit.

52. (New) A system comprising:

a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor;

an illumination unit comprising:

at least one of a first LED and a first laser configured to project NIR light in a light pattern toward the scene; and

at least one of a second LED and a second laser configured to project an NIR light beam toward the scene;

at least one optical element, wherein the NIR light passes through the at least one optical element;

a processor configured to:

output a first image comprising light reflected from the scene in response to the projection of the NIR light beam;

detect a face in the first image;

output a second image comprising a reflective light pattern reflected from the scene in response to the projection of the NIR light in the light pattern;

analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object;

determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image;

authenticate the detected face based on the depth information and the material classification.

53. (New) The system of claim 52, wherein the NIR camera is sensitive to light in a wavelength range of approximately 700 nm to 1100 nm.

54. (New) The system of claim 52, wherein the light pattern comprises a grid, a dot matrix, a pseudo-random, a random, or a stripe configuration.

55. (New) The system of claim 52, wherein the reference image used for material classification is stored in at least one data storage device of the processor or retrieved from a remote server.

56. (New) The system of claim 52, wherein the material classification is performed using a machine learning model trained to distinguish human skin from nonbiological materials.

57. (New) The system of claim 52, wherein the processor is configured to capture multiple reflective light patterns over time and determine depth information.

58. (New) The system of claim 52, wherein the processor is configured to determine unsuccessful authentication by detecting depth information inconsistent with the three-dimensional object.

59. (New) The system of claim 56, wherein the processor is configured to determine unsuccessful authentication by determining a material classification consistent with a nonbiological material.

60. (New) The system of claim 52, wherein the NIR camera and the illumination unit are integrated into a mobile device housing.
61. (New) The system of claim 52, wherein the optical element comprises at least one of a diffractive optical element (DOE) and a lens.
62. (New) The system of claim 52, wherein the optical element comprises a plurality of lenses and a diffractive optical element (DOE).
63. (New) The system of claim 52, wherein the optical element is configured to modify the light from the illumination unit into a random light pattern, a periodic light pattern, or a combination thereof.
64. (New) A mobile device comprising:
- a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, and wherein the NIR camera is sensitive to light in a wavelength range of approximately 700 nm to 1100 nm;
 - an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene;
 - at least one optical element, wherein the NIR light passes through the at least one optical element;
 - a processor configured to:
 - output a first image comprising light reflected from the scene and detect a face in the first image;
 - output a second image comprising a reflective light pattern;
 - analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object;

determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image;

authenticate the detected face based on the depth information and the material classification.

65. (New) The system of claim 64, wherein the material classification is performed using a machine learning model trained to distinguish human skin from nonbiological materials.

REMARKS

I. Status of the Claims

Claims 1-35 are cancelled.

Claims 36-65 are new.

Support can be found at least in Paragraphs [0017]-[0022], [0035], [0038], [0050], [0053], [0056], [0070], [0073], and [0136] of the published application (US 2025/0252783).

Claims 36-65 are currently pending and under examination. No new matter is entered herein.

II. Examiner Interview

Applicant thanks Examiner Tran for the Examiner Interview with Applicant's representative Zach Wiersma (Reg. No. 77,808) on August 7, 2025. During the interview, the clarity rejections and drawing objections were discussed. No agreement on patentability was reached during the interview.

III. Drawing Objections

The drawings stand objected to under 37 CFR 1.83(a).

Without commenting on the appropriateness of the objections, claims 1-35 are cancelled, thereby rendering the objections moot.

IV. Rejections under 35 U.S.C. § 112

Claims 1, 3-4, 8-15, 17-35 stand rejected under 35 U.S.C. 112(a) as allegedly failing to comply with the written description requirement.

Without commenting on the appropriateness of the rejections, claims 1-35 are cancelled, thereby rendering the rejections moot.

V. Double Patenting Rejections

Claims 1, 3-4, 8-15, and 17-35 stand rejected on the ground of nonstatutory double patenting as allegedly being unpatentable over claims 21-50 of copending Application No. 18/468,352.

Applicant respectfully requests reconsideration of these rejections in view of the present response, as well as the status of the cited reference.

CONCLUSION

Applicant requests reconsideration and allowance of all pending claims. The Commissioner is hereby authorized to charge Deposit Account No. 01-2384 for any outstanding fees incurred in connection with this Response. Please reference docket number 27843-2771.

Respectfully submitted,

/Christopher M. Goff/

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CMG/EMF/ZSW

VIA EFS



UNITED STATES
PATENT AND TRADEMARK OFFICE

P.O. Box 1450
Alexandria, VA 22313 - 1450
www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	10/14/2025 10:23:27 AM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	72692273	FILING DATE	03/24/2025
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 5

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-10-14 RCE 27843-2771.pdf	3	Request for Continued Examination (RCE)	1631 KB
2025-10-14 Track One Request 27843-2771.pdf	2	Track One Request	181 KB
2025-10-14 Response to Final Office Action 27843-2771.pdf	9	-	162 KB
2025-10-14 Response to Final Office Action 27843-2771-A.NE.pdf	(1-1) 1	Response After Final Action	105 KB
2025-10-14 Response to Final Office Action 27843-2771-CLM.pdf	(2-7) 6	Claims	115 KB

2025-10-14 Response to Final Office Action 27843- 2771-REM.pdf	(8-9)	2	Applicant Arguments/Remarks Made in an Amendment	147 KB
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Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-10-14 RCE 27843- 2771.pdf	C6C2C0162F29581A5BC4EF1F8439209E28965470B140D65977 3475D6BB93876B0A3B5843E02D207228D65B4549A8D404B9D 160C84AD3F81F15C255B11192F611
2025-10-14 Track One Request 27843-2771.pdf	ADD28E99A03DEBDF1A9B144DDD72041C9DDF2A0071036CE 0A6F7329252EB2BB2488579C6DD0EF0619C3F92D5132662059 38520220080E55EE9072A0F02918699
2025-10-14 Response to Final Office Action 27843-2771.pdf	90874960CB37E528C814332FB02A99F2FF7AE80938570F8396 6563CD3A220B67B2F1B221B8A74A0DEAFB4F04B52C3E648C FFA53C9CAC5A2586034AB5788496F2
2025-10-14 Response to Final Office Action 27843-2771- A.NE.pdf	A6588CDA2CCF05216EC64B5A88ACC663CC698570482C4314 8EEDBC79DAA8B1D6866026E13626DF3678C97BE2FA7273B3 BD99BEA92286A8141ACDCED12BC41CA4
2025-10-14 Response to Final Office Action 27843-2771- CLM.pdf	5BA9FFEF349F2F5FE339CFDF37DE75C0769393E0FC5D9F242 9084B5B8A286B4E580D526DC4BE4ED1DAA2265ACAECF1414 AC28E4F503645ED0B107918172851F2
2025-10-14 Response to Final Office Action 27843-2771- REM.pdf	476A5AC67C7A2D631A39B188B51854189F8D848EC7F227219 0247DB62C1D6B431E1838D971380923131D8999C7BBA89AF8 9FB44AB68572F9E96D174CAC542E5F

This Acknowledgement Receipt evidences receipt on the noted date by the USPTO of the indicated documents, characterized by the applicant, and including page counts, where applicable. It serves as evidence of receipt similar to a Post Card, as described in MPEP 503.

New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.

Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

PATENT APPLICATION FEE DETERMINATION RECORD Substitute for Form PTO-875	Application or Docket Number 19/088,110	Filing Date 03/24/2025	☐ To be Mailed
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ENTITY: ☒ LARGE ☐ SMALL ☐ MICRO

APPLICATION AS FILED - PART I

FOR	(Column 1) NUMBER FILED	(Column 2) NUMBER EXTRA	RATE (\$)	FEE (\$)
☐ BASIC FEE (37 CFR 1.16(a), (b), or (c))	N/A	N/A	N/A	
☐ SEARCH FEE (37 CFR 1.16(k), (i), or (m))	N/A	N/A	N/A	
☐ EXAMINATION FEE (37 CFR 1.16(o), (p), or (q))	N/A	N/A	N/A	
TOTAL CLAIMS (37 CFR 1.16(i))	minus 20 = *		x \$200 =	
INDEPENDENT CLAIMS (37 CFR 1.16(h))	minus 3 = *		x \$ 600 =	
☐ APPLICATION SIZE FEE (37 CFR 1.16(s))	If the specification and drawings exceed 100 sheets of paper, the application size fee due is \$310 (\$155 for small entity) for each additional 50 sheets or fraction thereof. See 35 U.S.C. 41(a)(1)(G) and 37 CFR 1.16(s).			
☐ MULTIPLE DEPENDENT CLAIM PRESENT (37 CFR 1.16(j))				
* If the difference in column 1 is less than zero, enter "0" in column 2.			TOTAL	

APPLICATION AS AMENDED - PART II

	(Column 1)		(Column 2)	(Column 3)	RATE (\$)	ADDITIONAL FEE (\$)
AMENDMENT	10/14/2025		CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*	30	Minus	** 30	= 0
	Independent (37 CFR 1.16(h))	*	3	Minus	*** 3	= 0
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						0
AMENDMENT			CLAIMS REMAINING AFTER AMENDMENT	HIGHEST NUMBER PREVIOUSLY PAID FOR	PRESENT EXTRA	
	Total (37 CFR 1.16(i))	*		Minus	**	=
	Independent (37 CFR 1.16(h))	*		Minus	***	=
	☐ Application Size Fee (37 CFR 1.16(s))					
	☐ FIRST PRESENTATION OF MULTIPLE DEPENDENT CLAIM (37 CFR 1.16(j))					
TOTAL ADD'L FEE						
* If the entry in column 1 is less than the entry in column 2, write "0" in column 3.						LIE
** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 20, enter "20".						/SHIRELL M CARMICHAEL/
*** If the "Highest Number Previously Paid For" IN THIS SPACE is less than 3, enter "3".						
The "Highest Number Previously Paid For" (Total or Independent) is the highest number found in the appropriate box in column 1.						

This collection of information is required by 37 CFR 1.16. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, VA 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. **SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, VA 22313-1450.**

If you need assistance in completing the form, call 1-800-PTO-9199 and select option 2.



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090
76656	7590	10/22/2025	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	
7700 Forsyth Boulevard			PAPER NUMBER	
Suite 1800			2668	
St. Louis, MO 63105			NOTIFICATION DATE	
			DELIVERY MODE	
			10/22/2025	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

uspatents@atllp.com

<i>Decision Granting Request for Prioritized Examination (Track I)</i>	Application No. 19/088,110	Applicant(s) SCHILLEN et al.	
	Examiner JoAnne L Burke	Art Unit OPET	AIA (FITF) Status Yes
1. THE REQUEST FILED <u>14 October 2025</u> IS GRANTED . The above-identified application has met the requirements for prioritized examination A. ☐ for an original nonprovisional application (Track I). B. ☒ for an application undergoing continued examination (RCE). 2. The above-identified application will undergo prioritized examination. The application will be accorded special status throughout its entire course of prosecution until one of the following occurs: A. filing a <u>petition for extension of time</u> to extend the time period for filing a reply; B. filing an <u>amendment to amend the application to contain more than four independent claims, more than thirty total claims</u>, or a multiple dependent claim; C. filing a <u>request for continued examination</u> ; D. filing a notice of appeal; E. filing a request for suspension of action; F. mailing of a notice of allowance; G. mailing of a final Office action; H. completion of examination as defined in 37 CFR 41.102; or I. abandonment of the application. Telephone inquiries with regard to this decision should be directed to JoAnne Burke at (571) 272-458. In his/her absence, calls may be directed to Petition Help Desk at (571) 272-3282.			
/JOANNE L BURKE/ Lead Paralegal Specialist, OPET			

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 76656

Thu Oct 23 05:10:54 EDT 2025

Dear Patent Center Customer:

Correspondence Address:

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MISSOURI, 63105
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 76656, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

Disclaimer:

The list of documents shown below are provided as a courtesy and is not part of the official file wrapper. The content of the images shown in the Image File Wrapper is the official record.

Application	Document	Mailroom Date	Attorney Docket No
19088110	TRACK1.GRANT	10/22/2025	27843-2771 202595US04

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

Please do not reply to this email as it was sent from an unmonitored mailbox.

Sincerely,

The Patent Center Team



UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE
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NOTICE OF ALLOWANCE AND FEE(S) DUE

76656 7590 11/21/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER	
TRAN, PHUOC	
ART UNIT	PAPER NUMBER
2668	

DATE MAILED: 11/21/2025

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	02/23/2026

THE APPLICATION IDENTIFIED ABOVE HAS BEEN EXAMINED AND IS ALLOWED FOR ISSUANCE AS A PATENT. PROSECUTION ON THE MERITS IS CLOSED. THIS NOTICE OF ALLOWANCE IS NOT A GRANT OF PATENT RIGHTS. THIS APPLICATION IS SUBJECT TO WITHDRAWAL FROM ISSUE AT THE INITIATIVE OF THE OFFICE OR UPON PETITION BY THE APPLICANT. SEE 37 CFR 1.313 AND MPEP 1308.

THE ISSUE FEE AND PUBLICATION FEE (IF REQUIRED) MUST BE PAID WITHIN THREE MONTHS FROM THE MAILING DATE OF THIS NOTICE OR THIS APPLICATION SHALL BE REGARDED AS ABANDONED. THIS STATUTORY PERIOD CANNOT BE EXTENDED. SEE 35 U.S.C. 151. THE ISSUE FEE DUE INDICATED ABOVE DOES NOT REFLECT A CREDIT FOR ANY PREVIOUSLY PAID ISSUE FEE IN THIS APPLICATION. IF AN ISSUE FEE HAS PREVIOUSLY BEEN PAID IN THIS APPLICATION (AS SHOWN ABOVE), THE RETURN OF PART B OF THIS FORM WILL BE CONSIDERED A REQUEST TO REAPPLY THE PREVIOUSLY PAID ISSUE FEE TOWARD THE ISSUE FEE NOW DUE.

HOW TO REPLY TO THIS NOTICE:

I. Review the ENTITY STATUS shown above. If the ENTITY STATUS is shown as SMALL or MICRO, verify whether entitlement to that entity status still applies.

If the ENTITY STATUS is the same as shown above, pay the TOTAL FEE(S) DUE shown above.

If the ENTITY STATUS is changed from that shown above, on PART B - FEE(S) TRANSMITTAL, complete section number 5 titled "Change in Entity Status (from status indicated above)".

For purposes of this notice, small entity fees are 40% the amount of undiscounted fees, and micro entity fees are 20% the amount of undiscounted fees.

II. PART B - FEE(S) TRANSMITTAL, or its equivalent, must be completed and returned to the United States Patent and Trademark Office (USPTO) with your ISSUE FEE and PUBLICATION FEE (if required). If you are charging the fee(s) to your deposit account, section "4b" of Part B - Fee(s) Transmittal should be completed. If an equivalent of Part B is filed, a request to reapply a previously paid issue fee must be clearly made, and delays in processing may occur due to the difficulty in recognizing the paper as an equivalent of Part B.

III. All communications regarding this application must give the application number. Please direct all communications prior to issuance to Mail Stop ISSUE FEE unless advised to the contrary.

IMPORTANT REMINDER: Maintenance fees are due in utility patents issuing on applications filed on or after Dec. 12, 1980. It is patentee's responsibility to ensure timely payment of maintenance fees when due. More information is available at www.uspto.gov/PatentMaintenanceFees.

PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

76656 7590 11/21/2025

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	02/23/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
TRAN, PHUOC	2668	382-118000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

1 _____

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☐ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☐ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☐ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☐ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. _____

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature _____

Date _____

Typed or printed name _____

Registration No. _____



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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090

76656 7590 11/21/2025
Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

EXAMINER

TRAN, PHUOC

ART UNIT	PAPER NUMBER
----------	--------------

2668

DATE MAILED: 11/21/2025

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 30 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. The United States Patent and Trademark Office (USPTO) collects the information in this record under authority of 35 U.S.C. 2. The USPTO's system of records is used to manage all applicant and owner information including name, citizenship, residence, post office address, and other information with respect to inventors and their legal representatives pertaining to the applicant's/owner's activities in connection with the invention for which a patent is sought or has been granted. The applicable Privacy Act System of Records Notice for the information collected in this form is COMMERCE/PAT-TM-7 Patent Application Files, available in the Federal Register at 78 FR 19243 (March 29, 2013).

<https://www.govinfo.gov/content/pkg/FR-2013-03-29/pdf/2013-07341.pdf>

Routine uses of the information in this record may include disclosure to:

- 1) law enforcement, in the event that the system of records indicates a violation or potential violation of law;
- 2) a federal, state, local, or international agency, in response to its request;
- 3) a contractor of the USPTO having need for the information in order to perform a contract;
- 4) the Department of Justice for determination of whether the Freedom of Information Act (FOIA) requires disclosure of the record;
- 5) a Member of Congress submitting a request involving an individual to whom the record pertains, when the individual has requested the Member's assistance with respect to the subject matter of the record;
- 6) a court, magistrate, or administrative tribunal, in the course of presenting evidence, including disclosures to opposing counsel in the course of settlement negotiations;
- 7) the Administrator, General Services Administration (GSA), or their designee, during an inspection of records conducted by GSA under authority of 44 U.S.C. 2904 and 2906, in accordance with the GSA regulations and any other relevant (i.e., GSA or Commerce) directive, where such disclosure shall not be used to make determinations about individuals;
- 8) another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c));
- 9) the Office of Personnel Management (OPM) for personnel research purposes; and
- 10) the Office of Management and Budget (OMB) for legislative coordination and clearance.

If you do not furnish the information requested on this form, the USPTO may not be able to process and/or examine your submission, which may result in termination of proceedings, abandonment of the application, and/or expiration of the patent.

Notice of Allowability	Application No. 19/088,110	Applicant(s) SCHILLEN et al.	
	Examiner PHUOC TRAN	Art Unit 2668	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☒ This communication is responsive to Amendment filed 10/14/2025.
☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☒ The allowed claim(s) is/are 36-65. As a result of the allowed claim(s), you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov**.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some* c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
- Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).**
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

- | | |
|---|--|
| 1. ☐ Notice of References Cited (PTO-892) | 5. ☐ Examiner's Amendment/Comment |
| 2. ☐ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date _____. | 6. ☒ Examiner's Statement of Reasons for Allowance |
| 3. ☐ Examiner's Comment Regarding Requirement for Deposit
of Biological Material _____. | 7. ☐ Other _____. |
| 4. ☐ Interview Summary (PTO-413),
Paper No./Mail Date _____. | |

/PHUOC TRAN/
Primary Examiner, Art Unit 2668

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

Reasons for Allowance

2. The following is an examiner's statement of reasons for allowance: None of the prior art whether taken singly or in combination discloses the combination of features "a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor; an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene; at least one optical element, wherein the NIR light passes through the at least one optical element; a processor configured to: output a first image comprising light reflected from the scene and detect a face in the first image; output a second image comprising a reflective light pattern; analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object; determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image; authenticate the detected face based on the depth information and the material classification" required by independent claim 36;

"a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor; an illumination unit comprising: at least one of a first LED and a first laser configured to project NIR light in a light pattern toward the scene; and at least one of a second LED and a second laser configured to project an NIR light beam toward the scene; at least one optical element, wherein the NIR light passes through the at least one optical element; a processor configured to: output a first image comprising light reflected from the scene in response to the projection of the NIR light beam; detect a face in the first image; output a second image comprising a reflective light pattern reflected from the scene in response to the projection of the NIR light in the light pattern; analyze the reflective light pattern to determine

whether the detected face includes depth information consistent with a three-dimensional object; determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image; authenticate the detected face based on the depth information and the material classification” required by independent claim 52;

“a near infrared (NIR) camera configured to capture a scene, the NIR camera comprising a CMOS sensor, and wherein the NIR camera is sensitive to light in a wavelength range of approximately 700 nm to 1100 nm; an illumination unit comprising at least one of an LED and a laser, the illumination unit configured to project NIR light in a light pattern toward the scene; at least one optical element, wherein the NIR light passes through the at least one optical element; a processor configured to: output a first image comprising light reflected from the scene and detect a face in the first image; output a second image comprising a reflective light pattern; analyze the reflective light pattern to determine whether the detected face includes depth information consistent with a three-dimensional object; determine a material classification of the detected face based on a comparison of light intensity distributions in the second image and in a reference image; authenticate the detected face based on the depth information and the material classification” required by independent claim 64. The remaining claims require similar combination of features.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled “Comments on Statement of Reasons for Allowance.”

Conclusion

3. Any inquiry concerning this communication or earlier communications from the examiner should be directed to PHUOC TRAN whose telephone number is (571)272-7399. The examiner can normally be reached 9am-5pm.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at <http://www.uspto.gov/interviewpractice>.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Vu Le can be reached at 571-272-7332. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: <https://patentcenter.uspto.gov>. Visit <https://www.uspto.gov/patents/apply/patent-center> for more information about Patent Center and <https://www.uspto.gov/patents/docx> for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/PHUOC TRAN/
Primary Examiner, Art Unit 2668

Issue Classification 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC						
Symbol					Type	Version
G06V	/	40	/	40	F	2022-01-01
G06T	/	7	/	50	I	2017-01-01
G06V	/	10	/	25	I	2022-01-01
G06V	/	10	/	54	I	2022-01-01
G06V	/	40	/	165	I	2022-01-01
G06V	/	40	/	172	I	2022-01-01
G06V	/	10	/	60	I	2022-01-01
G06V	/	10	/	82	I	2022-01-01
G06V	/	10	/	145	I	2022-01-01
G06V	/	10	/	764	I	2022-01-01
G01S	/	17	/	894	I	2020-01-01
G06V	/	10	/	141	I	2022-01-01
G06V	/	40	/	168	I	2022-01-01
G06F	/	21	/	32	I	2013-01-01
G06T	/	2207	/	30201	A	2013-01-01

CPC Combination Sets				
Symbol	Type	Set	Ranking	Version
/				

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/PHUOC TRAN/ Primary Examiner, Art Unit 2668 (Primary Examiner) _____ (Date) 31 October 2025		O.G. Print Claim(s) 1	O.G. Print Figure 2

Issue Classification 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

INTERNATIONAL CLASSIFICATION			
CLAIMED			
G06V	/	40	/ 40
NON-CLAIMED			
/		/	

US ORIGINAL CLASSIFICATION	
CLASS	SUBCLASS

CROSS REFERENCES(S)						
CLASS	SUBCLASS (ONE SUBCLASS PER BLOCK)					

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/PHUOC TRAN/ Primary Examiner, Art Unit 2668 (Primary Examiner) _____ (Date) 31 October 2025		O.G. Print Claim(s) 1	O.G. Print Figure 2

Issue Classification 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

☐ Claims renumbered in the same order as presented by applicant
 ☐ CPA
 ☐ T.D.
 ☐ R.1.47

CLAIMS															
Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original	Final	Original
	1		10		19		28	2	37	11	46	20	55	29	64
	2		11		20		29	3	38	12	47	21	56	30	65
	3		12		21		30	4	39	13	48	22	57		
	4		13		22		31	5	40	14	49	23	58		
	5		14		23		32	6	41	15	50	24	59		
	6		15		24		33	7	42	16	51	25	60		
	7		16		25		34	8	43	17	52	26	61		
	8		17		26		35	9	44	18	53	27	62		
	9		18		27	1	36	10	45	19	54	28	63		

NONE (Assistant Examiner) _____ (Date) _____		Total Claims Allowed: 30	
/PHUOC TRAN/ Primary Examiner, Art Unit 2668 (Primary Examiner) _____ (Date) 31 October 2025		O.G. Print Claim(s) 1	O.G. Print Figure 2

Search Notes 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

CPC - Searched*		
Symbol	Date	Examiner
G06V40/40, G01S17/894, G06F21/32, G06T7/50, G06V10/141, G06V10/145, G06V10/25, G06V10/54, G06V10/60, G06V10/764, G06V10/82, G06V40/165, G06V40/168, G06V40/172, G06T2207/30201	05/29/2025	PT

CPC Combination Sets - Searched*		
Symbol	Date	Examiner

US Classification - Searched*			
Class	Subclass	Date	Examiner

* See search history printout included with this form or the SEARCH NOTES box below to determine the scope of the search.

Search Notes		
Search Notes	Date	Examiner
Inventor name search	05/29/2025	PT
S.N. 18/264,326 refs. checked	05/29/2025	PT
S.N. 18/468,352 refs. checked	05/29/2025	PT
See search history printout(s)	05/29/2025	PT
See search history printout(s)	08/08/2025	PT
See search history printout(s)	10/31/2025	PT

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<i>Search Notes</i> 	Application/Control No. 19/088,110	Applicant(s)/Patent Under Reexamination SCHILLEN et al.
	Examiner PHUOC TRAN	Art Unit 2668

Interference Search			
US Class/CPC Symbol	US Subclass/CPC Group	Date	Examiner
	See search history printout(s)	10/31/2025	PT

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Bibliographic Data

Application No: 19/088,110

Foreign Priority claimed: ☒ Yes ☐ No

35 USC 119 (a-d) conditions met: ☒ Yes ☐ No ☐ Met After Allowance

Verified and Acknowledged:

/PHUOC TRAN/

Examiner's Signature

Initials

Title:

OPTICAL SKIN DETECTION FOR FACE UNLOCK

FILING or 371(c) DATE	CLASS	GROUP ART UNIT	ATTORNEY DOCKET NO.
03/24/2025	382	2668	27843-2771 202595US04
RULE			

APPLICANTS

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INVENTORS

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Christian LENNARTZ, Ludwigshafen am Rhein, GERMANY

CONTINUING DATA

This application is a CON of 18468352 09/15/2023 PAT 12456334

18468352 is a CON of 18264326 08/04/2023 PAT 12361760

18264326 is a 371 of PCT/EP22/53941 02/17/2022

FOREIGN APPLICATIONS

EPO EP21157800.0 02/18/2021

IF REQUIRED, FOREIGN LICENSE GRANTED**

04/25/2025

STATE OR COUNTRY

GERMANY

ADDRESS

Patent Docket Department

Armstrong Teasdale LLP

7700 Forsyth Boulevard

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St. Louis, MO 63105

UNITED STATES

FILING FEE RECEIVED

\$8,665

PE2E SEARCH - Search History (Prior Art)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
L1	282	((("SCHILLEN") near3 ("Peter")) OR ("GUTHIER") near3 ("Benjamin")) OR ("SCHICK") near3 ("Friedrich")) OR ("GUENTHER") near3 ("Manuel")) OR ("DIESELBERG") near3 ("Lars")) OR ("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2024/09/07 03:49 PM
L2	537832	(G06V40/40 or G01S17/894 or G06F21/32 or G06T7/50 or G06V10/141 or G06V10/145 or G06V10/25 or G06V10/54 or G06V10/60 or G06V10/764 or G06V10/82 or G06V40/165 or G06V40/168 or G06V40/172 or G06T2207/30201).cpc.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	OR	ON	ON	2024/09/07 03:50 PM
L3	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authentikat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:03 PM
L4	1	1 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authentikat\$4 OR identif\$8 OR	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:04 PM

L5	2	verifi\$8)).clm. 2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME project\$4 SAME (illuminat\$4 OR light) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:05 PM
L6	0	2 AND (image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:08 PM
L7	0	(image SAME (face OR facial) SAME skin SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND (project\$4 SAME structur\$4 SAME (illuminat\$4 OR light))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:09 PM
L10	329	(image SAME (face OR facial OR skin) SAME (three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:14 PM

L11	120	OR skin)) 2 AND (image SAME (face OR facial OR skin) SAME (three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR identif\$8 OR verifi\$8)).clm. AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial OR skin))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:15 PM
L12	14	11 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:18 PM
L13	24	10 AND ((project\$4 OR structur\$4) SAME (illuminat\$4 OR light) SAME (face OR facial) SAME skin)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:19 PM
L14	10	13 NOT 12	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:26 PM
L15	106	11 NOT 13	(US-PGPUB; USPAT; USOCR; EPO; DERWENT)	ADJ	ON	ON	2024/09/07 04:29 PM
L16	2	("20190213309" OR "2020082155").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2024/09/07 05:27 PM
L17	305	((("SCHILLEN") near3 ("Peter")) OR (("GUTHIER") near3 ("Benjamin")) OR (("SCHICK") near3 ("Friedrich")) OR (("GUENTHER") near3 ("Manuel")) OR (("DIESELBERG") near3 ("Lars")) OR (("LENNARTZ") near3 ("Christian"))).INV.	(US-PGPUB; USPAT; USOCR; EPO; JPO; DERWENT)	OR	ON	ON	2025/05/29 11:50 AM
L18	3	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 11:55 AM

L19	8	OR light) SAME (structur\$4 OR pattern) SAME (classif\$8 OR authentikat\$4 OR identif\$8 OR verifi\$8)).clm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:00 PM
L20	3	("20150374277" "20160242853" "20190213309" "20200082155" "20200211180" "20210049391" "20220222464" "20240119130").pn. OR ("12288421").urpn. AND (PGPB USPT USOC).dbnm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:02 PM
L21	11	17 AND (image SAME (face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern)) .clm.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:06 PM
L22	11	17 AND (infrar\$4 SAME laser) AND ((face OR facial) SAME (depth OR three-dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:12 PM
L23	11	17 AND (infrar\$4 SAME (camera OR sensor)) AND ((vertical SAME laser) OR VCSEL) AND ((face OR facial) SAME (depth OR three- dimension\$4 OR threedimension\$4 OR 3dimension\$4 OR 3- dimension\$4 OR 3d OR 3-d) SAME (illuminat\$4 OR light) SAME (structur\$4 OR pattern))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:16 PM
L24	5	21 AND 22	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:16 PM
		17 AND (infrar\$4 SAME	(US-PGPUB; USPAT;	ADJ	ON	ON	2025/05/29

		(camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	USOCR)				12:20 PM
L25	0	20 AND (infrar\$4 SAME (camera OR sensor)) AND (((vertical SAME laser) OR VCSEL) SAME (infrar\$4 OR NIF) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:26 PM
L26	3	20 AND ((vertical SAME laser) OR VCSEL) AND (infrar\$4 OR NIF) AND (pseudo\$11 OR random\$4 OR periodic)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:27 PM
L27	0	("20240037201" OR "20240005703" OR "12288421").pn. AND (((vertical SAME laser) OR VCSEL) SAME ((infrar\$4 OR NIF OR pseudo\$11 OR random\$4 OR periodic)))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:39 PM
L28	0	("20240037201" OR "20240005703" OR "12288421").pn. AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL)) AND ((infrar\$4 OR NIF) SAME laser) AND ((infrar\$4 OR NIF OR laser) SAME (pseudo\$11 OR random\$4 OR periodic))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L29	3	("20240037201" OR "20240005703" OR "12288421").pn.	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:43 PM
L30	0	29 AND ((first OR second) SAME ((vertical SAME laser) OR VCSEL))	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:44 PM
L31	3	29 AND ((vertical SAME laser) OR VCSEL)	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:45 PM
L33	3	29 AND ((infrar\$4 OR NIR) SAME laser) AND ((infrar\$4 OR NIR OR laser) SAME (pseudo\$11 OR	(US-PGPUB; USPAT; USOCR)	ADJ	ON	ON	2025/05/29 12:56 PM

L34	9	random\$4 OR periodic)) ("20150374277" OR "20160242853" OR "20160335483" OR "20190213309" OR "20200082155" OR "20200211180" OR "20210049391" OR "20220222464" OR "20240119130").pn.	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/05/29 12:59 PM
L35	14	1 AND ((vertical WITH laser) OR vcsel).clm. AND ((vertical WITH laser) OR vcsel)	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/08/07 01:55 PM
L36	30	(Artificial Intelligence) More like doc: US-20240331450-A1 Text: (US-9358350-B2 OR US-9940709-B2 OR US-10368795-B2 OR US-9672415-B2 OR WO-2019056310-A1 OR US-10922395-B2 OR US-10990805-B2 OR CN-111164610-B OR CN-114676403-A OR US-11741747-B2 OR US-12288421-B2 OR US-12361760-B2 OR US-20240331450- A1 OR US- 20240037201-A1 OR CN-120375419-A OR CN-116897349-A OR EP-4546296-A2 OR KR-20230146540-A OR EP-4295252-A1 OR JP- 2024512247-A).did.	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:08 AM
L37	30	(Artificial Intelligence) More like doc: US-12288421-B2 Text: (US-9358350-B2 OR US-9940709-B2 OR US-10368795-B2 OR US-9672415-B2 OR WO-2019056310-A1 OR US-10922395-B2 OR US-10990805-B2 OR CN-111164610-B OR CN-114676403-A OR US-11741747-B2 OR US-12288421-B2 OR US-12361760-B2 OR CN-120375419-A	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:08 AM

L38	30	OR US-20240331450-A1 OR US-20240037201-A1 OR EP-4546296-A2 OR EP-4295252-A1 OR KR-20230146540-A OR CN-116897349-A OR JP-2024512247-A).did. 36 OR 37	(US-PGPUB; USPAT; USOCR; FIT (AP, AT, AU, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, MC, MD, MY, NL, NO, NZ, OA, PH, PL, PT, RO, RS, RU, SE, SG, SI, SK, SU, TH, TN, TR, TW, UA, VN, WO); FPRS; EPO; JPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/08/08 11:13 AM
L39	50	(Artificial Intelligence) Similar to: 19/088,110 with 0 CPC Selections and 0 Text Selections Text: (US-20240005703-A1 OR US-12288421-B2 OR US-20240331450-A1 OR EP-4546296-B1 OR US-20240037201-A1 OR KR-20230146540-A OR CN-120375419-A OR US-20150262024-A1 OR US-9256794-B2 OR WO-2012087245-A2 OR WO-2025162970-A1 OR CN-109086694-B OR US-10311298-B2 OR WO-2023242875-A1 OR US-11967184-B2 OR WO-2023156317-A1 OR US-20150304535-A1 OR CN-112101186-A OR US-9848113-B2 OR US-20250078567-A1 OR TW-I723529-B OR WO-2015127313-A1 OR US-20250061180-A1 OR US-20160019421-A1 OR	(US-PGPUB; USPAT; USOCR; FIT (AU, AP, AT, BE, BG, BR, BY, CA, CH, CN, CS, CU, CZ, DD, DE, DK, EA, EE, EP, ES, FI, FR, GB, HR, HU, ID, IE, IL, IS, IT, JP, KR, LT, LU, LV, MA, OA, RU, SU, WO, MC, MD, MY, NL, NO, NZ, PH, PL, PT, RO, RS, SE, SG, SI, SK, TH, TN, TR, TW, UA, VN))	OR	ON	ON	2025/10/22 09:41 AM

		US-20150347833-A1 OR US-20200394390-A1 OR US-11068701-B2 OR KR-102332320-B1 OR KR-20160145536-A OR WO-2025036967-A1 OR US-20130202161-A1 OR JP-WO2018079031-A1 OR JP-2019139433-A OR EP-3872658-B1 OR EP-3872658-A1 OR CN-109117700-A OR JP-7318691-B2 OR US-11631278-B2 OR CN-119836650-A OR CN-105874472-B OR CN-107182218-A OR US-12300035-B1 OR TW-201327413-A OR US-20170132456-A1 OR US-11948398-B2 OR US-8891838-B2 OR WO-2018073335-A1 OR CN-118840777-B OR US-20130321585-A1 OR US-9818023-B2).did.					
L40	98	((infrar\$4 OR NIR) SAME camera) AND ((infrar\$4 OR NIR) SAME laser SAME project\$4) AND (light\$4 SAME reflect\$4 SAME detect\$4 SAME (face OR facial)) AND (light\$4 SAME pattern SAME (depth OR three-dimension\$4 OR 3d OR 3-d)) AND ((authenticat\$4 OR classif\$8) SAME (face OR facial))	(US-PGPUB; USPAT; USOCR; EPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/10/31 05:53 PM
L41	2	39 AND 40	(US-PGPUB; USPAT; USOCR; EPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/10/31 05:53 PM
L42	83	40 AND (laser SAME illuminat\$4 SAME project\$4)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/10/31 05:56 PM
L43	77	42 AND (laser SAME illuminat\$4 SAME project\$4 SAME light\$4 SAME pattern)	(US-PGPUB; USPAT; USOCR; EPO; DERWENT; IBM_TDB)	ADJ	ON	ON	2025/10/31 05:58 PM
L44	21	40 NOT 43	(US-PGPUB; USPAT; USOCR; EPO;	ADJ	ON	ON	2025/10/31 06:00 PM

			DERWENT; IBM_TDB)				
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PE2E SEARCH - Search History (Interference)

Ref #	Hits	Search Query	DBs	Default Operator	Plurals	British Equivalents	Time Stamp
N1	2	((infrar\$4 OR NIR) SAME camera SAME laser SAME project\$4 SAME light\$4 SAME pattern SAME reflect\$4 SAME detect\$4 SAME (face OR facial) SAME (depth OR three-dimension\$4 OR 3d OR 3-d) SAME (authenticat\$4 OR classif\$8)).clm.	(US-PGPUB; USPAT)	ADJ	ON	ON	2025/10/31 06:04 PM

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 76656

Fri Nov 21 05:06:24 EST 2025

Dear Patent Center Customer:

Correspondence Address:

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MISSOURI, 63105
UNITED STATES

This is a courtesy notification regarding the following USPTO patent application(s) associated with your Customer Number, 76656, that have new outgoing correspondence. This correspondence is now available for viewing in Patent Center.

The official date of notification of the outgoing correspondence will be indicated on the form (e.g., PTOL-90) accompanying the correspondence.

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The list of documents shown below are provided as a courtesy and is not part of the official file wrapper. The content of the images shown in the Image File Wrapper is the official record.

Application	Document	Mailroom Date	Attorney Docket No.
19088110	NOA	11/21/2025	27843-2771 202595US04

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If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

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Sincerely,

The Patent Center Team



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www.uspto.gov

ELECTRONIC ACKNOWLEDGEMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	12/16/2025 09:11:57 AM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	73609242	FILING DATE	03/24/2025
CUSTOMER #	76656	FIRST NAMED INVENTOR	Peter SCHILLEN
CORRESPONDENCE ADDRESS	-	AUTHORIZED BY	Christopher Goff

Documents

TOTAL DOCUMENTS: 1

DOCUMENT	PAGES	DESCRIPTION	SIZE (KB)
2025-12-16 Issue Fee Transmittal 27843-2771.pdf	1	Issue Fee Payment (PTO-85B)	112 KB

Digest

DOCUMENT	MESSAGE DIGEST(SHA-512)
2025-12-16 Issue Fee Transmittal 27843-2771.pdf	E7268D278DD990AF508B257355D8AEADDEC843F90849946B26 4D6828C8988DE02283D5D5BC709D68C93B8EF20DBDEA8E64 0759D540D3A5AB42DD7A826805BB6EC

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New Applications Under 35 U.S.C. 111

If a new application is being filed and the application includes the necessary components for filing date (see 37 CFR 1.53(b)-(d) and MPEP 506), a Filing Receipt (37 CFR 1.54) will be issued in due course and the date shown on this Acknowledgement Receipt will establish the filing date of the application

National Stage of an International Application under 35 U.S.C. 371

If a timely submission to enter the national stage of an international application is compliant with the conditions of 35 U.S.C. 371 and other applicable requirements a Form PCT/DO/EO/903 indicating acceptance of the application as a national stage submission under 35 U.S.C. 371 will be issued in addition to the Filing Receipt, in due course.

New International Application Filed with the USPTO as a Receiving Office

If a new international application is being filed and the international application includes the necessary components for an international filing date (see PCT Article 11 and MPEP 1810), a Notification of the International Application Number and of the International Filing Date (Form PCT/RO/105) will be issued in due course, subject to prescriptions concerning national security, and the date shown on this Acknowledgement Receipt will establish the international filing date of the application.



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PATENT AND TRADEMARK OFFICE

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ELECTRONIC PAYMENT RECEIPT

APPLICATION #	RECEIPT DATE / TIME	ATTORNEY DOCKET #
19/088,110	12/16/2025 09:11:57 AM Z ET	27843-2771 202595US04

Title of Invention

OPTICAL SKIN DETECTION FOR FACE UNLOCK

Application Information

APPLICATION TYPE	Utility - Nonprovisional Application under 35 USC 111(a)	PATENT #	-
CONFIRMATION #	4090	FILED BY	Barbara McGregor
PATENT CENTER #	73609242	AUTHORIZED BY	Christopher Goff
CUSTOMER #	76656	FILING DATE	03/24/2025
CORRESPONDENCE ADDRESS	-	FIRST NAMED INVENTOR	Peter SCHILLEN

Payment Information

PAYMENT METHOD	PAYMENT TRANSACTION ID	PAYMENT AUTHORIZED BY
DA / 012384	E2025BF012245815	Barbara McGregor

FEE CODE	DESCRIPTION	ITEM PRICE(\$)	QUANTITY	ITEM TOTAL(\$)
1501	UTILITY ISSUE FEE	1290.00	1	1290.00
TOTAL AMOUNT:				\$1,290.00

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New Applications Under 35 U.S.C. 111

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PART B - FEE(S) TRANSMITTAL

Complete and send this form, together with applicable fee(s), by mail or fax, or via the USPTO patent electronic filing system.

By mail, send to: Mail Stop ISSUE FEE
Commissioner for Patents
P.O. Box 1450
Alexandria, Virginia 22313-1450

By fax, send to: (571)-273-2885

INSTRUCTIONS: This form should be used for transmitting the ISSUE FEE and PUBLICATION FEE (if required). Blocks 1 through 5 should be completed where appropriate. All further correspondence will be mailed to the current correspondence address as indicated unless corrected below or directed otherwise in Block 1, by (a) specifying a new correspondence address; and/or (b) indicating a separate "FEE ADDRESS" for maintenance fee notifications. **Because electronic patent issuance may occur shortly after issue fee payment, any desired continuing application should preferably be filed prior to payment of this issue fee in order not to jeopardize copendency.**

CURRENT CORRESPONDENCE ADDRESS (Note: Use Block 1 for any change of address)

76656 7590 11/21/2025

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

Note: A certificate of mailing can only be used for domestic mailings of the Fee(s) Transmittal. This certificate cannot be used for any other accompanying papers. Each additional paper, such as an assignment or formal drawing, must have its own certificate of mailing or transmission.

Certificate of Mailing or Transmission

I hereby certify that this Fee(s) Transmittal is being deposited with the United States Postal Service with sufficient postage for first class mail in an envelope addressed to the Mail Stop ISSUE FEE address above, or being transmitted to the USPTO via the USPTO patent electronic filing system or by facsimile to (571) 273-2885, on the date below.

(Typed or printed name)
(Signature)
(Date)

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090

TITLE OF INVENTION: OPTICAL SKIN DETECTION FOR FACE UNLOCK

APPLN. TYPE	ENTITY STATUS	ISSUE FEE DUE	PUBLICATION FEE DUE	PREV. PAID ISSUE FEE	TOTAL FEE(S) DUE	DATE DUE
nonprovisional	UNDISCOUNTED	\$1290	\$0.00	\$0.00	\$1290	02/23/2026

EXAMINER	ART UNIT	CLASS-SUBCLASS
TRAN, PHUOC	2668	382-118000

1. Change of correspondence address or indication of "Fee Address" (37 CFR 1.363).

☐ Change of correspondence address (or Change of Correspondence Address form PTO/AIA/122 or PTO/SB/122) attached.

☐ "Fee Address" indication (or "Fee Address" Indication form PTO/AIA/47 or PTO/SB/47; Rev 03-02 or more recent) attached. **Use of a Customer Number is required.**

2. For printing on the patent front page, list

(1) The names of up to 3 registered patent attorneys or agents OR, alternatively,

(2) The name of a single firm (having as a member a registered attorney or agent) and the names of up to 2 registered patent attorneys or agents. If no name is listed, no name will be printed.

1 Armstrong Teasdale LLP

2 _____

3 _____

3. ASSIGNEE NAME AND RESIDENCE DATA TO BE PRINTED ON THE PATENT (print or type)

PLEASE NOTE: Unless an assignee is identified below, no assignee data will appear on the patent. If an assignee is identified below, the document must have been previously recorded, or filed for recordation, as set forth in 37 CFR 3.11 and 37 CFR 3.81(a). Completion of this form is NOT a substitute for filing an assignment.

(A) NAME OF ASSIGNEE

(B) RESIDENCE: (CITY and STATE OR COUNTRY)

TRINAMIX GMBH

LUDWIGSHAFEN AM RHEIN, GERMANY

Please check the appropriate assignee category or categories (will not be printed on the patent): ☐ Individual ☒ Corporation or other private group entity ☐ Government

4a. Fees submitted: ☒ Issue Fee ☐ Publication Fee (if required)

4b. Method of Payment: (Please first reapply any previously paid fee shown above)

☒ Electronic Payment via the USPTO patent electronic filing system ☐ Enclosed check ☐ Non-electronic payment by credit card (Attach form PTO-2038)

☒ The Director is hereby authorized to charge the required fee(s), any deficiency, or credit any overpayment to Deposit Account No. 012384

5. Change in Entity Status (from status indicated above)

☐ Applicant certifying micro entity status. See 37 CFR 1.29

☐ Applicant asserting small entity status. See 37 CFR 1.27

☐ Applicant changing to regular undiscounted fee status.

NOTE: Absent a valid certification of Micro Entity Status (see forms PTO/SB/15A and 15B), issue fee payment in the micro entity amount will not be accepted at the risk of application abandonment.

NOTE: If the application was previously under micro entity status, checking this box will be taken to be a notification of loss of entitlement to micro entity status.

NOTE: Checking this box will be taken to be a notification of loss of entitlement to small or micro entity status, as applicable.

NOTE: This form must be signed in accordance with 37 CFR 1.31 and 1.33. See 37 CFR 1.4 for signature requirements and certifications.

Authorized Signature /Christopher M. Goff/

Date December 16, 2025

Typed or printed name Christopher M. Goff

Registration No. 41,785



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APPLICATION NO.	ISSUE DATE	PATENT NO.	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	01/20/2026	12530925	27843-2771 202595US04	4090

76656 7590 01/07/2026

Patent Docket Department
Armstrong Teasdale LLP
7700 Forsyth Boulevard
Suite 1800
St. Louis, MO 63105

ISSUE NOTIFICATION

The projected patent number and issue date are specified above. The patent will issue electronically. The electronically issued patent is the official patent grant pursuant to 35 U.S.C. § 153. The patent may be accessed on or after the issue date through Patent Center at <https://patentcenter.uspto.gov/>. The patent will be available in both the public and the private sides of Patent Center. Further assistance in electronically accessing the patent, or about Patent Center, is available by calling the Patent Electronic Business Center at 1-888-217-9197.

The USPTO is implementing electronic patent issuance with a transition period, during which period the USPTO will mail a ceremonial paper copy of the electronic patent grant to the correspondence address of record. Additional copies of the patent (i.e., certified and presentation copies) may be ordered for a fee from the USPTO's Certified Copy Center at <https://certifiedcopycenter.uspto.gov/index.html>. The Certified Copy Center may be reached at (800)972-6382.

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b) (application filed on or after May 29, 2000)

The Patent Term Adjustment is 0 day(s). Any patent to issue from the above-identified application will include an indication of the adjustment on the front page.

If a Continued Prosecution Application (CPA) was filed in the above-identified application, the filing date that determines Patent Term Adjustment is the filing date of the most recent CPA.

Applicant will be able to obtain more detailed information by accessing the Patent Center (<https://patentcenter.uspto.gov>).

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Application Assistance Unit (AAU) of the Office of Patents Stakeholder Experience (OPSE), Stakeholder Support Division (SSD) at (571)-272-4200.

INVENTOR(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional inventors):

Peter SCHILLEN, Ludwigshafen am Rhein, GERMANY;
Benjamin GUTHIER, Ludwigshafen am Rhein, GERMANY;
Friedrich SCHICK, Ludwigshafen am Rhein, GERMANY;
Manuel GUENTHER, Zuerich, SWITZERLAND;
Lars DIESELBERG, Karlsruhe, GERMANY;

APPLICANT(s) (Please see PATENT CENTER site <https://patentcenter.uspto.gov> for additional applicants):

trinamiX GmbH, Ludwigshafen Am Rhein, GERMANY;

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TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - Correspondence Notification for Customer Number 76656

Thu Jan 08 07:07:33 EST 2026

Dear Patent Center Customer:

Correspondence Address:

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7700 Forsyth Boulevard
Suite 1800
St. Louis, MISSOURI, 63105
UNITED STATES

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Application	Document	Mailroom Date	Attorney Docket No.
19088110	ISSUE.NTF	01/07/2026	27843-2771 202595US04

To view your correspondence online, please sign in to [Patent Center](#) and then select Workbench/View correspondence. To update your email address(es), select Manage/Manage customer numbers.

If you have any questions, please contact the [Patent Electronic Business Center](#) (EBC) at ebc@uspto.gov or 866-217-9197 Monday – Friday, 6 a.m. to midnight ET.

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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
19/088,110	03/24/2025	Peter SCHILLEN	27843-2771 202595US04	4090
76656	7590	01/20/2026	EXAMINER	
Patent Docket Department			TRAN, PHUOC	
Armstrong Teasdale LLP			ART UNIT	
7700 Forsyth Boulevard			PAPER NUMBER	
Suite 1800			2668	
St. Louis, MO 63105			NOTIFICATION DATE	
			DELIVERY MODE	
			01/20/2026	
			ELECTRONIC	

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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APPLICATION NO.	ISSUE DATE	PATENT NO.
19/088,110	20-Jan-2026	12530925

Patent Docket Department
Armstrong Teasdale LLP
St. Louis, MO 63105

EGRANT NOTIFICATION

Your electronic patent grant (eGrant) is now available, which can be accessed via Patent Center at <https://patentcenter.uspto.gov>

The electronic patent grant is the official patent grant under 35 U.S.C. 153. For more information, please visit <https://www.uspto.gov/electronicgrants>

TO: uspatents@atllp.com
FROM: noreply@uspto.gov
CC: patentcenter_eofficeaction@uspto.gov
SUBJECT: USPTO: Patent Electronic System - eGrant Notification for Customer Number 76656

Tue Jan 20 06:05:49 EST 2026

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Correspondence Address:

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Application	Patent	Document	Mailroom Date	Attorney Docket No.
19088110	12530925	EGRANT.NTF	01/20/2026	27843-2771 202595US04

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